

Acknowledgement

As I bring this project to completion, I am profoundly grateful for the support and encouragement I have received throughout this journey. From the initial brainstorming sessions to the final execution, I have been fortunate to receive the assistance of many remarkable individuals whose contributions have been invaluable.

First and foremost, I would like to express my heartfelt thanks to my lecturers for their unwavering dedication and inspiring guidance. Their expertise and encouragement have significantly enriched my knowledge and skills. I am especially grateful to Ms. Shanika, whose exceptional mentorship has been pivotal in shaping my understanding and ensuring the success of this project.

I would also like to extend my appreciation to my classmates and peers, whose collaboration and constructive feedback have been an incredible source of motivation. Your insights and support have encouraged me to aim higher and embrace challenges with determination.

A special thank you goes to my family, whose steadfast belief in my abilities and constant encouragement have provided me with the strength and confidence to persevere. Your love and support have been the foundation of all my achievements.

Finally, I want to thank ESOFT Metro Campus for their assistance and for providing the resources needed to complete this assignment successfully. I also want to express my gratitude to everyone who has contributed to my academic and personal development.

Activity 01

1.1 Scope and Objectives

Enomy-Finances aims to design and implement a computer system to meet their increment of client demands by supporting their staff and clients. The primary objectives include:

1. To develop a User-Friendly System

- Developing an intuitive, user-friendly interface with a reduced learning curve for all staff and clients.
- Make navigation easy between modules while incorporating with error avoidance mechanisms to enhance the use experience of the facility.
- Attach graphical and visual tools to represent financial data in an understandable manner for the clients, so that it helps them for decision-making purposes.

2. To Integrate Modern Technology to enhance accessibility

- Shift to a web-based, scalable platform to handle both, remote and mobile access by clients and staff.
- Ensure cross-platform compatibility, including desktop and mobile devices.
- Implement a cloud infrastructure with minimum latency and reduced downtimes.

3. To enhance client Data Storage protection and processing

- Combine advanced data encryption techniques with authentication protocols to secure sensitive information.
- Use financial regulatory compliances in order to establish trust from the clients and other stakeholders.
- -Employ database management techniques optimized for high transaction volumes without degradation in performance.

4. To Ensure System Reliability and Resilience

- Design robust mechanisms for error handling that prevent any disruption of services and ensures operational continuity.
- Establish a failsafe against network and hardware failures by introducing auto-failover systems and data caching mechanisms.
- Testing the system periodically to detect vulnerabilities and keep updating it often

5. To ensure scalability and customizable solutions

- Planning the design in such a way that it can meet further scalability and extensions if needed.
- Designing modular components that can be updated or replaced without affecting overall system performance.

6. To Empower Clients with Financial Insights

- Leverage data analytics to provide actionable insights, such as predicting savings growth or investment risks.
- Provide scenario analysis tools that will enable clients to simulate different financial outcomes.
- Provide transparent breakdowns of all financial transactions regarding fees, profits, and taxes to make better decisions.

1.2 Economical and financial challenges for this project

- ✓ **Designing a cost effective system** – Balancing of the cost of infrastructure updates, new softwares and devices, and their maintenance along while delivering a high-quality, advanced and future-proof solution.
- ✓ **Implementing suitable transaction fee while maintaining profitability**- Offering fee in a comparatively lower range in order to attract client while also ensuring a cost of profit for the organization as means to cover operational costs.

- ✓ **Managing Budget constraints-** Allocating client's budget in an effective manner to ensure all requirements are met.
- ✓ **Maximizing Return on Investment (ROI)** - Ensuring that the project provides a reasonable ROI within a suitable timespan in order to guarantee a sustainable long-time growth.
- ✓ **Compliance and Regulatory costs-** Adhering to financial industry standards requires investment in legal and procedural areas.

1.3 Description of System Investigation methods.

1. Interviews

Conduct in-depth interviews with the key stakeholders in order to properly identify workflows, challenges and requirements

Advantages

- Provides an in-detail analysis of the daily operations of the client
- Facilitates open-ended discussions which gives the opportunity to uncover unarticulated needs.

Disadvantages

- Time consuming
- Responses may be subjected to influence or personal biases.

2. Observations

Observe staffs during their work hours to gain a firsthand insight of the current system and processes. This can help to identify bottlenecks and vulnerabilities which should be improved.

Advantages

- Offers an un-biased understanding of the system background
- Highlights the difference between reported issues and actual practices.

Disadvantages

- May disrupt normal working methods

- Observer being bias or misunderstanding the technical procedures may lead to inaccurate conclusions.

3. Questionnaires

Distribute surveys to staffs and clients to gather quantitative data. This can give an insight into user satisfaction and feature preferences.

Advantages

- Efficient to collect feedback from a larger crowd.
- Data can be easily analyzed.

Disadvantages

- Only a limited amount of data can be obtained due to a limited depth of response.

4. Document Analysis

Reviewing existing documents about the system such as user manuals, error logs and reports. This can be used to analyse the system performance and recurring issues.

Advantages

- Provides a historical perspective of the system.

Disadvantages

- Outdated and incomplete documents may lead to false assumptions
- Time consuming

5. Surveys

Assemble a small group of end users and discuss specific aspects of the system like the user interface and access control practices.

Advantages

- Provide a diverse perspectives
- Encourages creative inputs and collaborative feedbacks.

Disadvantages

- Group dynamics may influence individual opinions.

6. Board meetings

Conduct meetings with stakeholders to brainstorm and prioritize features.

Advantages

- Builds stakeholder alignment and buy-in for the project.

Disadvantages

- Discussions could deviate from the agenda if not properly moderated.

1.4 User and System Requirements

User Requirements:

The following are the user requirements for the system which addresses the core functionality of currency conversion, savings, and investment calculations, presentation of data, and administration of data. The Currency Conversion Module should be able to handle the six stipulated currencies of GBP, USD, EUR, BRL, JPY, and TRY, and other currencies providing refreshed, up-to-the-minute exchange rates in each currency. It should allow users to input an amount in one currency and determine the equivalent amount in another, considering that the transactions have limitations ranging from 300 to 5000 units of the currency that should be converted. The solution should also implement a computation of charges according to a multi-tier structure that would depend on the amount in conversion. The Savings and Investment Module in the system should enable the client to input an initial lump sum and monthly investment, and an investment type in order to generate an investment quote.

The system shall provide custom investment quotes in periods of 1, 5, and 10 years each, with estimated returns for a minimum and a maximum, total profit, fees, and taxes at each period. This system should also offer comprehensive error handling to address invalid inputs and system failures. Furthermore, the system must be user-friendly, delivering both textual and graphical representations of data, with currency values presented to two decimal places. This will help the users simply interpret the facts supplied to them. Another requirement is that the system should securely store information regarding the users, including personal contact information, transaction history, and saved quotes.

Secure systems like encrypted databases for data storage must be implemented in order to store data.

Another important requirement would be the Accessibility of the system, which might include shifting

from the current client-server approach to a web-based application. This would provide the ability for both workers and clients to use the system from anywhere, giving expanded reach and ease.

System Requirements:

The project will require an improved hardware infrastructures for data processing, storage, and scalability of transactions. High-speed network connections is also necessary to access the most current updates, such as currency conversion rates, in real time, and security methods like encrypted databases for compliance with data protection regulations.

The software should be developed as a web-based or hybrid application. The system should be able to run on many devices ensuring a responsive design and better accessibility for the clients and staff.

On the back-end, this should be supported by robust, scalable technologies, and on the front-end, the system should be implemented in current frameworks thereby to offer responsiveness and ease to users.

The user and transaction data storage database should be a robust and scalable relational database management systems. The system should integrate with APIs to fetch real-time currency exchange rates and to provide secure user authentication and role-based access control mechanisms to secure sensitive data. The system should also contain robust error logging that logs all the problems occurring within the system, including the activities of users. This set up should be in such a way that it logs in both the database and local file system in case of network failure to ensure that no diagnostic information is lost.

Additionally, security is a very important aspect of the system; it has to follow the industry standards of data encryption for secure data transmission. Regular vulnerability assessment and penetration testing should be done to identify and fix any security issues in the earliest time possible in order to ensure operational continuity. Lastly, the system should be scalable, using a modular design approach that allows for easy future updates would give a solution.

1.5 Software Development Life Cycle

Software development life cycle (SDLC) is an organized process that is used to design, build, and test good-quality software. It is a technique that defines the complete practice of software development step-by-step. The purpose of the SDLC life cycle model is to create high-quality, maintainable software that meets the user's needs. SDLC in software engineering models specifies the plan for each stage so that each stage of the software development model can do its role efficiently to provide the product at a cheap cost within a specific time frame that fulfills client needs. In this article we shall see Software Development Life Cycle (SDLC) in full.

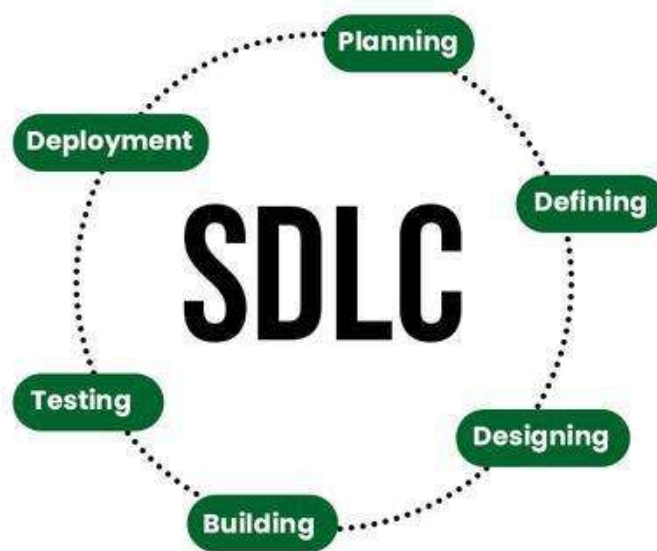


Figure 1: Steps of SDLC

There are 06 steps in SDLC. They are;

01. Planning – Requirement analysis is performed by the developers of the organization. This results of this analysis forms the building blocks of the project.

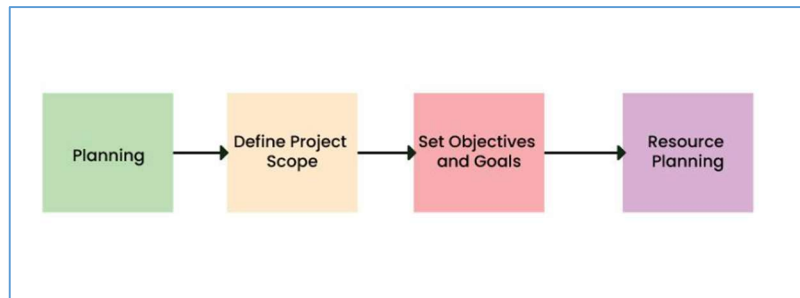


Figure 2: Planning phase of SDLC

02. Defining Requirements – All the requirements needed for the software are specified. These requirements are then approved by users and stakeholders. This is done by utilizing Software Requirement Specification document.

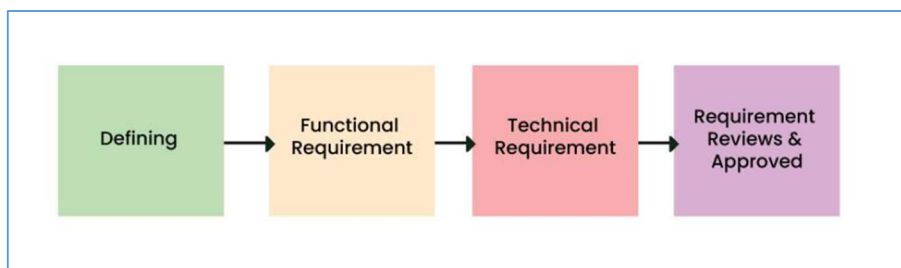


Figure 3: Defining Requirements phase of SDLC

03. Designing – Software designers come up with a best architecture design by using the SRS document. Multiple designs are presented in the Design Document Specification (DDS). These designs are checked by analysts and stakeholders, and the most appropriate one is chosen.

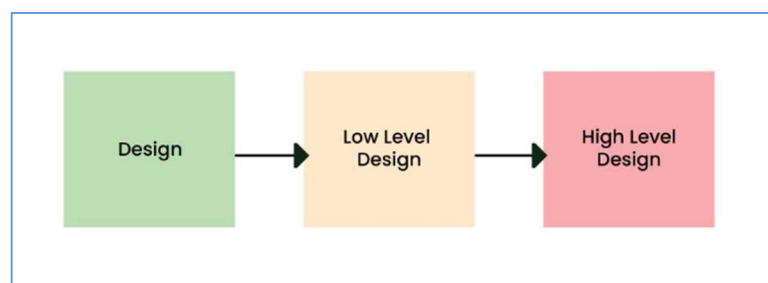


Figure 4: Designing phase of SDLC

04. Developing – The fundamental development of the product starts here. Developers use a specific programming code as per the DDS design. Conventional programming tools like compilers, interpreters, debuggers, etc. are also put into use at this stage.

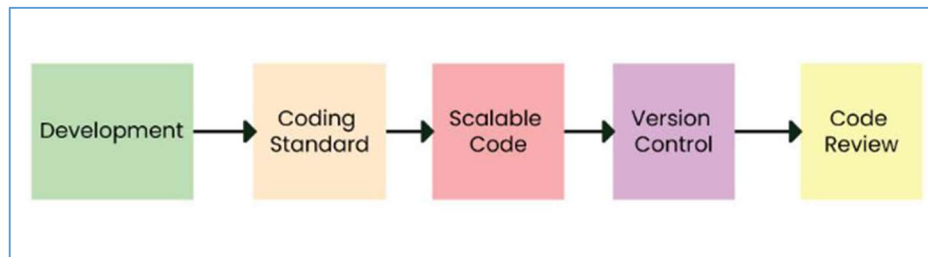


Figure 5: Developing stage of SDLC

05. Testing and Integration- After the development of the product, testing of the software is necessary to ensure its smooth execution. At this stage, all the probable flaws are tracked, fixed, and retested.

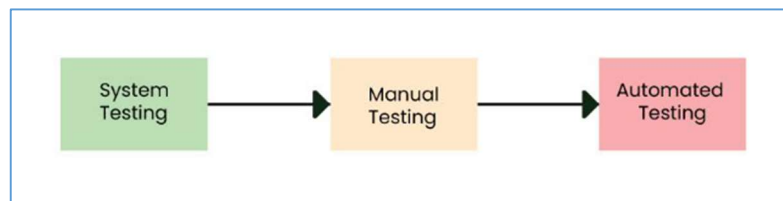


Figure 6: Testing stage of SDLC

06. Deployment and Maintenance- After detailed testing, the conclusive product is released in phases and tested in a real industrial environment. After retrieving beneficial feedback, the company releases it as it is or with improvements.

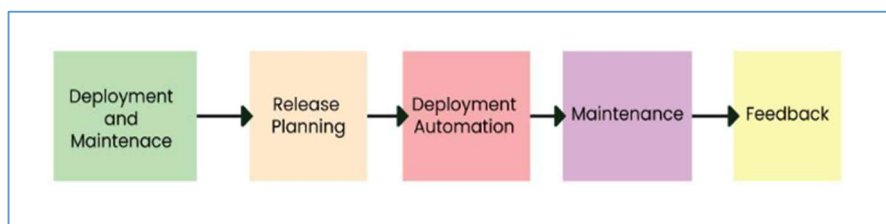


Figure 7: Deployment and Maintenance phase of SDLC.

(www.geeksforgeeks.com, 2024)

1.6 Software Development Lifecycle Models

From the feasibility study to application maintenance, the Software Development Life Cycle (SDLC) is a project management model that outlines the steps in the creation of an information system. There are various SDLC models, each of which follows a certain order of stages to guarantee effective software development. These models provide specific frameworks and approaches for executing the stages of the SDLC. SDLC models can be divided into two types;

- a) Sequential Models
- b) Iterative Models

1.6.1 Sequential Model

A. Waterfall Model

This is a linear sequential model where phrases proceed one after the other in a steady downward flow. Each phrase starts after the previous phrase is completed. Each phrase provides an output that acts as the input for the following phrase. The waterfall model does not allow us to go back to the previous phrase to make any kind of changes. They act according to a fixed plan and changes in the plan will be very costly. This model is suitable for projects where the requirements are clear and unlikely to change.

❖ Advantages

- Simple and easy to understand and use
- Easy to manage due to the rigidity of the model.
- Phases are processed and completed one at a time.
- Works well for smaller projects with well-understood requirements.
- Properly defined stages.
- Clear milestones.
- Arranging tasks is simple.
- Well documented processes and results.

❖ Disadvantages

- No working software is produced until late during the life cycle.
- High possibilities of risk and uncertainty.

- Doesn't suit complex and object-oriented projects.
- Poor model for long and ongoing projects.
- Not suitable for the projects where requirements are likely to change.
- Measuring progress within stages is difficult.
- Cannot accommodate changing requirements.
- Adjusting scope during the life cycle can end a project. (www.tutorialspoint.com, 2012)

❖ Risk Management

In the Waterfall model, successful risk management is incorporated by several essential components. Risk identification is carried out throughout the project planning phase by engaging stakeholders to share their viewpoints and identify potential hazards based on project needs, scope, and restrictions. Risk assessment incorporates qualitative and quantitative evaluations. Qualitative methods can rate the possibility and impact of hazards and quantitative methods assigns numerical values, allowing risks to be ordered by importance. To handle these risks, risk mitigation contains planning tactics such as changing timelines, adding contingency time, and reallocating resources, with particular accountability assigned to a risk owner. Risk monitoring is a continuous process that analyzes project progress and re-assesses risks controls according to new or emerging threats while ensuring stakeholders are informed about mitigation measures, assessment updates, and risk statuses. Finally, contingency planning provides setting up time and financial reserves to manage unexpected events effectively, ensuring the project remains on schedule despite unforeseen problems.

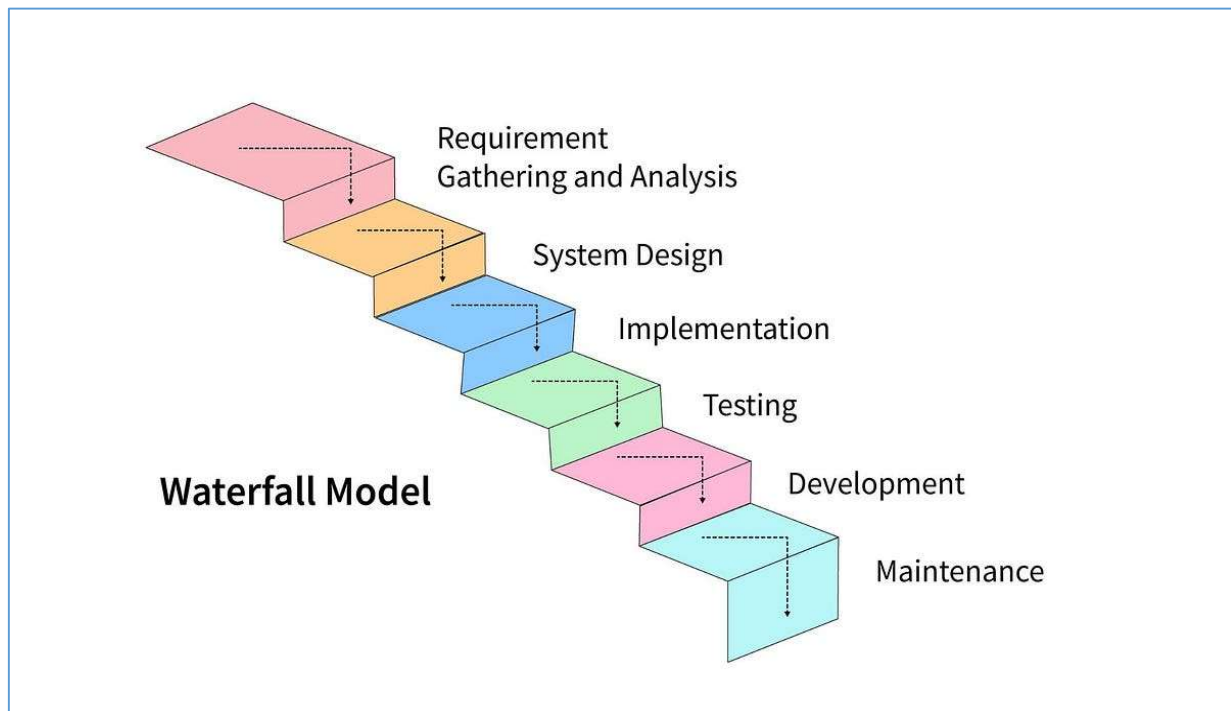


Figure 8: Waterfall Model

B. V- Model

This is an extension of waterfall model. This model greatly emphasizes on testing. Similar to waterfall model, this also follows the downward flow, only after the completion of one phase, we are allowed to go to the next phase. But the key feature is; it associates each phase with a testing phrase also. This maps each developmental stage to the type of testing needed for that particular stage. V-model is suitable for small to medium sized projects where requirement is clear and stable.

❖ Advantages

- Highly-disciplined and Phases are completed one at a time.
- Suitable for smaller projects where requirements are very well understood.
- Simple and easy to understand and use.
- Easy to manage due to the rigidity of the model.

❖ Disadvantages

- High possibility of risk and uncertainty.
 - Not suitable for complex and object-oriented projects.
 - Poor model for long and ongoing projects.
 - Not suitable for the projects where requirements are likely to change.
 - Once an application is in the testing stage, it is difficult to go back and change a functionality.
 - No working software is produced until late during the life cycle.
- (www.tutorialspoint.com, 2012)

❖ Risk Management

In this methodology, risk management is critical at every point. During the Requirements Phase, risks such as unclear, incomplete, or changing requirements, along with misunderstanding amongst stakeholders, must be carefully identified and mitigated through assessments and validations. In the Design Phase, potential dangers such as technological challenges, integration issues, and external dependencies are tested. Using prototypes will help to validate the feasibility of the system design and lowers risks early on. During implementation, frequent code reviews is important to find and fix problems early, thereby minimizing errors and inconsistencies. In the Unit Testing Phase, a complete test strategy must be designed to uncover and prevent their recurrence in later stages. Lastly, during Integration, evaluating integration-related risks are mitigated them by fully evaluating module interfaces is a must.

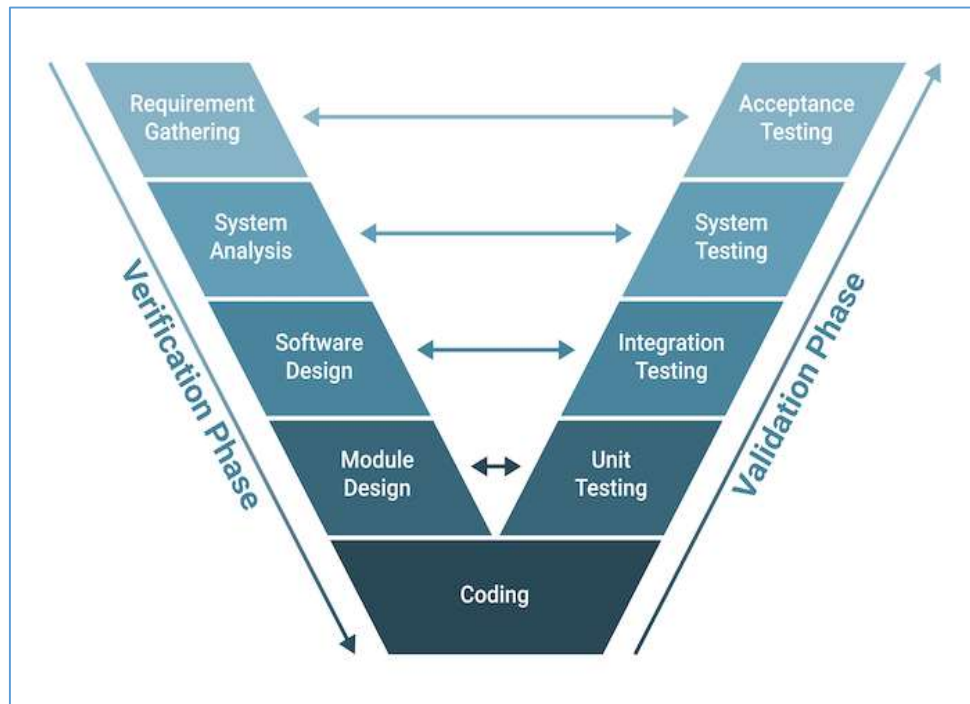


Figure 9: V Model

1.6.2 Iterative Models

A. Agile Model

Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. Every iteration involves cross functional teams working simultaneously on various areas. At the end of the iteration, a working product is displayed to the customer and important stakeholders. The agile method is based on a value system with 04 components called the “Agile Manifesto”. The 04 components are;

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to changes over following a project plan.

The most popular Agile methods include Rational Unified Process, Scrum, Crystal Clear, Extreme Programming, Adaptive Software Development, Feature Driven Development, and Dynamic Systems Development Method (DSDM).

❖ Advantages

- Very realistic approach to software development.
- Promotes teamwork and cross training.
- Rapidly developing and demonstrating functionality.
- Minimum Resource requirements.
- Suitable for fixed or changing requirements projects.
- Delivers early partial working solutions.
- Good model for organizations that change steadily.
- Minimal rules and documentation can be easily employed.
- Enables concurrent development and delivery within an overall planned context.
- Little or no planning required.
- Easy to manage.
- Gives flexibility to developers.

❖ Disadvantages

- Not suitable for handling complex dependencies.
- More risk of sustainability, maintainability and extensibility.
- An overall plan, an agile leader and agile PM practice is a must.
- Strict delivery management is need to meet the deadline.
- Depends heavily on customer interaction, so if customer is not clear, team can be driven in the wrong direction.
- There is a very high individual dependency, since there is minimum documentation.
- Transfer of technology to new team members may be quite challenging due to lack of documentation. (www.tutorialspoint.com, 2012)

❖ Risk Management

In Agile method, risk management focuses on managing risks and guaranteeing project success through continual cooperation and adaptability. Uncertain needs provide a risk resulting to unclear specifications, scope creep, and budgets. To mitigate this, early identification of requirements and a defined change management approach are crucial. Inadequate testing is another danger, as Agile promotes speed and adaptability, this may result in insufficient testing and undetected defects. A complete testing plan helps ensure quality of testing. Agile risk management also stresses in collaborative decision-making. Teams share ownership of risks and accountability for responses. Balancing risks and rewards is important. Therefore teams need to understand risk appetite and make informed decisions by comparing opportunities against potential negatives. Agile teams also prioritize effective risk response techniques, integrating risk management into workflows to proactively address and manage risks throughout the project.

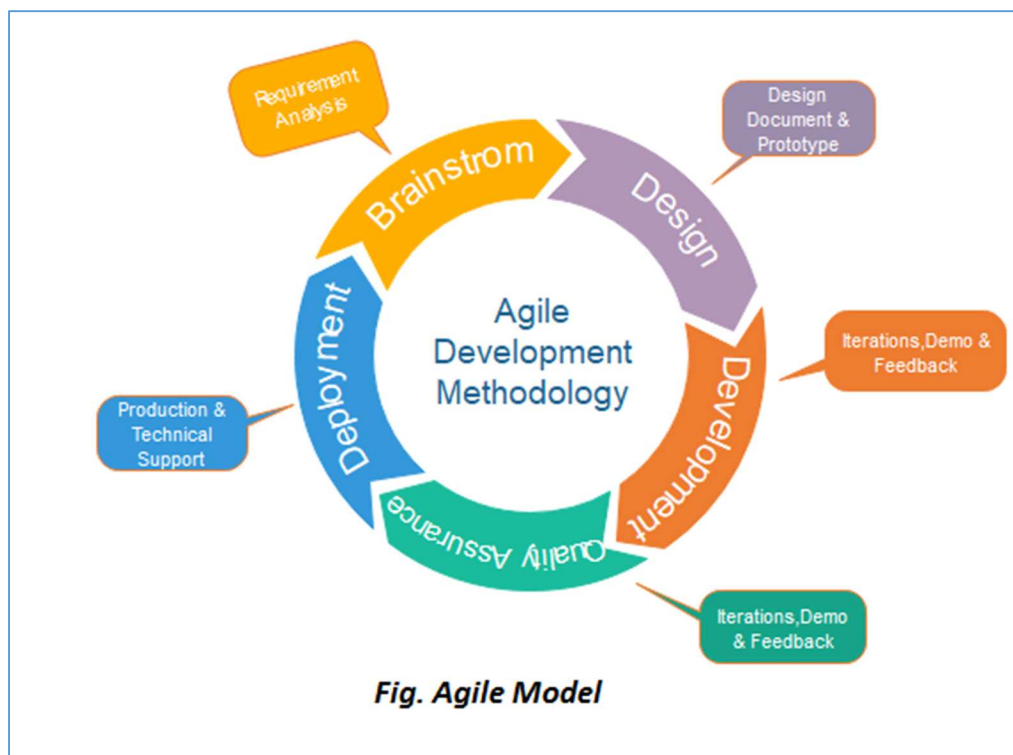


Figure 10: Agile Model.

B. Spiral Method

The Spiral Model is a risk-driven software development process that combines iterative development with systematic risk management. It has a number of phases, or "spirals," in which each loop consists of planning, risk analysis, development, and testing. The concept allows for continuous refinement of the result in several cycles, thus being suitable for large, complex projects with developing requirements. The concentration it places on risk assessment with each iteration aids in the early detection and control of potential issues in the development process.

❖ Advantages

- Changing requirements can be accommodated.
- Allows extensive use of prototypes.
- Accurate capture of Requirements.
- System is seen by the users earlier.
- Development can be divided into smaller parts and the risky parts can be developed earlier which helps in better risk management.

❖ Disadvantages

- Complex management.
- End of the project may not be known early.
- Not suitable for small or low risk projects and could be expensive for small projects.
- Process is complex
- Spiral may go on indefinitely.
- Large number of intermediate stages requires excessive documentation.

(www.tutorialspoint.com, 2012)

❖ Risk Management

- Risk Identification: Early risk identification and management are emphasized under the Spiral Model. The project team assesses potential risks and develops management strategies throughout the planning stage. This ensures that potential issues are resolved before they have an impact on the project's budget or schedule.

- **Risk Assessment:** After hazards are identified, they are thoroughly assessed to ascertain how they might affect the project. This assessment considers factors like the risk's propensity to materialize, the impact's intensity, and the resources required to lessen it.

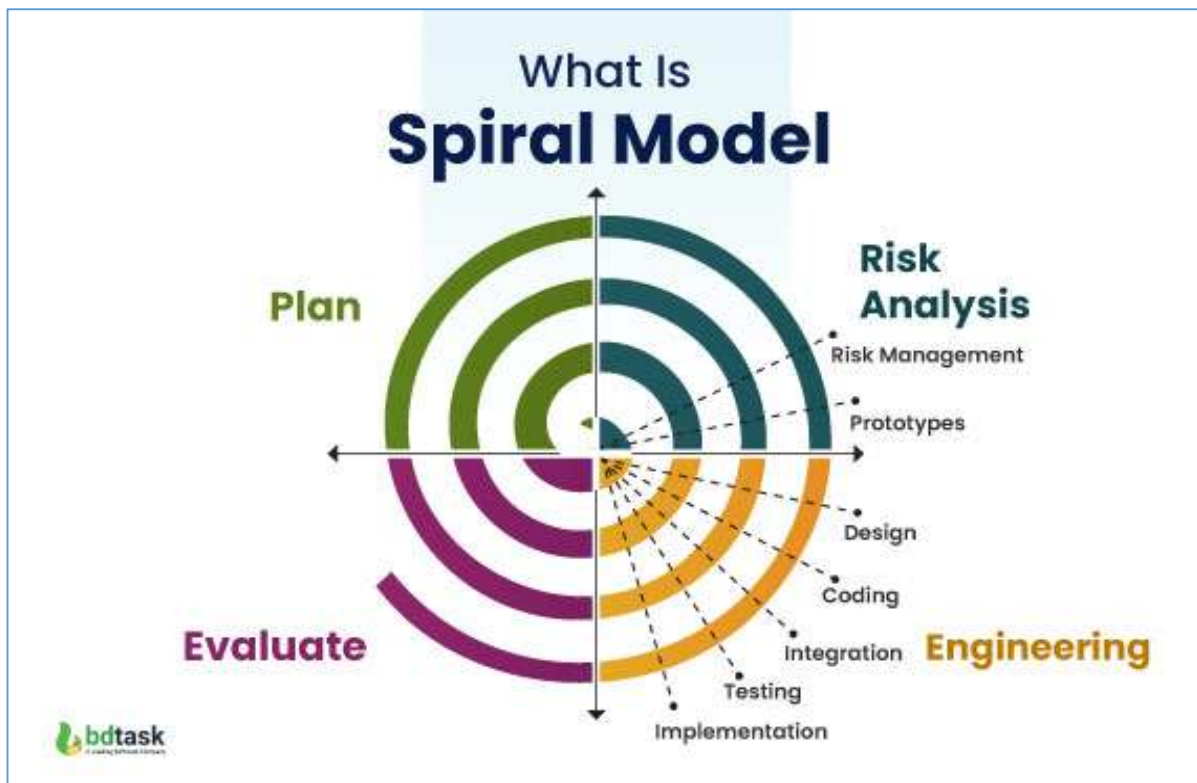


Figure 11: Spiral Model

1.7 Selected SDLC Model for Ecomy Finances

The Waterfall Model is particularly suitable for the Enomy-Finances project as it addresses all three main parts of the project: complexity, scalability, and well-defined requirements. Firstly, the project contains well defined and clear requirements, therefore changes in requirements in the future is highly unlikely; this aspect itself makes the implementation of waterfall model appropriate for this project. Further, the organization's system should be a safe and robust one and also should be able to safeguard sensitive data regarding finances like key functions of currency conversions, investment management, and storing records. Given the criticality of these activities, the orderly and sequential nature followed in the Waterfall Model ensures that each phase-requirements collection, system design, implementation, testing, deployment, and maintenance, is adequately completed before moving to the next. This conscious approach removes risks related to insufficient or poorly integrated components. Such specific requirements of the project, like limitations on transactions and calculations of returns on investment and error-handling protocols, couple well with strengths in pre-planning and heavy documentation that is present in the Waterfall Model. Also, client availability for consultation at every phase-be it planning or designing-solves the conventional drawback of this model in flexibility issues. The emphasis on documentation further promotes compliance with regulations and enhances the reputation of the organization and trust of clients and stakeholders. Therefore, Waterfall Model will be ideal for the production of a system that meets the high standards that Enomy-Finances requires by offering a clear roadmap and allowing rigorous testing to assure security and dependability.

1.7.1 Benefits and Drawbacks of the chosen model

➤ Benefits

1. Clear Structure

The Waterfall Model presents a proper framework for development due to its linear and sequential nature. Each phase, from requirements gathering to deployment, is distinct and follows a logical order, This reduces the confusion and ensures that no step is overlooked. This structure makes it easier for the team to understand the project flow. And also helps to easily ensure that all requirements are aligned.

2. Defined Milestones

Well-defined deliverables at the end of every phase make the progress measurable and maintainable. As an example, completion of the requirements phase delivers a software specification document, while system architecture diagrams are delivered in the design phase. These milestones become necessary to track the progress and affords opportunities to review and confirm the efficiency of the work before proceeding further.

3. Clear and in-detail Documentation

Extensive documentation is one of the highlights of Waterfall Model. Every small thing is documented in great detail-from requirements to design specifications, test cases, and user manuals. That ensures all aspects of the project are well-documented, which will make it easier to keep in track with the project and also it's helpful for future teams to understand the system for maintenance or enhancements. This is especially important in a domain like financial services, where regulatory compliance and constant upgrades are critical.

4. Easy Management

The step-by-step process of waterfall method simplifies project management as the project is divided into manageable, understandable parts. This guides the project manager can allocate resources effectively, make estimations for timelines, and track the progress with much less ambiguity. Predictability, hence, renders the Waterfall Model suitable for projects where an orderly approach is practically a demand.

➤ Drawbacks

1. Inflexibility

One of the major drawbacks of the Waterfall Model is its inflexibility. It is very expensive and time-consuming to revisit a phase once it has been completed if it has to accommodate any changes. For example, if Enomy-Finances decides to add a new investment type after the design phase, this would necessitate revisiting previous stages, which leading to delays and budget overruns.

2. Delayed Feedback

Stakeholders will only be able to see the complete and functioning system at the end of the project.

Without in-between deliverables, assurance between the client's expectations and the end product may only emerge towards the tail-end of a project, making any corrections more difficult and costly to make.

3. Long Timeframe

Since each phase has to be completed for the next one to start, the development process may take a considerably longer time to finish. This sequential approach may not be suitable for dynamic environments where rapid deployment and iterative updates are required. For example, if rapid market changes affect currency exchange rates or investment trends, the system might not be agile enough during its development to adapt to such changes.

By understanding these benefits and drawbacks, it becomes rather clear that, while the Waterfall Model serves effectively for projects with stable and well-defined requirements, its limitations to handle changing needs and deliver early results have to be taken into consideration too.

1.7.2 Appropriateness of the chosen model

The Waterfall Model is particularly suited for the Enomy-Finances project, as it provides a systematic and structured method that emphasizes in-detail planning and careful execution. This is important for establishing a secure and strong system in the sensitive financial sector. Precise features like currency translation, investment management, and safe client data storage are required by Enomy-Finances. All these must also perform effortlessly and precisely in order to meet client expectations and regulatory standards. The model's dependence on a full requirements phase ensures that all functional and non-functional requirements are well understood and documented before development can begin. This reduces potential ambiguity and miscommunication. Also, the willingness of the client to present detailed specifications at the start and being available for clarifications greatly minimizes the limitations imposed by the model's rigidity, thereby allowing the developers to solve out any issues.

Moreover, The model focuses on precise design and intensive testing phases. This goes well with the advanced features required for the project, such as flowy error management, secure data processing, and strict compliance with financial rules. All such processes make sure that potential risks, vulnerabilities, and

failures within the system are identified and treated before its deployment, hence yielding a highly dependable and efficient system. In total, we can conclude that by following the Waterfall Model, the project will maintain a clear roadmap with specified milestones to ensure that the final output will meet the high standards expected in the financial industry, ensuring its long-term maintainability.

1.8 SDLC Design and Development tools.

The correct tool in this ever-changing world of software development may mean the difference between success and failure when it comes to designing and delivering a project. SDLC or the Software Development Life Cycle is a step-by-step procedure for developing software applications, and with the right tools for SDLC, it can highly enhance the overall development process with improved efficiency and effectiveness. An overview of different types of tools used for design and development is given below;

1. Integrated Development Environment (IDE)

An Integrated Development Environment (IDE) is a software tool that offers a complete setting for creating, debugging, testing, and deploying software. It generally comprises a code editor, compiler, debugger, and automation tools, enhancing development efficiency.

- ✓ **Visual Studio:** A versatile and widely used IDE. Supports multiple programming languages like c#, Python and JavaScript. It also contains debugging tools and integrated testing frameworks.
- ✓ **Eclipse:** Suitable for Java development as it offers extensive plugin support. It is also suitable for back-end development.
- ✓ **IntelliJ IDEA:** Ideal for Java or Kotlin. It is known for its intelligent code completion and support for web development frameworks.
- ✓ **Pycharm:** Specially designed for python development. Could be chosen if python is used for scripting or backend development.
- ✓ **NetBeans:** A solution suitable for Java and PHP development. Provides tools for quick prototyping and debugging.

2. Version Management

Version control is a system that monitors modifications to code over time, enabling developers to manage, communicate, and revert to earlier versions when necessary. It ensures team members may work on the same project without overwriting each other's modifications.

- ✓ **GitHub:** Offers robust version control features, collaboration tools, and a platform for hosting and reviewing code. Its integration with CI/CD pipelines simplifies deployment.
- ✓ **GitLab:** Similar to GitHub but includes built-in DevOps tools. This is suitable for end-to-end software development and deployment.
- ✓ **Bitbucket:** Excellent for integrations with Atlassian tools like JIRA for project management.

3. Database Management System (DBMS)

A DBMS is software that facilitates the creation, management, and interaction with databases. It facilitates the effective storage, organization, and retrieval of data, while preserving data integrity and security.

- ✓ **MySQL** - Reliable and efficient for structured data storage. Suitable for managing client records, transaction histories, and financial data.
- ✓ **PostgreSQL** – Known for its robustness and support for complex queries. Ideal for applications which require advanced data handling.
- ✓ **Microsoft SQL server**- An enterprise-level database system. Integrates well with applications developed in Visual Studio.
- ✓ **SQLite**- Suitable for prototyping or applications with minimal database requirements.
- ✓ **MongoDB** – A NoSQL database that offers flexibility in data structure.

4. Prototyping

Prototyping involves producing a visual representation or functioning model of a software program to illustrate functionality and design. It helps obtain client feedback early in development.

- ✓ **Figma**- A cloud-based design tool that facilitates collaboration, making it perfect for designing UI/UX wireframes and mockups for client presentations.
- ✓ **Adobe XD**- Offers powerful features for creating interactive prototypes and design systems.

- ✓ **Sketch-** Excellent tool for designing user interfaces, particularly for macOS users.
- ✓ **InVision-** Allows for the creation of clickable prototypes and gathering feedbacks
- ✓ **Balsamiq** - Focuses on low-fidelity wireframes. Useful for using in early stages of UI design.

5. Testing

Testing is the process of reviewing a software application to ensure it performs as intended, is free of bugs, and fulfils the quality requirements. It can comprise manual and automated testing of features, performance, and security.

- ✓ **JUnit-** A widely-used framework for unit testing in Java-based applications, ensuring that individual components work as intended.
- ✓ **Selenium-** Ideal for automated testing of web applications. Ensures the system behaves correctly across different browsers and devices.
- ✓ **Postman** – Useful for API testing.
- ✓ **Apache JMeter-** Suitable for performance and load testing. Ensures the system can handle the expected volume of data.
- ✓ **TestComplete-** An automated testing tool supporting web mobile and desktop applications.

1.8.1 Tools suitable for the Enomy Finances project

For the efficiency of the Enomy-Finances project, a blend of some selected tools will provide security, efficiency, and ease for the developer. **Visual Studio** will be used as the main IDE because of its versatility, strong debugging capabilities, and its support for most programming languages such as C# and Python. This makes it ideal for the construction of an effective financial system. Speaking of version control, **GitHub** is a solution that by default allows for easy collaboration, and it is also well integrated with the CI/CD pipelines, and has reliable versioning that makes development and deployment way faster. For the database side, **PostgreSQL** provides complex queries without troubles and enhances the integrity of sensitive data, and brings high scalability for financial applications.

Figma is highly ideal for developing the user interface because its cloud-based collaboration allows feedback in real time. Hence, providing faster iterations and more user-friendly prototypes. Quality can be ensured with tools such as **JUnit** and **Selenium**, which provide strong unit and automated testing for web components. Additionally, **Postman** and **Apache JMeter** can handle Application Programming Interface (API) testing and performance evaluation. This ensures the system can handle high loads with seamless functionality. With the usage of all these tools, the project can be smoothly executed and delivered within the deadline.

1.9 Risks related to software lifecycle project and software lifecycle models, and their risk management strategies.

1.9.1 Risks associated with the software lifecycle project and its risk management strategies

A risk, in terms of project management, is some uncertain occurrence or circumstance that may turn out to influence a project's objectives such as scope, timing, budget, or quality in a negative manner. One of the essential parts of securing the success of a project is to recognize and solve those risks. The strategy of risk management includes identification, understanding likelihood and consequence, assuming measures for the reduction, transfer, accepting, or avoiding of the threats.

Following is a table that demonstrates a few of the potential risks the Enomy-Finances project may run into, based on scope management issues, resource unavailability, and technical problems. The mentioned risk analysis also provides with a personalized solution in terms of mitigating effects in order not to disrupt project implementation.

Risk	Explanation	Risk Management Strategy
Scope creep	Addition of features and making changes for the initial requirements will lead to delays and increase the costs	- Clearly define and analyse the requirements. - Implement a change control process to evaluate and approve for changes.
Budget Issues	Overspending due to false estimations, unexpected expenses due to changes in requirements or any failures	- Create a detailed budget plan. - Allocate contingencies for unexpected expenses. - Monitor and analyse the costs.
Staff Issues	Lack of skilled staffs, Unavailability of key team members will lead to delays in project upgrades	- Ensure to have a proper team - Guide the staffs with relevant knowledge. - Have backup team members.
Resource Constraints	Lack of proper hardware, softwares or infrastructures to support the project can lead to performance compromises and project delays	- Conduct and document a resource needs assessment. - Acquire required resources in advance.
Technical Challenges	Issues related to integration of new technologies, compatibility problems, or unexpected bugs will affect the performance of the system.	- Perform a feasibility study before starting the project. - Use prototyping to identify issues. - Allocate time for thorough testing and debugging.
Data Security Risks	Potential breaches of sensitive data may lead to trust issues and reputational damages.	- Implement proper encryption and authentication controls. - Conduct security audits often.
Gaps in client communication	Misunderstandings of requirements or delayed feedbacks from the client can result in an incorrect system or missed deadlines	- Establish clear communication channels - Schedule regular meetings. - Keep the stakeholders informed for any project updates
Timeline overruns	Delays in project phases due to dependency on other tasks or unexpected issues.	- Create realistic project timelines - Use project management tools

Regulatory Compliance issues	Failure to comply with financial regulations could lead to legal and operational challenges.	- Stay updated with regulatory changes. - Implement compliance checks for the system
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1.9.2 Risks associated with the software lifecycle models and its risk management strategies

The software development lifecycle models contain a variety of risks that, if not addressed properly, will affect the success of the project. For example, the structured nature of Waterfall Model makes it vulnerable to flexibility concerns, while Agile's iterative nature makes it prone to risks related to scope creep, which is generated through frequent changes. Also, risk analysis cause cost and complexity challenges in the Spiral Model, and the very strict phase dependency of the V-Model results in delay. It is crucial to understand the above risks and adopt appropriate management solutions to ensure that the chosen lifecycle model matches the project's goals and limits.

Lifecycle Model	Key Risk	Description	Risk Management Strategy
Waterfall	Flexibility issues	The structure of the waterfall methods makes it difficult to incorporate changes especially after the requirement and design stages are passed.	- Conduct comprehensive requirement analysis - Keep track with the evolving technologies in the industry. - Ensure that the client is satisfied and clear about their requirements.
V- model	Phase dependency delays	Due to the strict sequence development and testing phases; if any one phase is delayed, it can affect the whole project timeline	- Establish clear timelines and dependencies for each phase. - Use automated testing tools.

			• Overlap compatible phases to accelerate the process.
Agile	Scope creep	Agile's iterative nature allows to apply frequent changes to the project, if needed. This can lead to unexpected expansion of the project scope which will increase costs and delayed timelines.	• Define a clear scope with prioritized deliverables. • Use sprint reviews and stakeholder meetings to limit expansions. • Implement a proper protocol to analyse and approve changes.
Spiral	Cost and Complexity	Spiral's iterative risk analysis and development can make the projects time consuming, expensive and overly complex, particularly for a small and low-risk projects	• Rank risks and focus on the high priority ones to reduce unnecessary iterations. • Balance costs with a cost-benefit analysis.

Activity 02

2.1 Feasibility Study

A feasibility study is an assessment that defines how appropriate the idea of the proposed project is likely to be, such as a new product line or technical system. The study will analyze the relevant factors surrounding the project, including technical, economic, and legal considerations, to decide if the project is worth investing time into. It may also analyse all the issues that could arise.

Feasibility studies also enables the organization with new business development, identification of how it would work, its challenges, competition, market analysis and the extent and source of funding required for expanding the business. They may help formulate a marketing strategy to attract the investors and bankers to assure that investment in a specific project or venture is a worthy decision.

By the end of a feasibility study, stakeholders should have a thorough understanding of all aspects of the project and can decide whether to proceed further with the project. For instance, a manufacturer may review a feasibility study to determine whether the expansion of a plant will pay off in increased revenue. The feasibility study can either be done by a company's management or by its directors who can further pass the process onto senior managers if the organization lacks deep profound expertise and/or has no time for performing the research. (Indeed, 2024)

2.1.1 Importance of Feasibility Study

Feasibility studies are vital in the analysis of the practicality and possible success of any development before spending considerable amounts of time, resources, and finances toward it. Feasibility studies enable an organization to recognize whether a project proposal is possible through determining their various parameters, including financial viability, technical feasibility, resource availability, and possible threats. The most important benefit of a feasibility study is that it finds valid reasons whether to forward or cancel a project idea so that decisions are made based on analysis rather than assumptions. This ensures that Projects face a minimization of possible failures through the early detection of possible issues and encouraging realistic expectations regarding a project's eventual outcome.

It also helps to focus on project teams' activities and coordination in terms of accomplishing certain goals. Giving clear directions through scope, objectives, and constraints of the project, aids in aligning processes

to the project vision. In addition, feasibility study reduces business possibilities examines options available and selected which is the most feasible. This extensive study ultimately leads not only to the saving of time and resources in traditional business decision making, but also in identifying unique solutions that may have been overlooked in other routes. The study will also assess existing and required resources (managing staff, financing, infrastructure, and technical sides).

Increased chances of projects succeeding are generated from conducting a feasibility study is yet another factor which can identify and sometimes mitigate possible risks. Apart from that, it also offers clues concerning various conditions like market conditions, competition, regulatory conditions, possible problems, etc. with the aim of enabling systems to exclusively deal with conditions that might hinder their development.

2.2 Feasibility Report

A feasibility report is a critical document that evaluates the practicality and viability of a proposed project. Its main purpose is to provide decision-makers with an objective analysis to determine whether the project should proceed. The report assesses various factors, including technical, economic, legal, operational, and scheduling aspects, to gauge the feasibility of the project. It enables stakeholders to make informed decisions, identify potential risks, and allocate resources effectively. A typical feasibility report contains essential components such as an executive summary, project description, market analysis, technical feasibility assessment, financial analysis, operational evaluation, legal compliance check, risk analysis, and concluding recommendations. By conducting a feasibility report, organizations ensure a thorough examination of the project's potential success and lay the groundwork for a successful venture. (Indeed.com, 2024)

2.2.1 Key contents of a Feasibility Report

- a) **BACKGROUND** – Introduction to the organization and the nature of their business, why they need a new system and other necessary information.
- b) **PROBLEM DEFINITION** – An overview of how the current system works and what are the problems observed in the system that needs improvements.

- c) **PROBLEM SOLUTION**- A description of what functionalities that will be newly included into the system.
- d) **SCOPE OF THE SYSTEM**- What functions will be automated by the new system (the boundary of the system)
- e) **OBJECTIVES OF THE SYSTEM**- The expected deliverables of the new system, and what improvements are to be made to the existing system.
- f) **LIST OF KEY STAKEHOLDERS AND THEIR ROLES**- An analysis of the entities which will be influenced by this project, like project sponsor, project managers, system analysts, software developers and testers, suppliers, customers and other stakeholders, along with their roles in this project.
- g) **FEASIBILITY CRITERIA**- Evaluation of Operational, Technical, Economic, Legal and Schedule feasibility of the new system.

i. Operational Feasibility

This is an investigation to identify the impact of the new system on company operations. Staff and clients. The new system must not influence a negative impact on the operation of the organization and its staffs and clients. This system should align with the organization's goals and user adoptions. Staff members that has any form of resistance should be identified, and necessary trainings should be provided.

ii. Technical Feasibility

Examines if the hardware and software needed for the new system can be acquired and used effectively without any forms of hindrance. Any possible risks associated with unavailable or unknown technologies must be identified. An examination of whether the developers and other technical staffs have the necessary competence to handle proposed technology must also be conducted.

iii. **Economic Feasibility**

This form of analysis examines if the new system will generate enough financial benefits. If financial benefits outweighs the costs, then the project is economically feasible. This is ascertained by a Cost/Benefit analysis. Costs include; initial development cost, hardware and software cost, ongoing costs and maintenance costs. Benefits include savings on staff reduction, cost savings by minimizing waste, increased sales revenue and etc.

Normally a Cost/Benefit analysis is performed for a number of years (atleast 04 years), starting from the initial year of operations. Next, the Return of Investment (ROI) and payback period must be considered.

iv. **Legal Feasibility**

Examines if the new system will function within the legal framework of the country/s where it will be operating. For instance, an e-commerce system that collects personal data of its clients must adhere to the privacy guidelines such as the UK Data Protection act 1998 and EU Data Protection regulations.

v. **Schedule Feasibility**

Analyzes if the project can be completed within the given time frame, in other words; if the time duration is sufficient to effectively manage workloads and deliver it before the launch day.

- h) **RISK ANALYSIS OF THE NEW SYSTEM-** Includes early analysis of the risks that can be encountered when planning and executing a project, and planning contingencies for those risks. Risks can include; developer incompetence, unclear requirements, resources unavailability and etc.
- i) **SCHEDULE-** Task breakdown with appropriate and realistic timelines.
- j) **RECOMMENDATIONS-** Making clear and well-supported recommendations regarding the best solution or cause of action. The recommendations should align with the evaluation of the feasibility criterias and must align with the objectives and scope of the project. (Lecturer Notes, 2024)

2.3 Impact of Feasibility Study in Enomy Finances

A feasibility study plays a vital role in deciding the success of the Economy-Finance project.

Feasibility study helps the project by thoroughly examining its viability across numerous dimensions, including financial, technical, operational, legal, schedule, and economic criteria. This complete assessment identifies potential risks, aligns the project with organizational objectives, and establishes a strong foundation for planning and execution.

Financial feasibility is crucial in ensuring the project stays under budget, maximizing resource allocation, and preventing overspending. It assesses funding sources, cost estimates, and the overall financial sustainability of the planned system by implementing proper financial analysis. Technical feasibility evaluates whether the current technological infrastructure can support the project's requirement, This also ensures if the proposed system meets the organization's needs, particularly for handling sensitive financial data and implementing complex functionalities like currency conversion and investment planning and generating quotes.

Operational feasibility focuses on how well the system will integrate into existing workflows and its usability for both staff and clients. This involves that the implementation of the system ensures that the system's user-friendly interface, efficient processing, and support for safe data handling correspond with operational requirements. Operational feasibility also ensures that no harm is caused for clients like difficulty in using the system, and staff such as high reduction of jobs, if the system is developed. Legal feasibility assures compliance with regulatory standards such as data protection laws and financial rules. This lowers the chance of legal difficulties. Schedule feasibility examines the practicality of the suggested deadlines to guarantee the project can be finished on schedule without compromising the quality of the system.

- ✓ Moreover, the economic feasibility of the project is analyzed by examining its potential for creating returns on investment, cost savings, and contributing to the organization's growth and competitiveness in the financial industry. This is measured using the Cost/Benefit analysis.

Therefore, it can be said that feasibility study helps reveal potential obstacles early in the process, such as resource limitations, technological constraints, or compliance issues, and provides practical recommendations to mitigate them. These reviews greatly minimizes uncertainty, improves informed

decision-making, and enhances the possibility of delivering a strong, secure, and user-focused solution.

Activity 03



Figure 12: Presentation Slide 01

Speaker notes:

Good Morning/Afternoon everyone! Firstly, thank you for coming.

My name is Isma Haseemdeen, I work as a junior software developer at Phonyt Digital Solutions. Today's presentation is about covering the application of an appropriate software development approach to plan the implementation of the application our client, of Enomy-Finances.

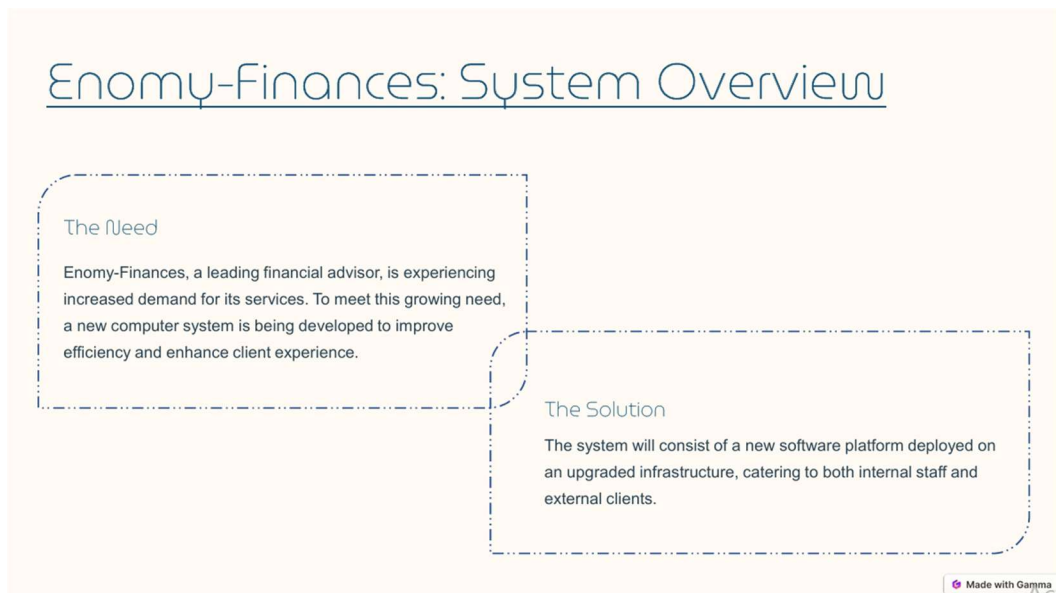


Figure 13: Presentation Slide 02

Speaker notes: Firstly, let me give you'll a short introduction about our client; Enomy Finances.

Enomy finances is an organization in the financial sector that provides advices and services related to mortgages, savings, and investments. They access client's incomes and outgoings and provide advice on budgeting and consolidating debt. They provide access to savings and investment opportunities and they also manage investment portfolios.

Economy finances have recently seen an increase in demand of their services. So they have decided to expand their services. As part of their expansion, it would like to commission a new computer system to support both their staffs and clients.

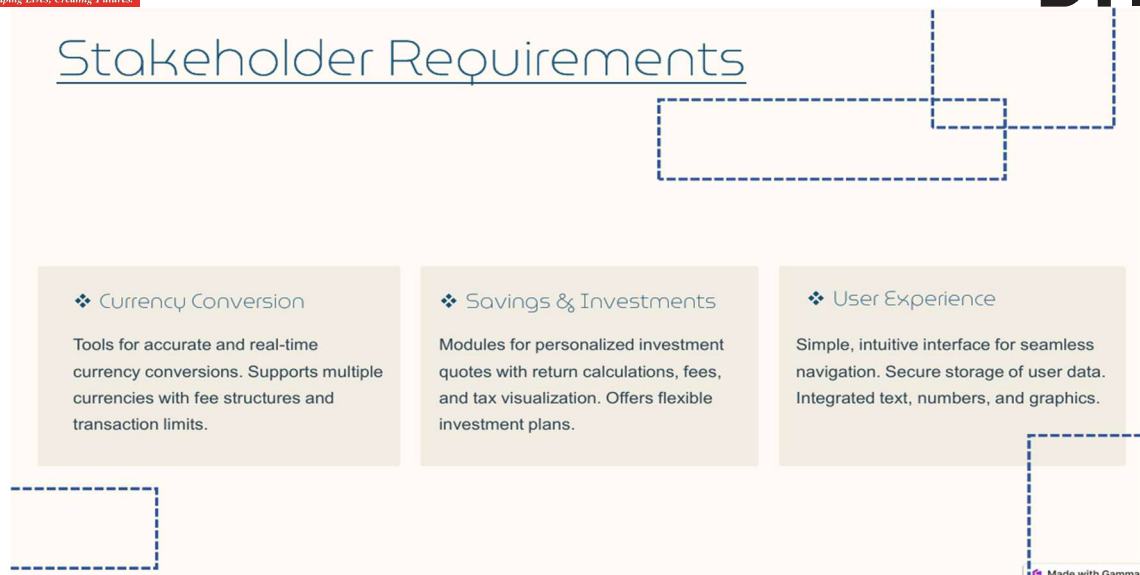


Figure 14: Presentation Slide 03

Speaker Notes: Next, let's look at the stakeholder requirements of this project. The client is expecting a currency conversion module, which includes tools that calculate cost of purchasing different types of currencies, supporting various types of currencies with a fee structure for each currency transaction and also implementing a transaction limit amount.

The client is also expecting a Savings and Investment module that should include flexible investment plans, and generating potential quotes client according to their incomes and outgoings. The quotes must contain information about return calculations, fee, monthly contribution and tax visualizations.

Along with those, our client is also expecting a simple and intuitive interface for easy navigation, a secured storage space to store sensitive data and support for users through appropriate use of textual, numeric and graphical data representation.

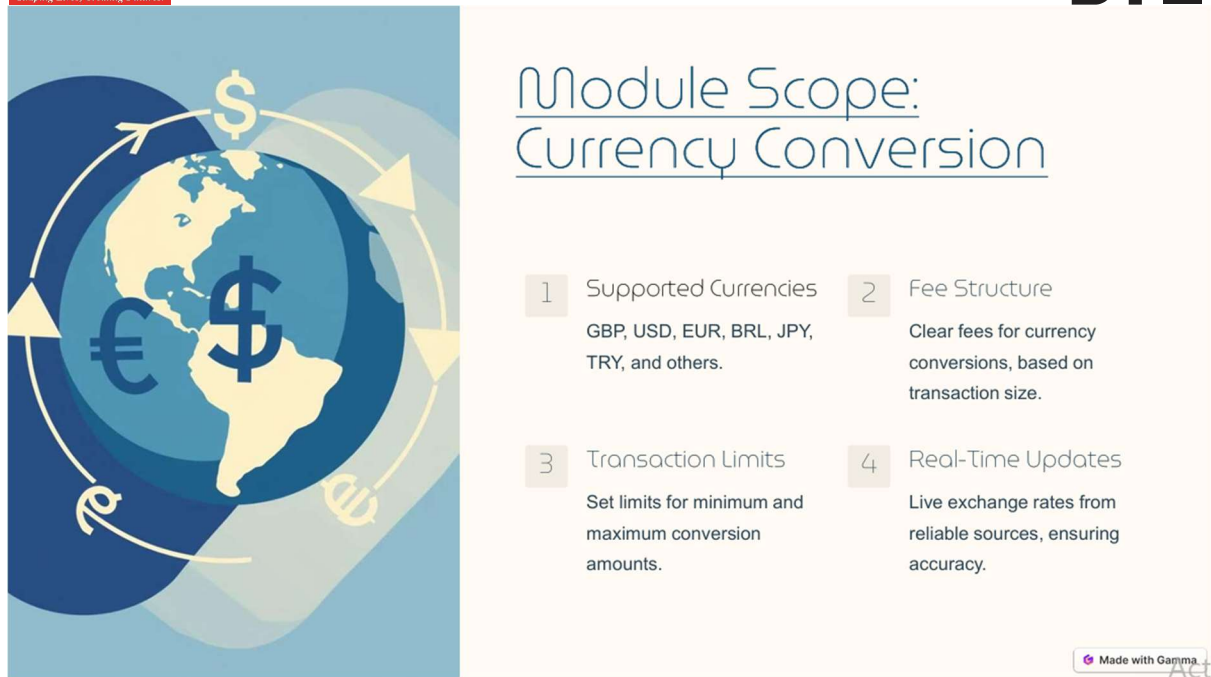


Figure 15: Presentation Slide 04

Speaker Notes: Let's now take an insight into the Currency conversion module, which is one of the key feature of this system. This is designed to support the financial operations for both staff and clients. This module will include a range of currencies like GRP, USD, EUR, BRL, JPY, TRY and etc that supports to make their transactions. Each transaction will be charged with a transaction fee which will depend on the transaction size. A minimum transaction limit and a maximum transaction limit should also be implemented, that should be taken into consideration when performing a transaction. Lastly, Real-time updates regarding the currency rates must be gathered from a reliable and trustworthy source, hence ensuring the accuracy of the information.

Savings and Investments Module

☐ Personalized Quotes

The module will generate tailored quotes based on individual client preferences and financial situations.

☐ Return Calculations

The system will calculate projected returns over different timeframes (1, 5, and 10 years), providing clients with a clear understanding of potential outcomes.

☐ Visualization of Fees and Taxes

The system will transparently display fees and taxes associated with investments, enabling clients to make informed decisions.

☐ Investment Plans

Clients will be able to view investment plans to get an overview of their desired type..



Figure 16: Presentation Slide 05

Speaker Notes: The Savings and Investments Module provides clients with personalized financial tools to manage their savings and investments effectively.

The modules includes Generation of personalized quotes for the clients according to the preferences and financial statuses. The systems will also guide the clients on return calculations, giving them a clear understanding of the potential outcomes of their investments. The system will also transparently display fees and taxes associated with the investments, thereby helping them to make informed decisions. Lastly the clients can also have a look at Investment Plans to get a knowledge of their desired type of Investment.

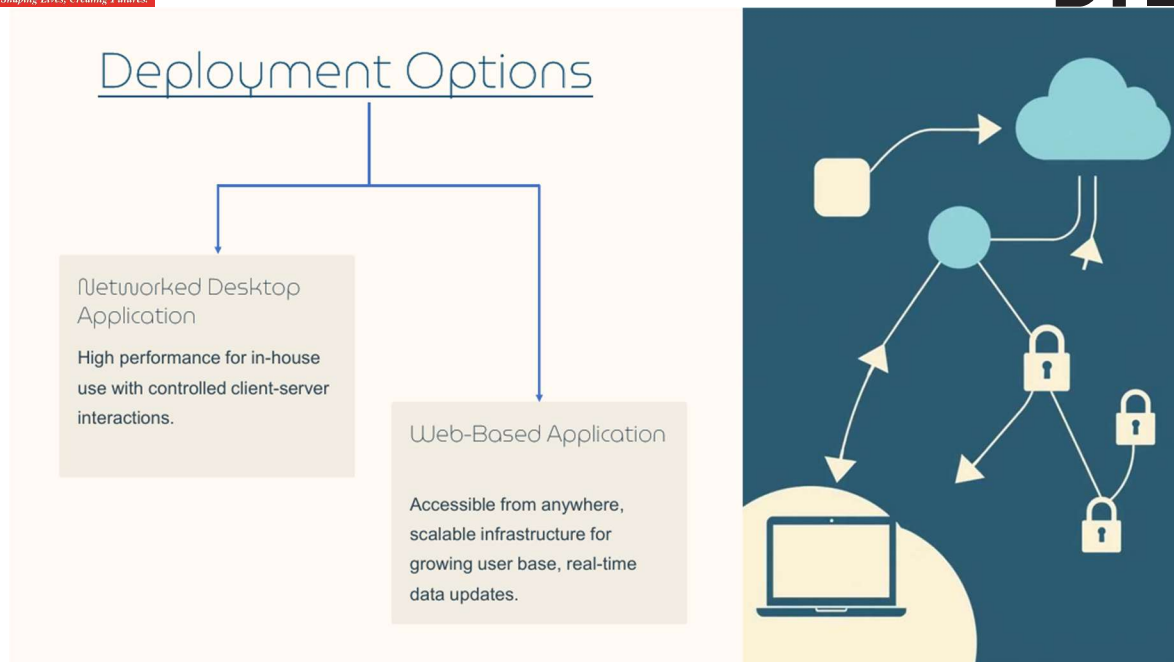


Figure 17: Presentation Slide 06

Speaker notes: According to the client, they wanted a computer system to expand their company's demands. Therefore, a suitable computer system for a company like Enomy Finances can be of two forms. Both of the forms have their own advantages

First is the Networked Desktop System. By deploying a system like this, high performance for in-house use and a controlled client-server interaction can be ensured which will result in a more secured environment. On the other hand, if we deploy a web-based application; the system will be accessible from anywhere building the availability. The system will also be scalable for growth and updates in terms of infrastructure and will have a faster access to real-time updates.

Both of the options will be ideal for the deployment of this system, with each having its own features that is efficient for the Enomy Finances system.

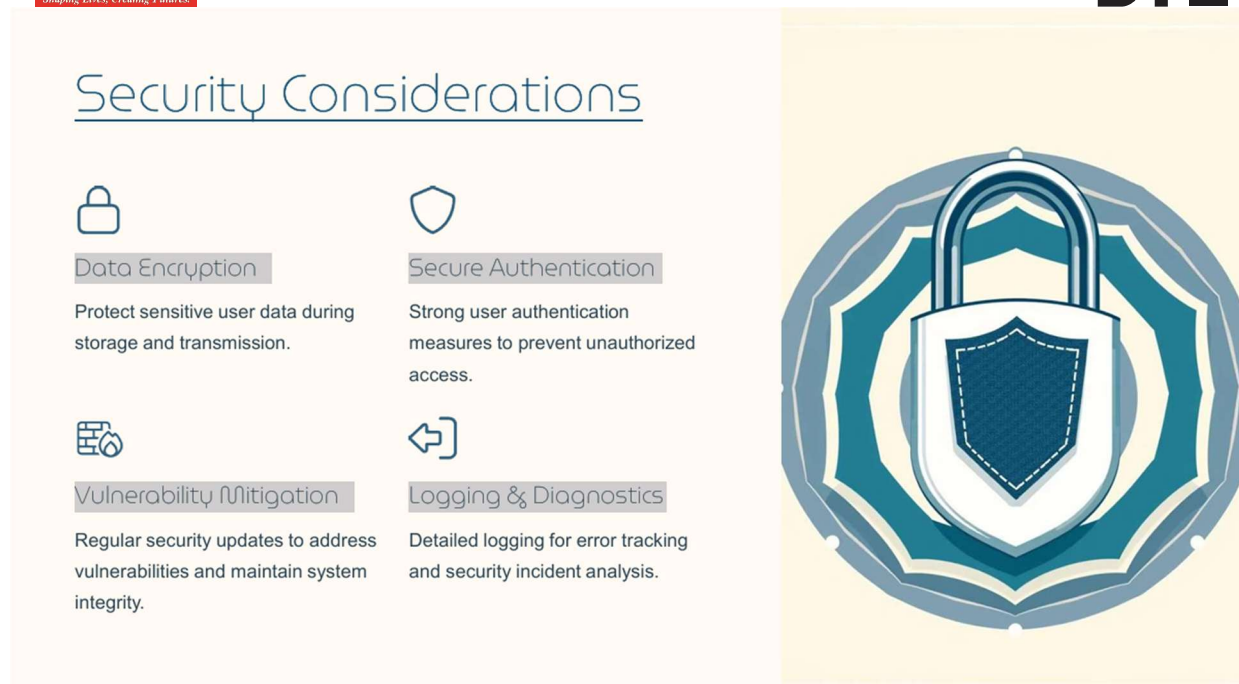


Figure 18: Presentation Slide 07

Speaker Notes: In order to protect the system from any forms of security related risks like malware attacks, unauthorized access and data breaches, certain security considerations must be taken into account. First is the encryption of Data. Encrypting sensitive data like client information, income records, and transaction records during both, at rest and at transit will protect the system from data leakages and ransomware attacks.

Following secure authentication procedures like implementing strong password practices, regularly updating passwords, following a two-factor authentication or multi-factor authentication procedure will secure the system's data from unauthorized accesses.

Regularly conducting audits and updating the security of the system will prevent the system from network vulnerabilities, thereby preventing operational downtime.

Finally, detailed logging for error tracking and security analysis will provide an extra layer of protection to the system

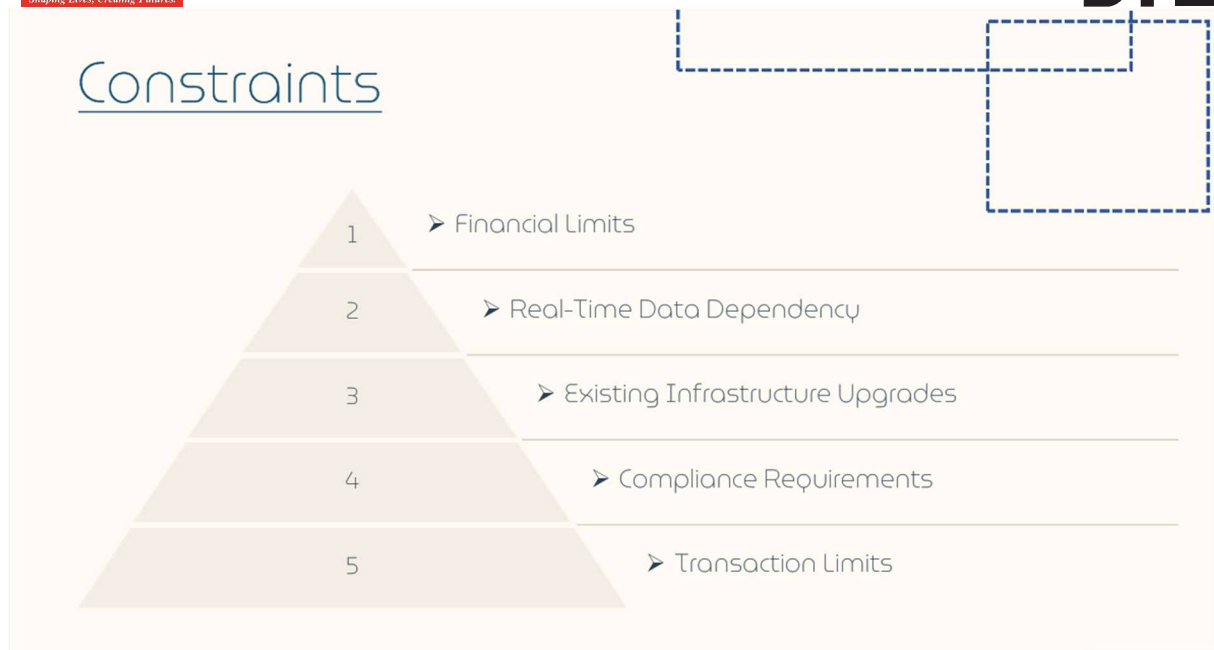


Figure 19: Presentation Slide 08

Speaker Notes: Up next, let's have a look at some of the constraints of this project. The first one is the financial limits. Which is, budget restrictions may impact the scope of the project and also reduce the quality and standard of the project.

Next is real-time data dependency. This system relies of real-time updates on currency rates. Therefore the system is affected by external sources. Any forms of delay or misinformation will lead to servere impacts in the system performance.

Infrastructure updates for the existing system will be quite challenging too. Enhancements might require smooth integration. It can also be very expensive and time consuming depending on the level of improvements needed.

Compliance requirements is another major constraint for this project. Adhering to financial regulations and data protection laws while also sticking to the requirements of the client is critical.

Finally, transaction limits. Predefined minimum and maximum transaction limits must be maintained for operational efficiency and risk management. The amounts must also be realistic and acceptable by the clients to ensure their satisfaction also.

Functional Requirements

✓ User Inputs

The system will accept user inputs such as currency amounts, investment preferences, and risk tolerance.

✓ Outputs

The system will generate outputs such as quotes, returns, profit, taxes, and fees in a user-friendly format.

✓ Data Integration

The system will integrate with real-time data sources for accurate currency conversions and investment data.

✓ Error Handling

The system will include robust error handling mechanisms to prevent disruptions and provide helpful diagnostics.

Act

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Figure 20: Presentation Slide 09

Speaker Notes: Let's discuss about some of the Functional Requirements of this project.

Firstly, the system will take inputs from the clients like currency amounts, investment preferences and risk tolerance to tailor financial solutions for each user.

Accordingly, the system should generate outputs such as quotes, returns, profits, taxes and fee in a simple and user-friendly format for easy interpretation.

The System should also connect with real-time reliable data resources to gain information about currency rates and investment data.

Lastly, a robust error-handling mechanisms should be in place to prevent disruptions and provide help when issues arise.

Non-Functional Requirements

- 1 **User-Friendly Interface**
The system will feature an intuitive and user-friendly interface, accessible to both tech-savvy and non-technical users.
- 2 **High Availability**
The system will be designed for high availability and reliability, ensuring uninterrupted service to clients.
- 3 **Fast Performance**
The system will deliver fast performance even under heavy load, ensuring a responsive and seamless user experience.
- 4 **Compliance**
The system will adhere to all relevant financial and data protection regulations.
- 5 **Maintainability**
The system will be designed for easy maintenance and updates, ensuring future adaptability and flexibility.
- 6 **Extensibility**
The system will be extensible, allowing for future expansion and integration with new features and services.

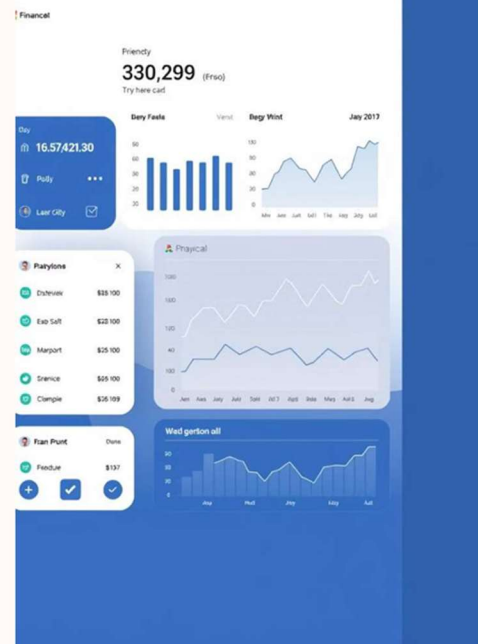


Figure 21: Presentation Slide 10

Speaker Notes: Finally, let's conclude this presentation by discussing about the Non-functional requirement of this project.

The first one is a user-friendly interface. The system is expected to have an intuitive interface that is accessible to both the technical and non-technical users, in order to attract clients from all sides.

The system should also contain high availability. In other words, it should be highly reliable and should be providing with an uninterrupted service for the clients.

Next, even under a heavy load; the systems should maintain to function fast and provide a smooth user experience for the clients.

The system should follow all the necessary financial and data protection regulations, ensuring security and compliance, and free from legal penalties.

The system should be easily maintainable and easy to upgrade. Thereby guaranteeing long-term flexibility and adaptability.

Finally, the system must be scalable, enabling future expansions and addition of new features to meet the evolving needs of the industry.

Thank you for your time!

Isma Haseemdeen / E233595

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Figure 22: Presentation Slide 11

Speaker Notes: With that, let's have a quick summary of today's presentation.

We started the presentation by giving an introduction to our client, Enomy finances. Then we discussed about the stakeholder requirements of this project, followed by a detailed insight into the currency conversion and savings and investment module, which are the key functions of this system. Then we discussed about alternative options to develop the system. Next we spoke about the security considerations that should be followed when designing the system, followed by the possible constraints of this project, and finally, we had a look at the functional and non-functional requirements of this project.

With that being said, let's wind up the presentation.

Thank you for your time and have a good day.

Activity 04

4.1 Software Behavioral Tools and Techniques (UML diagrams)

Software engineers utilize Unified Modelling Language (UML) Diagrams, a standardized visual representation, to show the various components of a system, including its structure, behavior, interactions, and linkages. Developers, designers, and stakeholders may communicate and comprehend complex systems more efficiently by using UML diagrams as their common language. They make it simpler to plan, examine, and develop software projects by assisting in the visualization and documentation of the architecture, design, and functionality of software systems. (lucidchart, n.d.)

4.1.1 Use case Diagrams

It is a UML use case diagram; such is a significant tool for defining requirements of systems/software with a focus on expected behavior and system functions (what) and not on the implementation (how). It visually depicts the one-to-many relationships among its actors, use cases, and the system, concentrating primarily on the users' view and the outside observable behavior of the system. Use case diagrams are not completely clear and high-level, but they usually describe the relationships without describing certain execution stages or the order in which they occur. They're typically established very early in the course of development; specifying system context, capturing requirements, validating design, guiding implementation, and even providing tests all on an architect-created use case are performed using these high-level diagrams. These diagrams would ordinarily be developed in common by an analyst and domain experts. (visualparadigm.com, n.d.)

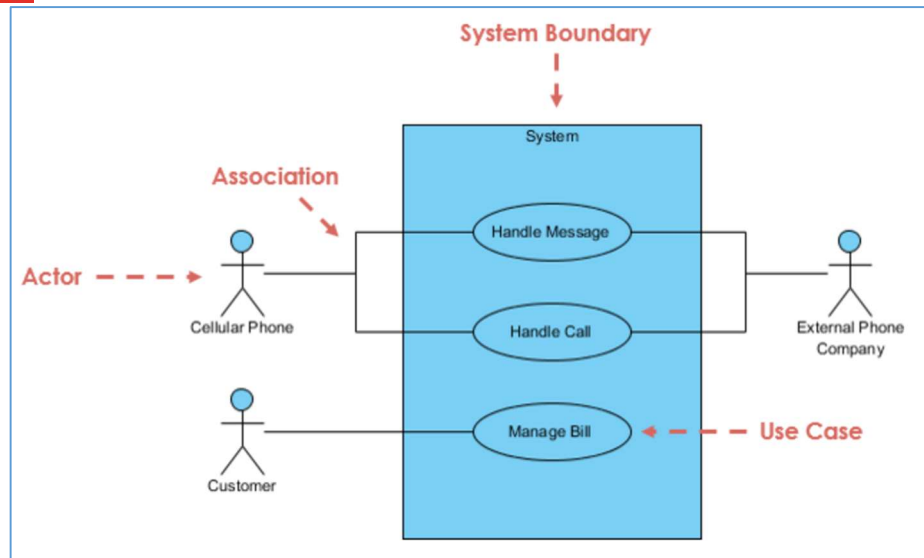


Figure 23: Elements of a Use Case Diagram. (Visualparadigm.com)

4.1.1.1 Use Case Diagram for Enomy Finances

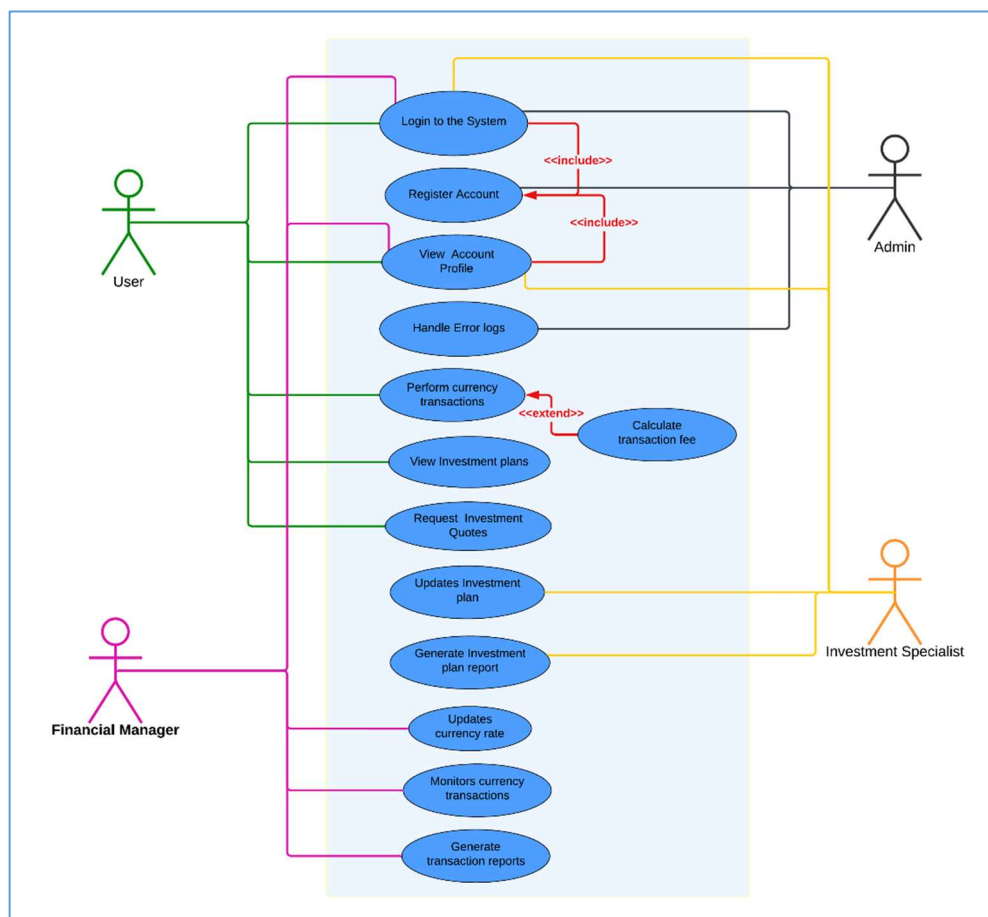


Figure 24: Use case diagram for Enomy Finances.

(Here user entity represents the clients who use the system for currency transactions, view investment plans and generate investment quotes)

The Enemy Finances Use Case Diagram is a visual representation of the system's functionalities and interactions involving various actors. It illustrates how different actors, including User, Financial Manager, Investment specialist and an Administrator, interact with the computerized system. Key use cases include Performing currency transactions, Generating investment quotes, Updating currency rates, Updating Investment plans, Viewing investment plans, Managing accounts, Error handling and Generating reports.

4.1.2 Activity Diagram.

Activity diagrams represent how activities are organized to provide a service at various levels of abstraction. They highlight the interaction and coordination between events or processes. This makes them very effective in modeling workflows within use cases, where activities may overlap or need synchronization, and also for expressing business processes that involve different use cases. These diagrams help in the identification of potential use cases by investigating the workflows of the business, set up pre- and post-conditions for context, model workflows within and between use cases, show complex workflow in object operations, and represent complex activities in high-level views.

(visualparadigm.com, n.d.)

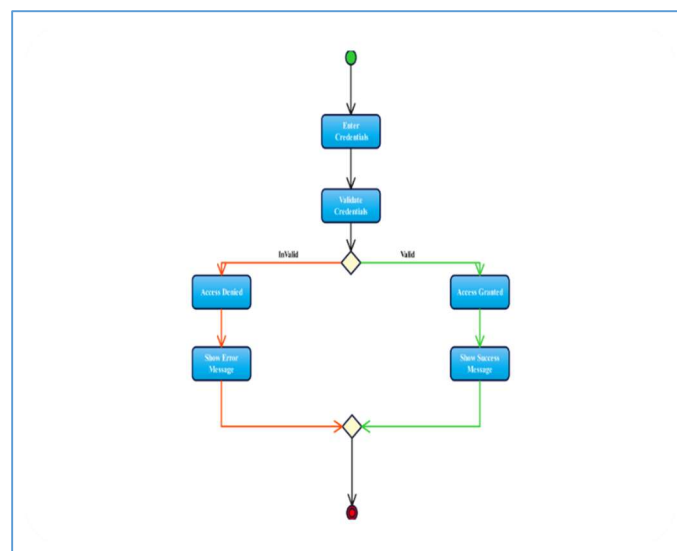


Figure 25: A simple activity diagram for login process.

4.1.2.1 Activity diagram for Enomy Finances.

A. Investment Quote Generation.

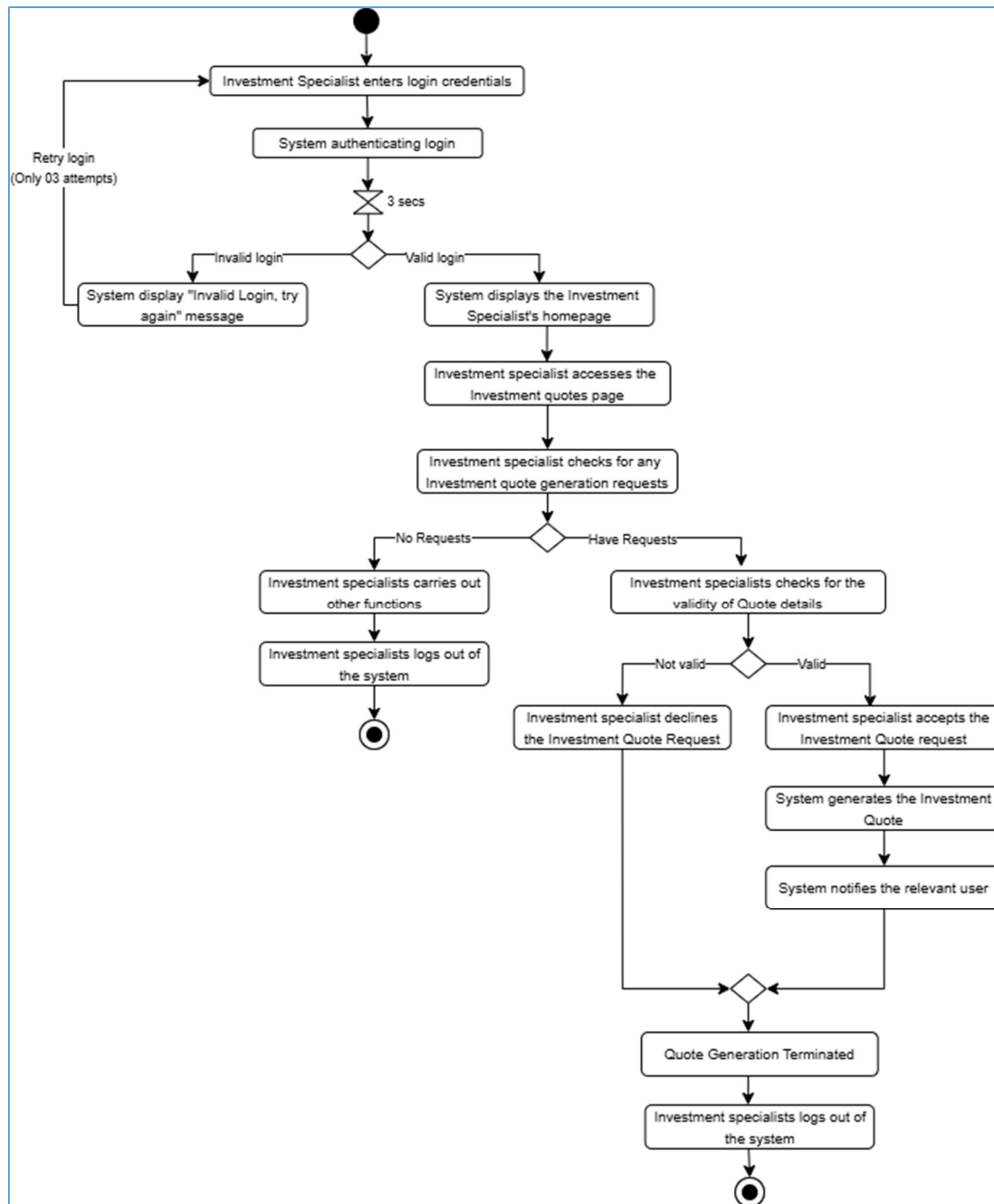


Figure 26: Activity Diagram for Investment Quote Generation of Enomy Finances System.

B. Currency Transaction (Here user represents the clients who use the system for currency transactions, view investment plans and generate investment quotes)

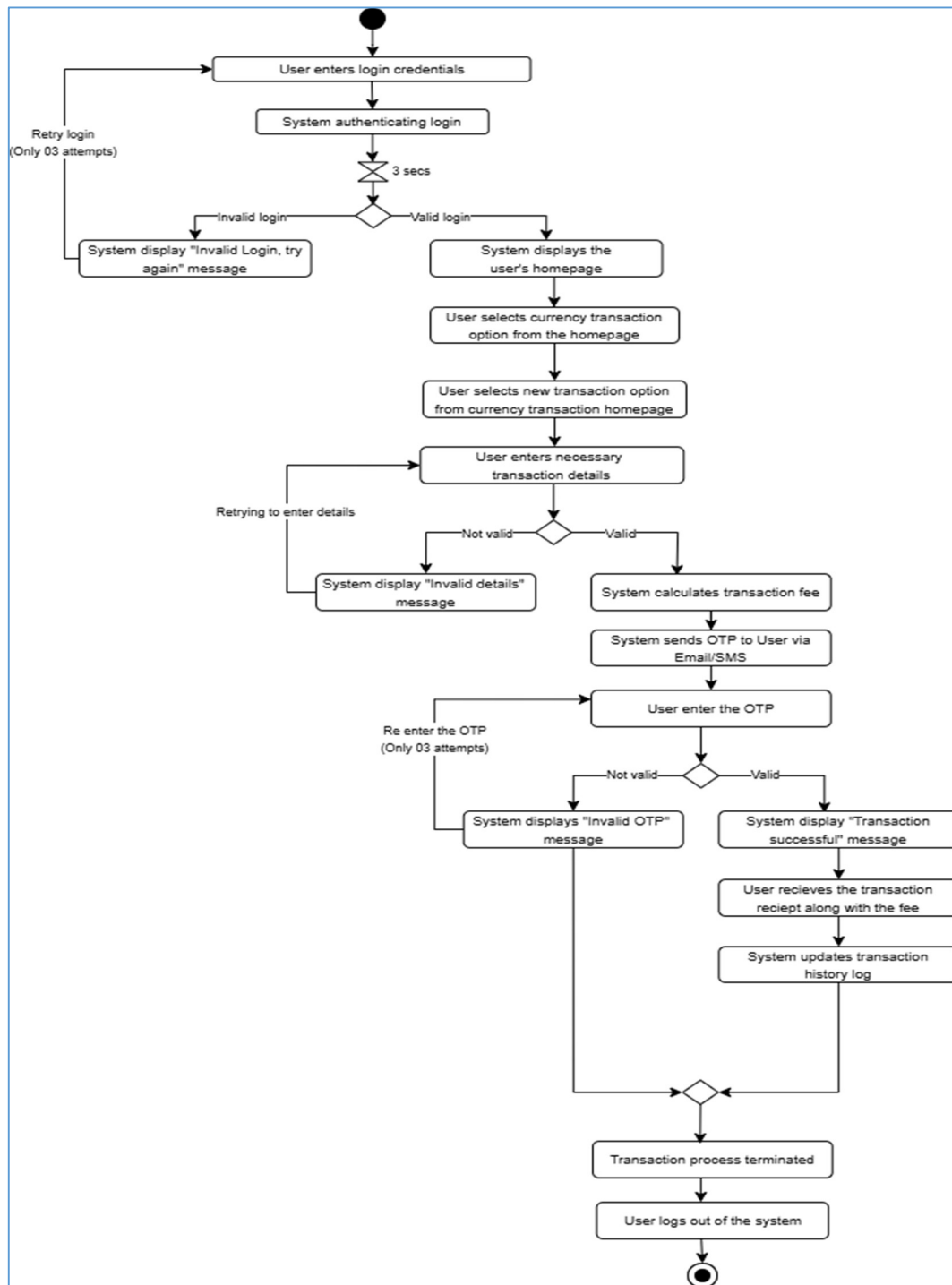


Figure 27: Activity diagram for users performing currency transaction.

C. Currency Rate Updation

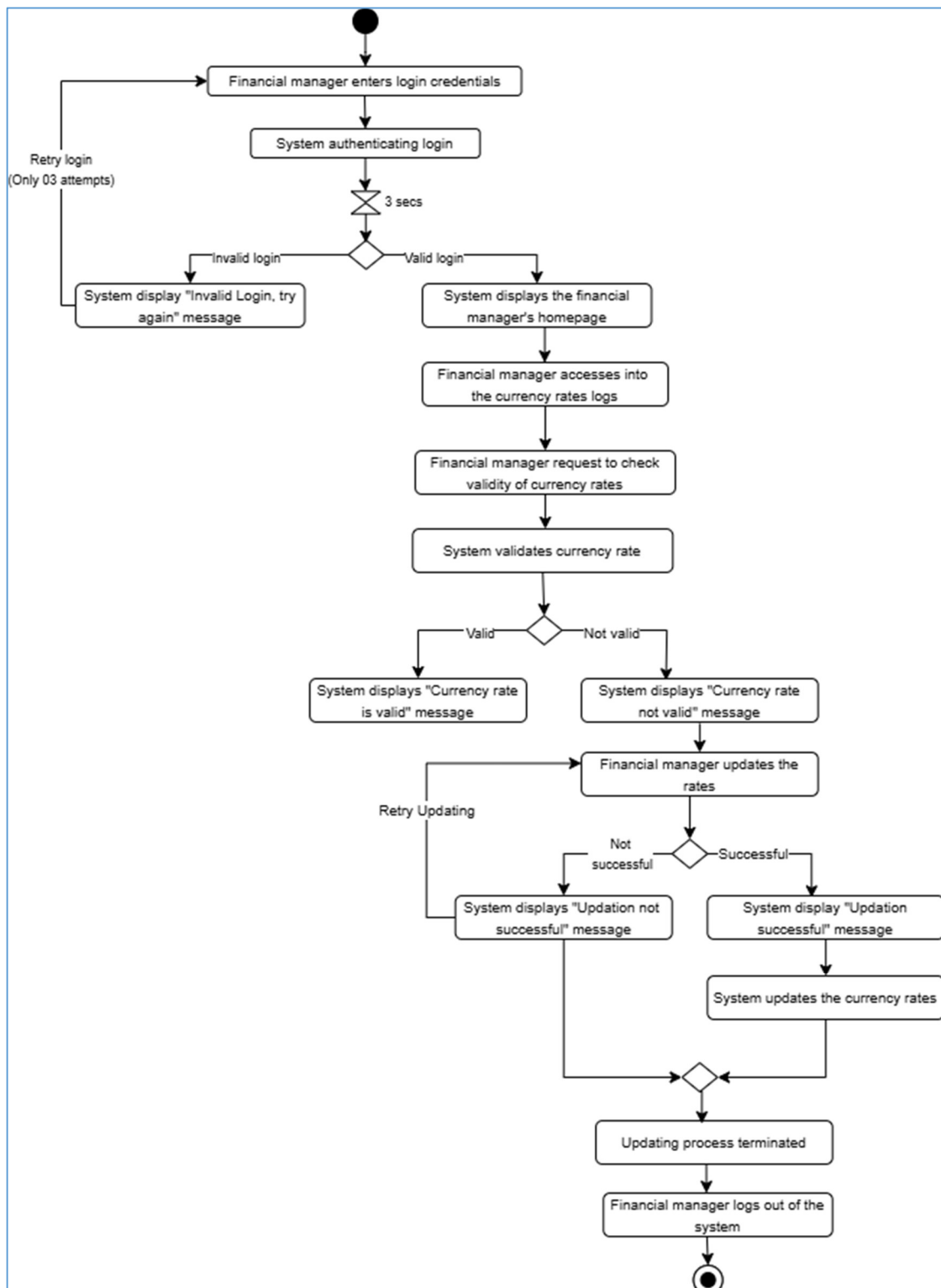


Figure 28: Activity diagram for financial manager updating currency rates.

D. Fixing Errors

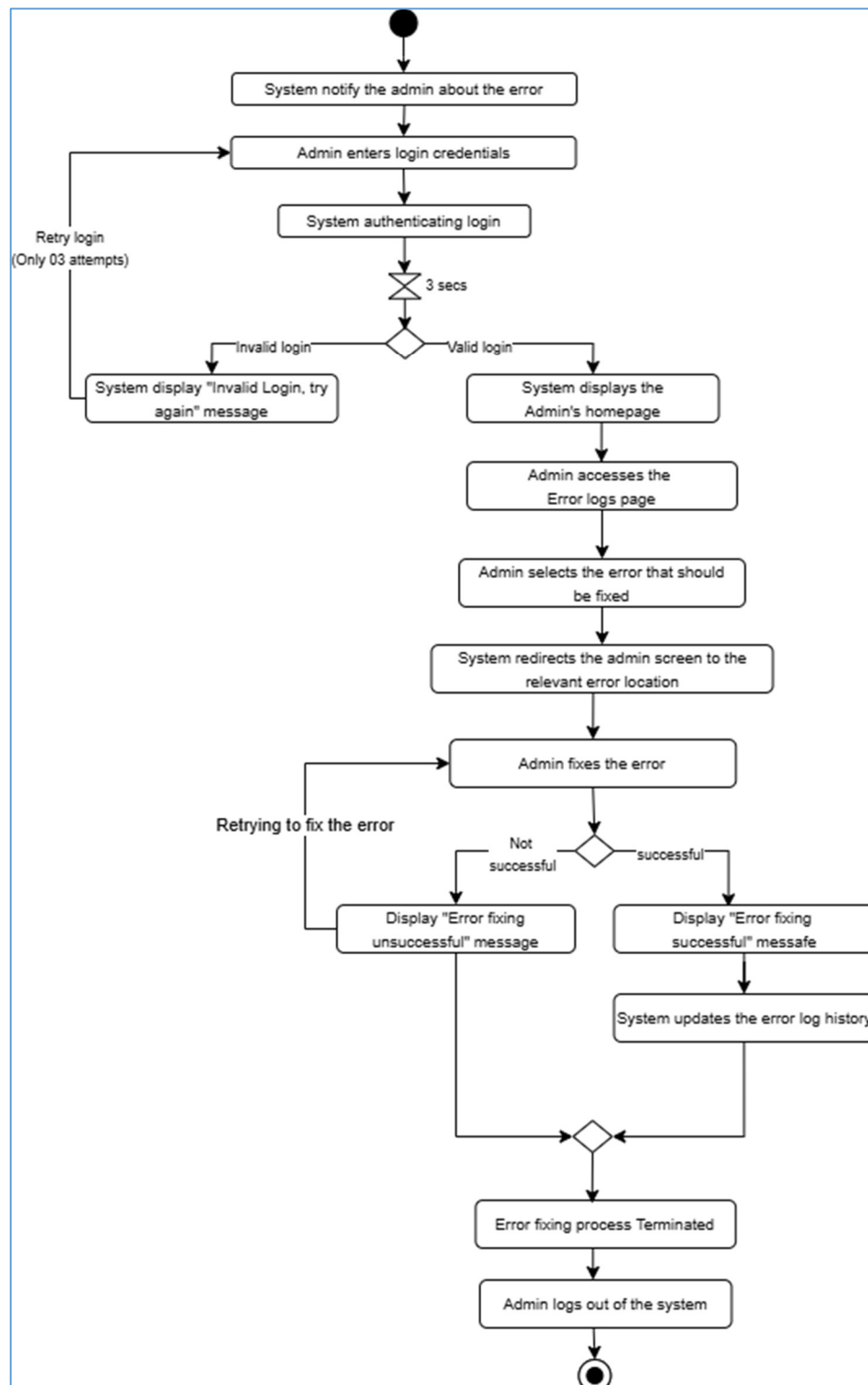


Figure 29: Activity diagram for admin fixing errors.

4.1.3 Data Flow Diagrams

A data flow diagram is a representation of the flow of information of any process or system. It makes use of standardized symbols, such as rectangles, circles, and arrows, with short text labels, to describe data inputs, outputs, storage sites, and the paths between each destination. Data flowcharts can be as simple as hand-drawn process overviews or complex, detailed, multi-level DFDs that drill progressively deeper into how the data is processed. They can be used to analyze an existing system or model a new one. Like all the best diagrams and charts, a DFD can frequently graphically "say" things that would be hard to describe in words, and they work for both technical and nontechnical audiences, from developer to CEO. That is why DFDs remain so popular after all these years. While they perform well for data flow software and systems, currently they are less appropriate to visualize interactive, real-time, or database-oriented software or systems. DFD diagrams are of 03 types; Level 0 or Context level DFD diagram, Level 1 DFD diagram and Level 2 DFD diagram.

(lucidchart, n.d.)

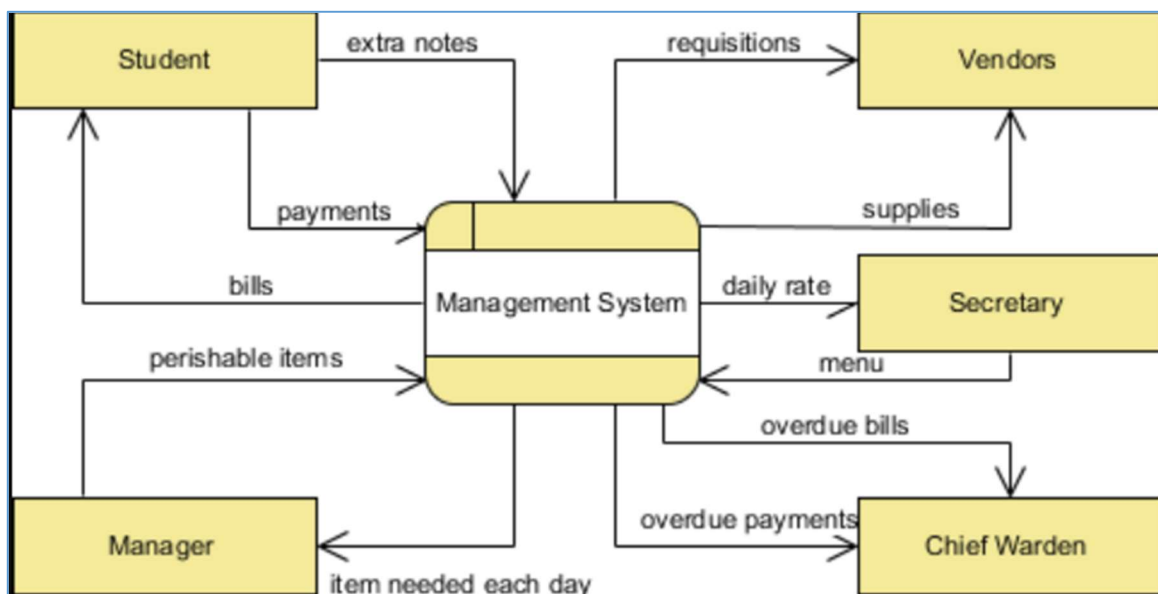


Figure 30: Zero level/Context level DFD diagram

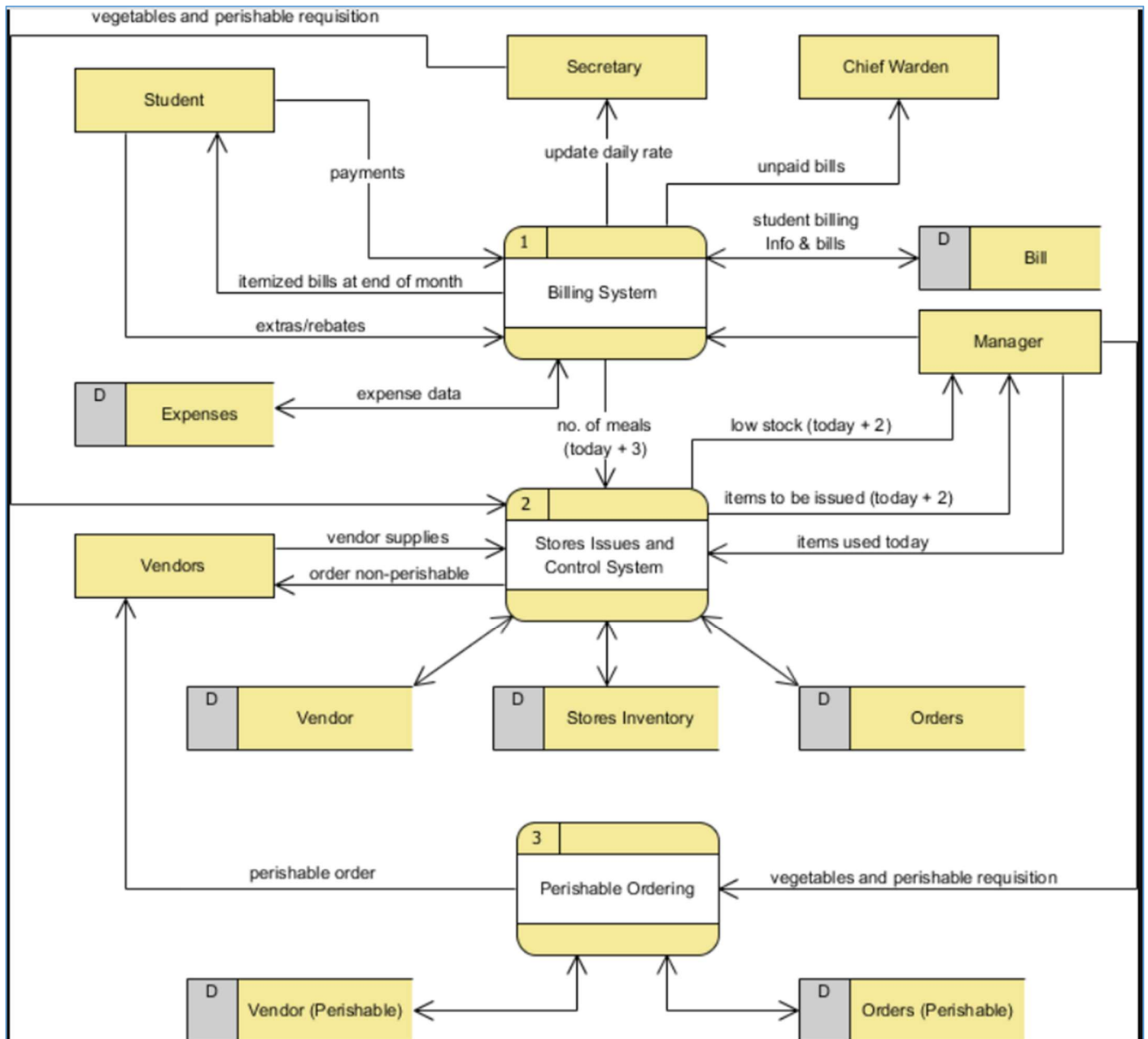


Figure 31: Level 1 DFD Diagram.

4.1.3.1 Zero/Context Level DFD Diagram for Enomy Finances.

(Here user represents the clients who use the system for currency transactions, view investment plans and generate investment quotes)

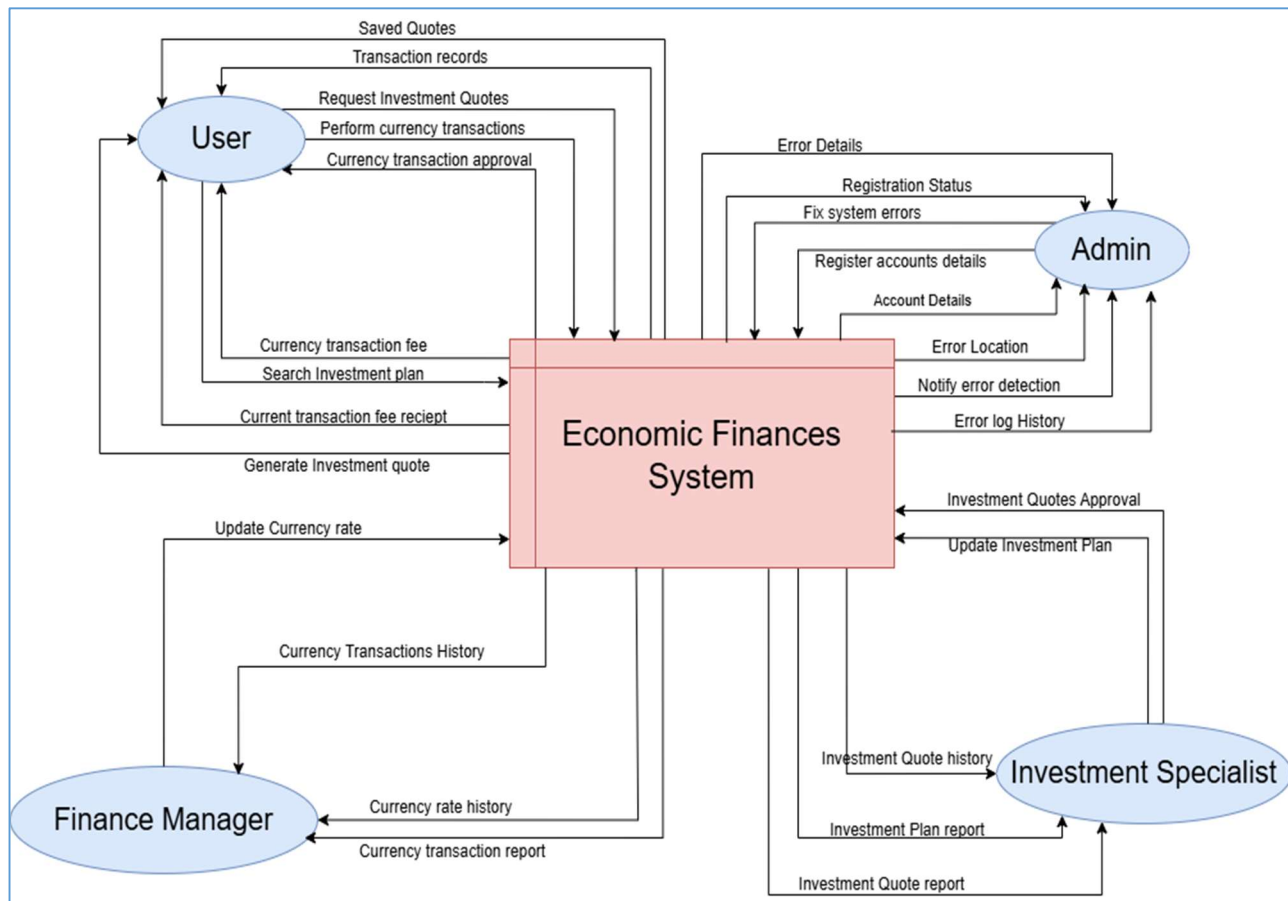


Figure 32: Level 0 DFD

4.1.3.2 Level 1 DFD Diagram for Enmony Finances

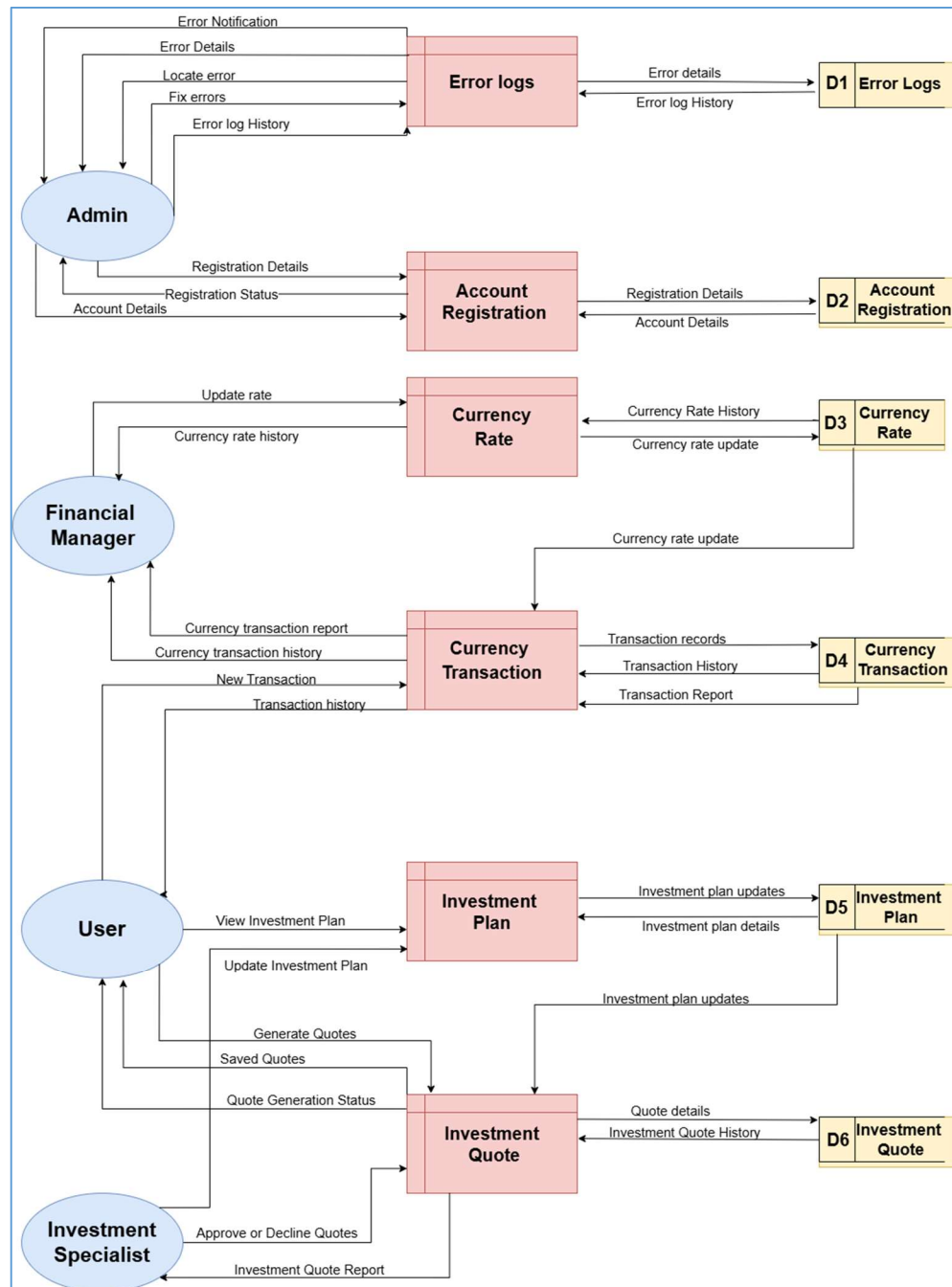


Figure 33: Level 1 DFD diagram.

(Here user represents the clients who use the system for currency transactions, view investment plans and generate investment quotes)

4.1.4 Finite State Machine

A Finite State Machine is used to model the dynamic behavior of a system by depicting the states of an object and the transitions between those states. These diagrams are particularly useful for modeling systems where an object can have several different states and can transition from one state to another based on events.

Key Components:

- **States:** Represent the conditions or situations during the life of an object. States are represented by rounded rectangles.
- **Transitions:** Indicate the movement from one state to another. Transitions are represented by arrows.
- **Events:** Triggers that cause a state transition. Events are usually written next to the transition arrow.
- **Initial State:** Represented by a filled circle, it indicates the starting point of the state machine.
- **Final State:** Represented by a filled circle within a circle, it indicates the end point of the state machine.

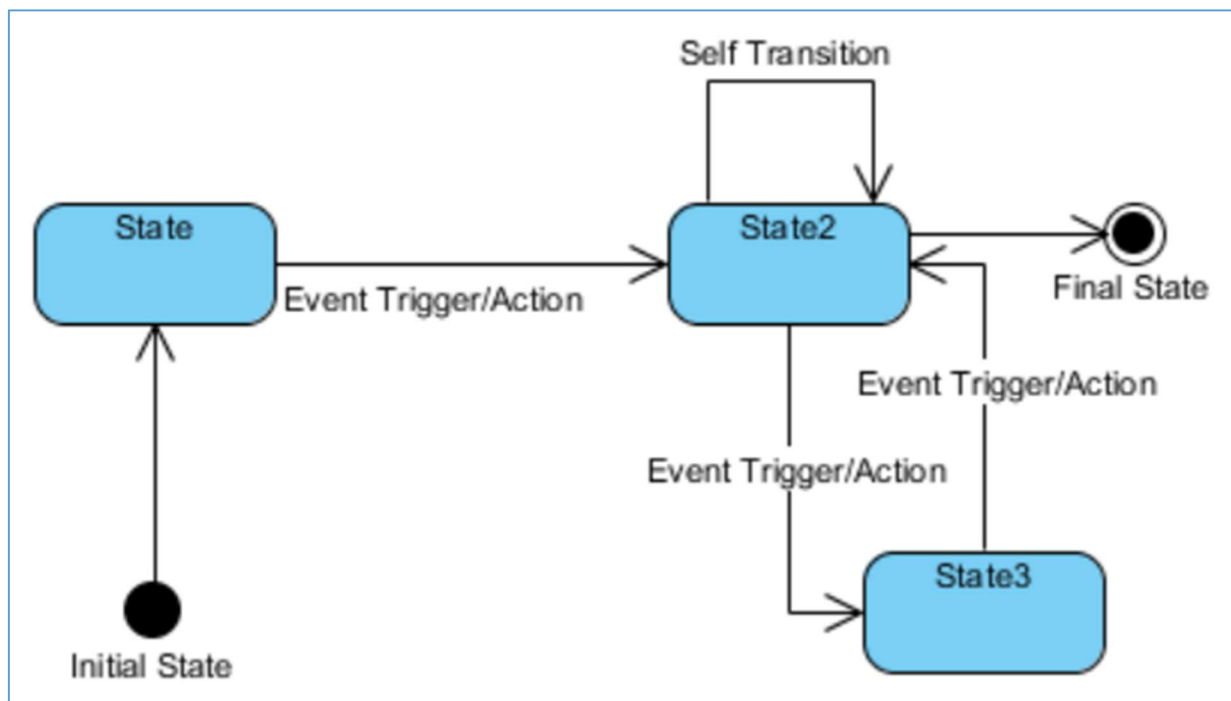


Figure 34: Finite State Machine Diagram.

4.1.4.1 Finite State Machine for Enomy Finance Currency Transaction.

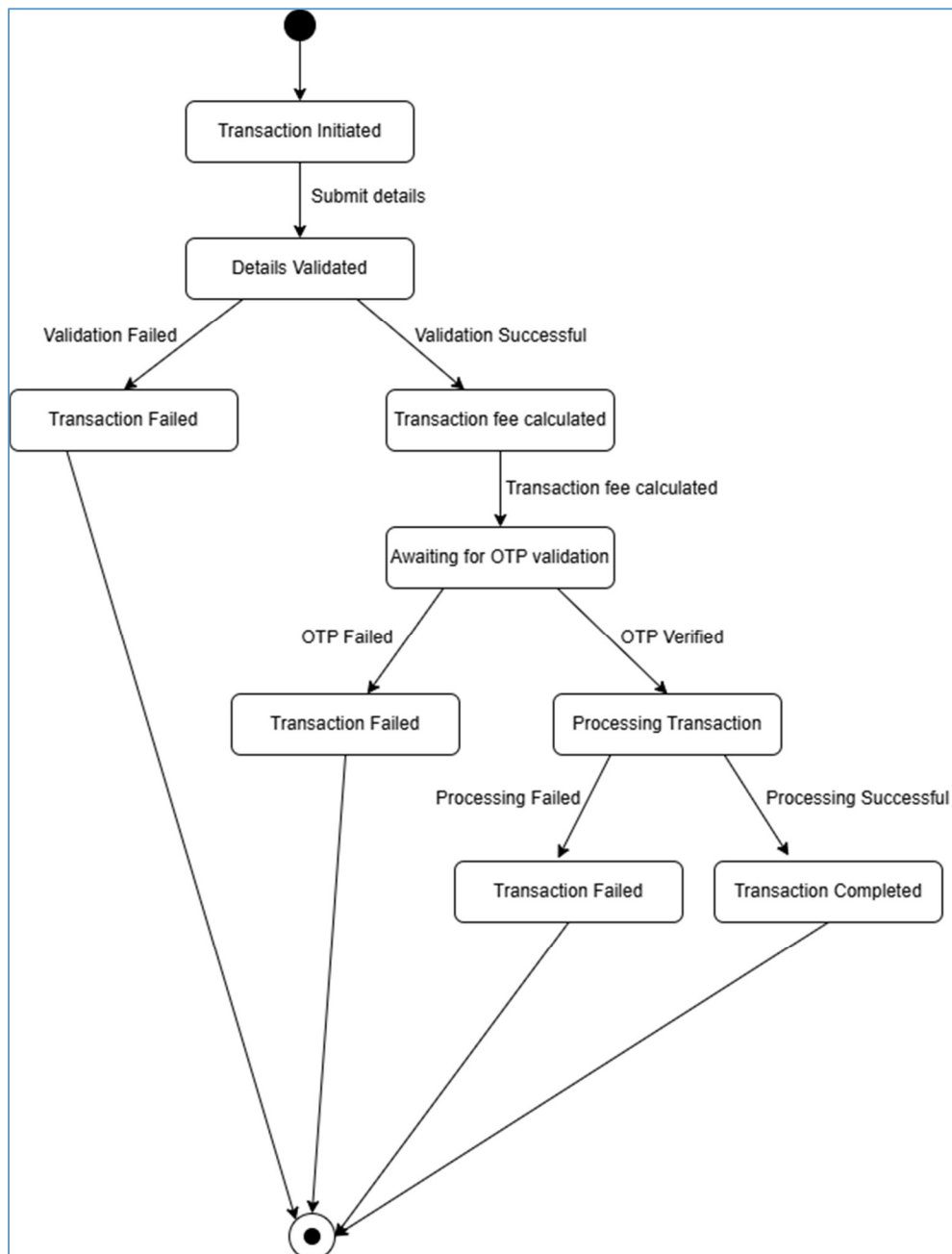


Figure 35: Finite State Machine for Currency Transaction.

4.1.5 Extended Finite State Machine

An Extended Finite State Machine Diagram (EFSM) builds upon the basic UML Finite State Machine Diagram by incorporating additional elements to handle more complex behaviors. EFSMs can include variables, guards, and actions that are dependent on conditions and events.

Additional Components:

- **Variables:** Data elements that can influence state transitions. These variables are part of the state context.
- **Guards:** Conditions that must be true for a transition to occur. Guards are written in square brackets next to the transition arrow.
- **Actions:** Operations performed during state transitions or while in a specific state. Actions can be written next to transitions or within states.

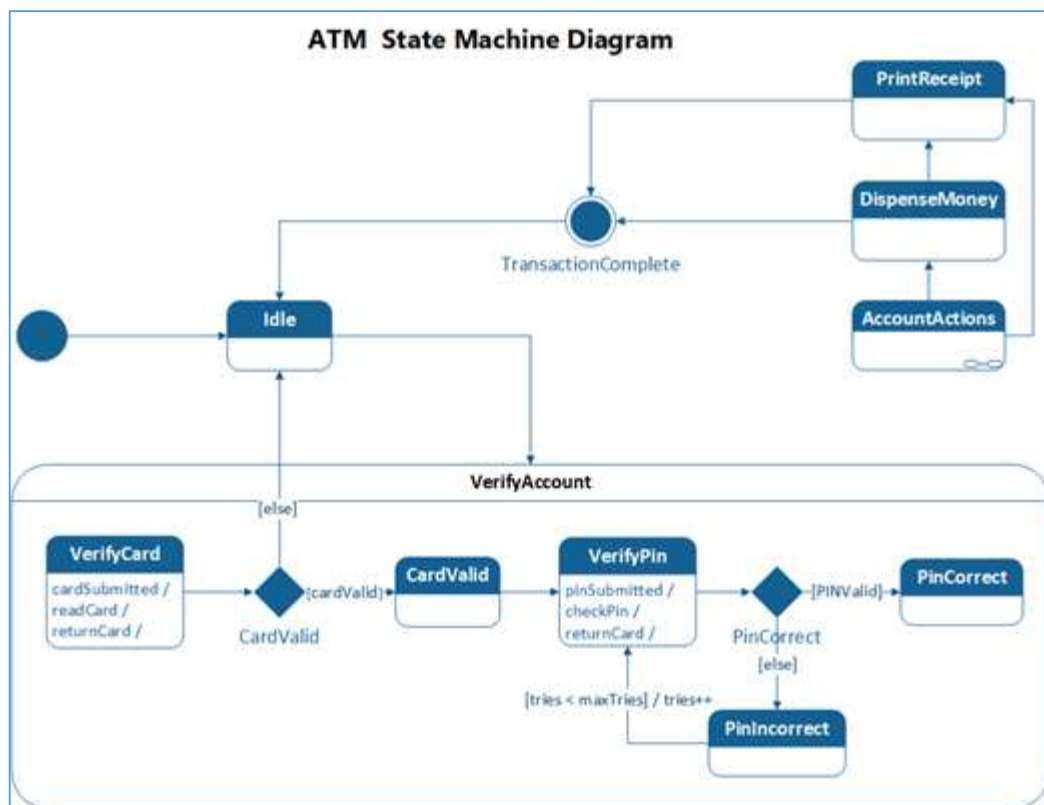


Figure 36: Extended Finite State Machine.

4.1.5.1 Extended Finite State Machine for Client of Enomy Finance.

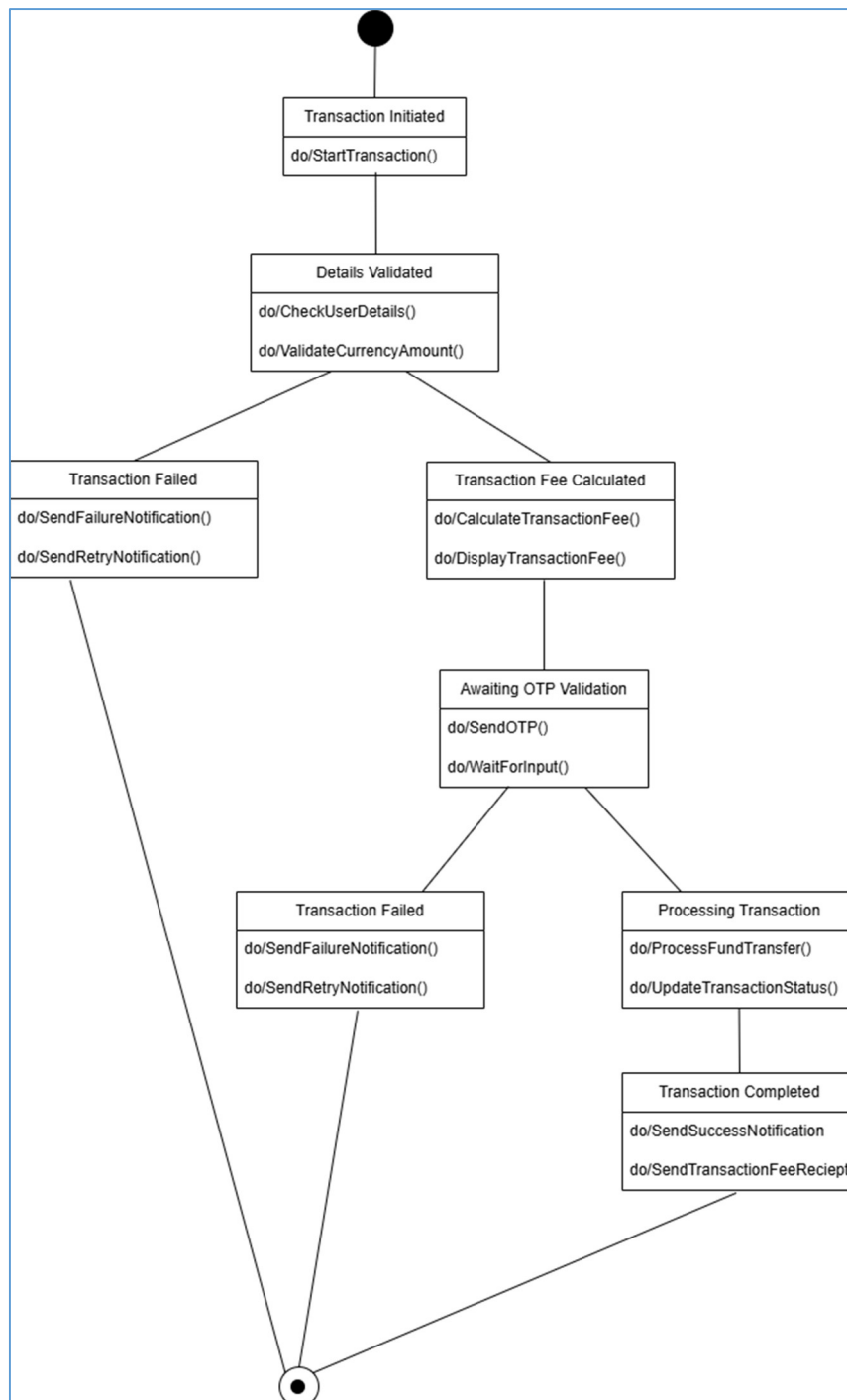


Figure 37: Extended FSM for Currency Transaction of Enomy Finances

4.1.6 Entity-Relationship Diagram (ERD)

An ER diagram, also known as ERDs or ER Models is one form of a flowchart representing how "entities"- people, objects, or concepts-relate to each other inside the system. Generally, ER diagrams have come into great usage for the creation or debugging of relational databases, be that in software engineering, business and information studies courses, or scholarship/research. They utilize a set of symbols-rectangles, diamonds, ovals, and connecting lines-which express the interconnected relationship of entities to relationships and vice-versa, with their respective attributes. Grammatically stated, it parallels grammatical structure, where the entities are nouns and the relationships are verbs.

(lucidchart.com, 2023)

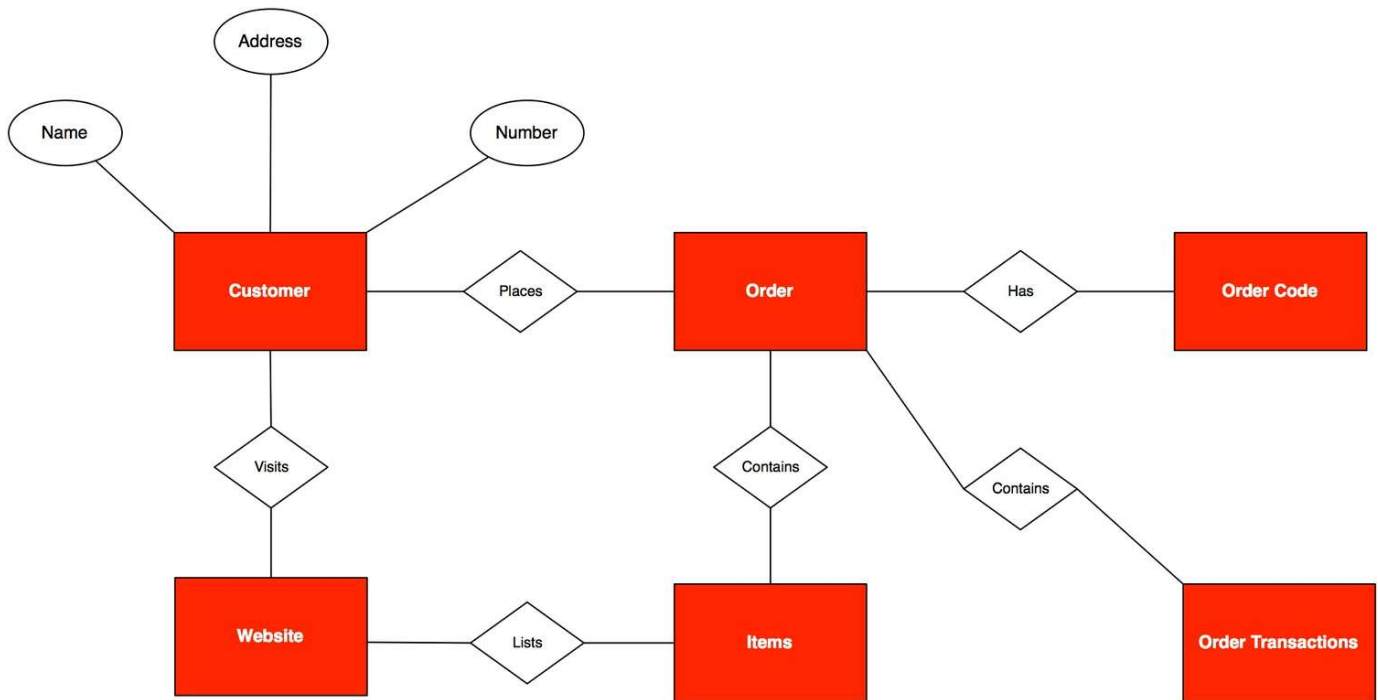
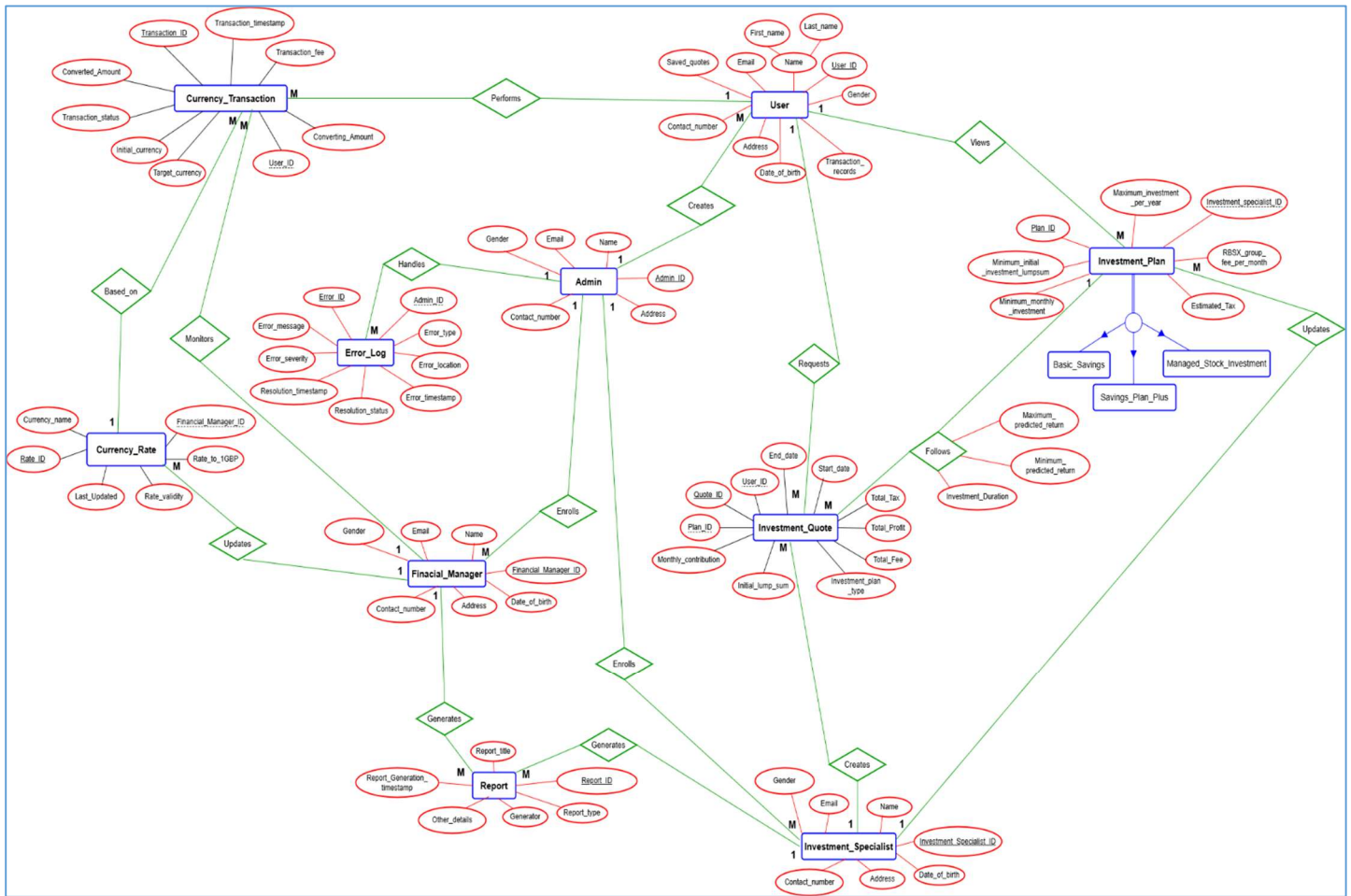


Figure 38: ERD for an order placement

4.1.6.1 Entity Relationship Diagram for Enomy Finances.



(Here user entity represents the clients who use the system for currency transactions, view investment plans and generate investment quotes)

4.2 Wireframes

Wireframes are basic schematics that outline how an application, website, or product is architected and how it functions. It focuses on core features of the screen layout, navigation bars, UX/UI elements, and interactions. Initial wireframes are of low fidelity and use placeholder text and basic shapes; this stresses utility over aesthetics, allowing teams to align on objectives.

Teams should start with a minimal design so that effective iteration, gathering of user feedback on core elements, and refining of user flows and templates unhindered by complex visuals can be carried out. This approach will ensure the product will provide an exemplary user experience.

(figma.com, n.d.)

4.2.1 Wireframes for the Enomy Finances System.

A. User Type page

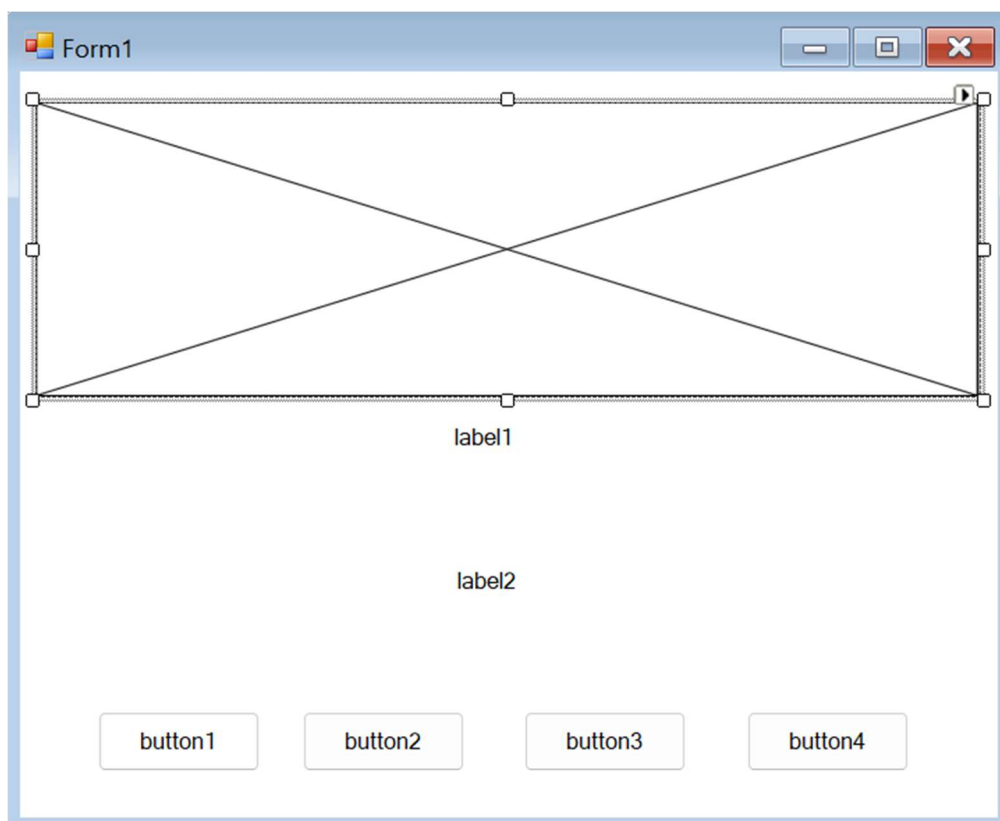


Figure 39: User Type Page Wireframe

B. Login Page

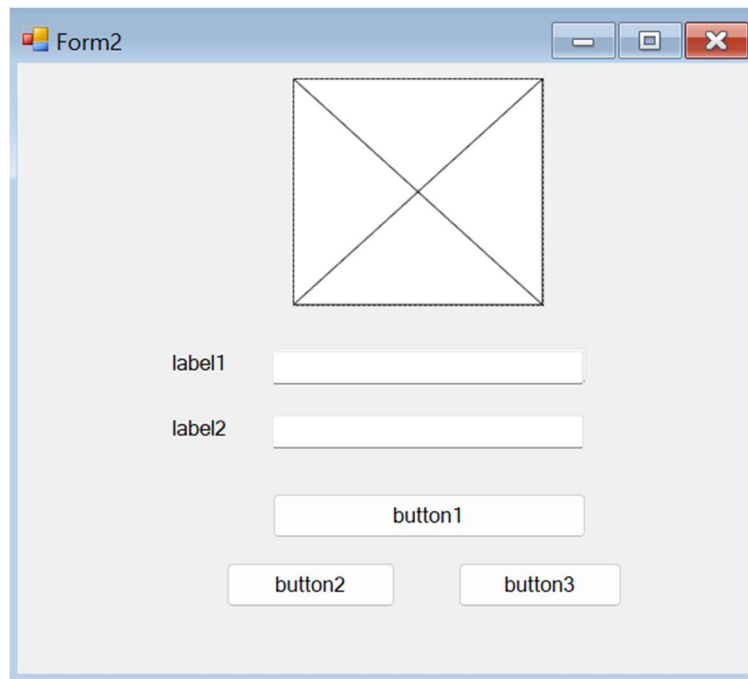


Figure 40: Login Page Wireframe.

C. Homepage

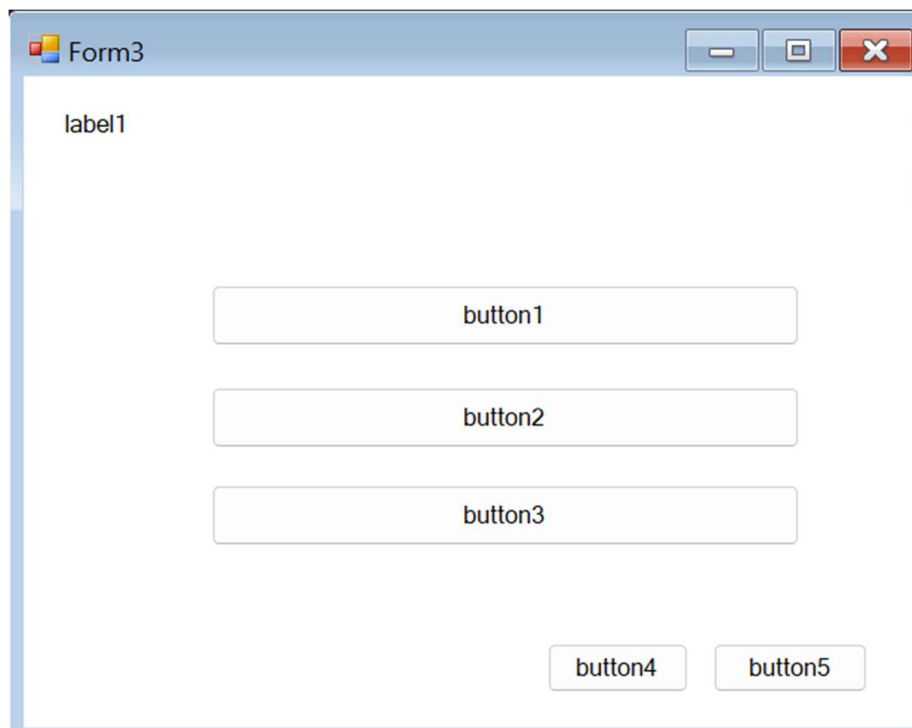


Figure 41: Homepage Wireframe.

D. Displaying History Page (Error logs, Transaction History and etc.)

Form5

label1

button5

label2

button1 button2

button3 button4

Figure 42: History Display Page Wireframe

E. Report Page Wireframe

Form6

label1

button1 button2 button5

button3 button4

Figure 43: Report Page Wireframe

F. Wireframe for Saved quotes, User account details and Currency Rate Details.

The wireframe shows a window titled "Form7" with standard Windows-style controls (minimize, maximize, close). Inside the window, the layout is as follows:

- A label "label2" is positioned in the top-left area.
- A button labeled "button1" is located in the top-right area, adjacent to a text input field.
- A table with 3 columns and 8 rows occupies the central portion of the window. The first row is taller than the others, possibly serving as a header.
- A button labeled "button2" is positioned in the bottom-left area.
- Buttons labeled "button3" and "button4" are positioned in the bottom-right area.

Figure 44: Wireframe for Details about Saved Quotes, User Accounts and Currency Rates.

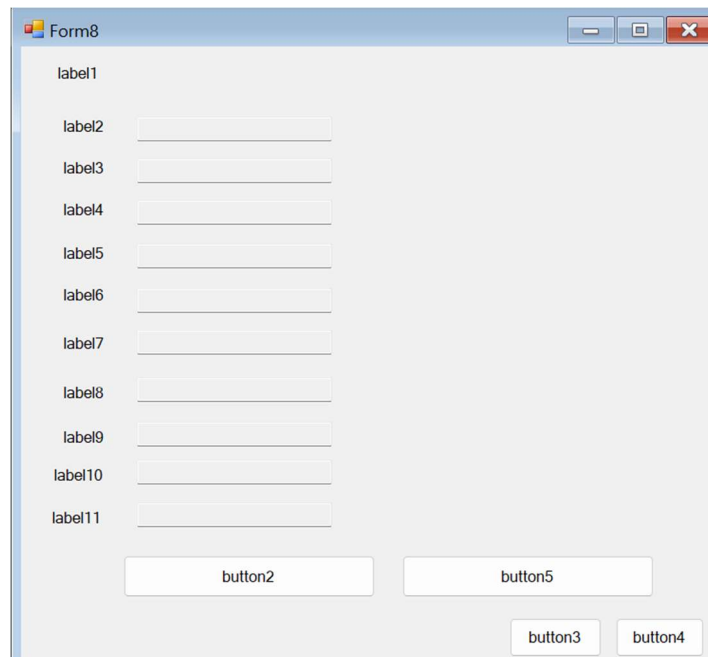
G. Currency Transaction Form

Figure 45: Currency Transaction Form Wireframe.

H. Investment Quote Generating Page

Figure 46: Investment Quote Generating Wireframe.

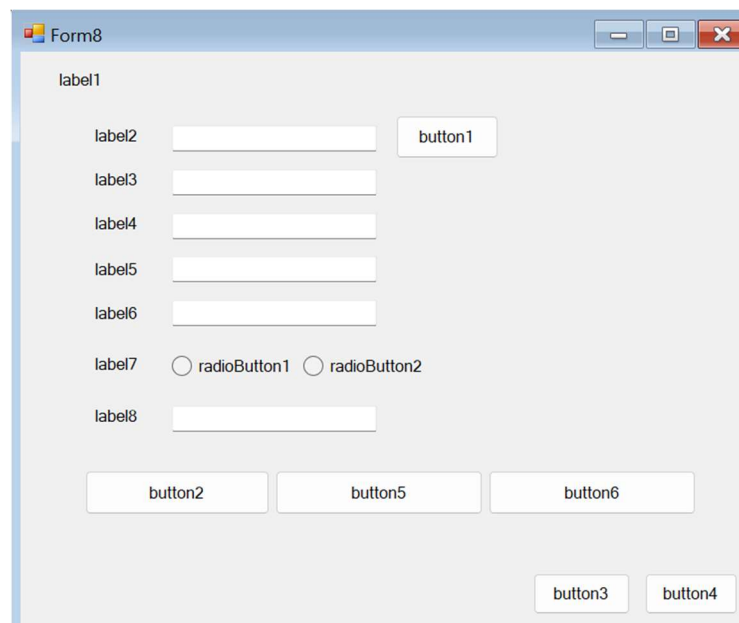
I. Investment Plan Details



The wireframe shows a form titled 'Form8' with a standard Windows-style title bar (minimize, maximize, close buttons). The form contains a vertical stack of labels from 'label1' to 'label11'. 'label1' is at the top, followed by 'label2' through 'label11'. Each label is followed by a text input field. Below the input fields, there are four buttons: 'button2' and 'button5' are positioned side-by-side, while 'button3' and 'button4' are positioned side-by-side below them.

Figure 47: Investment Plan Details Wireframe.

J. Registering Accounts



The wireframe shows a form titled 'Form8' with a standard Windows-style title bar. The form contains a vertical stack of labels from 'label1' to 'label8'. 'label1' is at the top, followed by 'label2' through 'label8'. 'label2' through 'label6' are each followed by a text input field. 'label7' is followed by two radio buttons labeled 'radioButton1' and 'radioButton2'. 'label8' is followed by a text input field. Below the input fields, there are six buttons: 'button2', 'button5', and 'button6' are positioned side-by-side, while 'button3' and 'button4' are positioned side-by-side below them. 'button1' is positioned to the right of the input field for 'label2'.

Figure 48: Registering Accounts form Wireframe

K. Error Logs (to fix errors)

Figure 49: Error Logs Wireframe.

L. Inserting, Updating or Deleting Currency Rate

Figure 50: Currency Rate form Wireframe.

All these set of wireframes provides an overview of how the system is to be designed while ensuring a seamless navigation, security, operational efficiency and adaptability.

4.3 Interface Designs for Enomy Finances

4.3.1 Common Interfaces for all users.

As soon as the users enter into the system, a screen will display a page where they will have to select their user type. Once they select their user type they will be directed to their relevant login page.



Figure 51: User Type Interface.

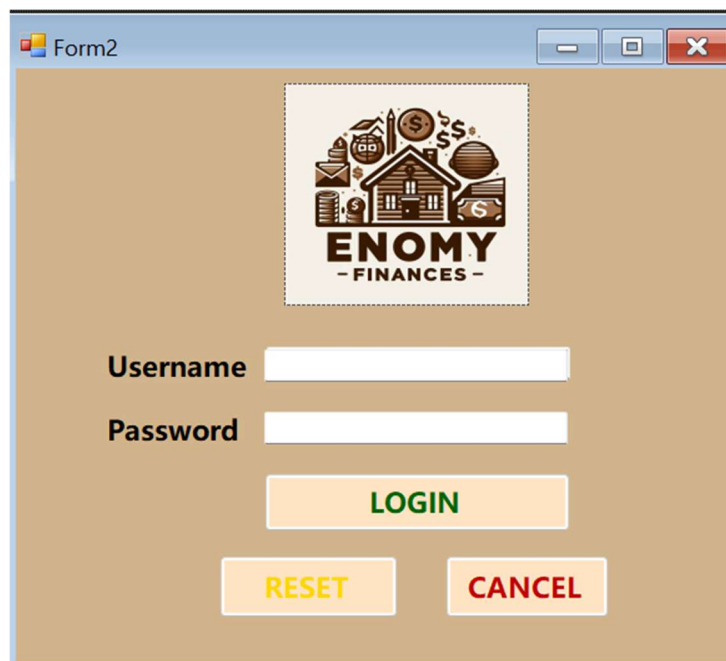


Figure 52: Login Interface.

4.3.2 Interfaces for Admin

After the Admin logs into the system by entering the correct username and password, the system will display the admin homepage. It contains the activities the admin is permitted to do, like dealing with the accounts of the users of the system and fixing errors. If the admin selects the user accounts option, it will show options like creating a new account or viewing existing accounts. The admin can select his/her preferred option. On the other hand, if the admin chooses the error logs option, it will also ask the admin to choose if he/she wants to fix an error or view error logs, which will then direct the admin to its respective pages.

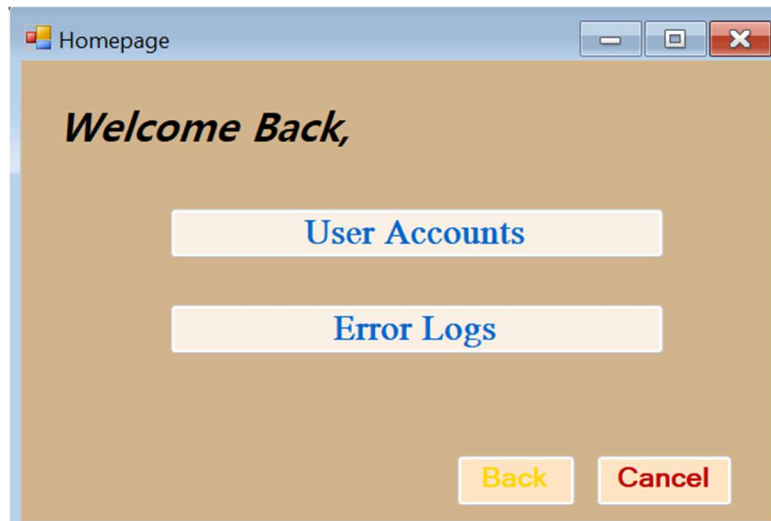


Figure 53: Admin Homepage.

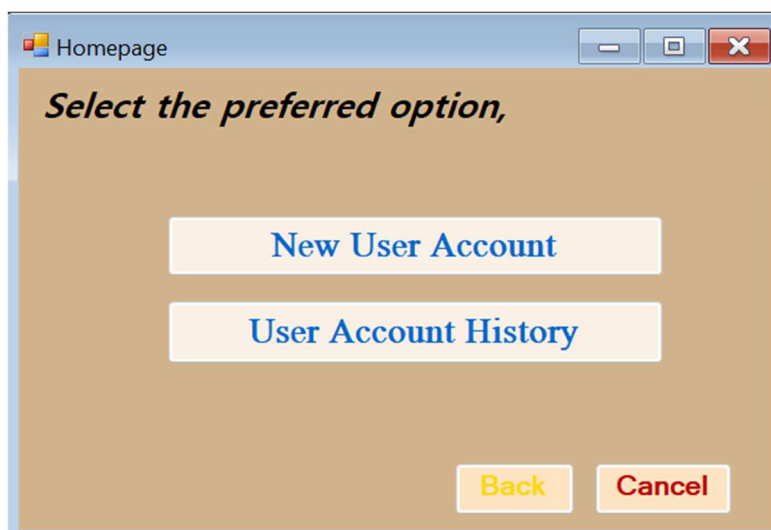


Figure 54: Homepage for User Account

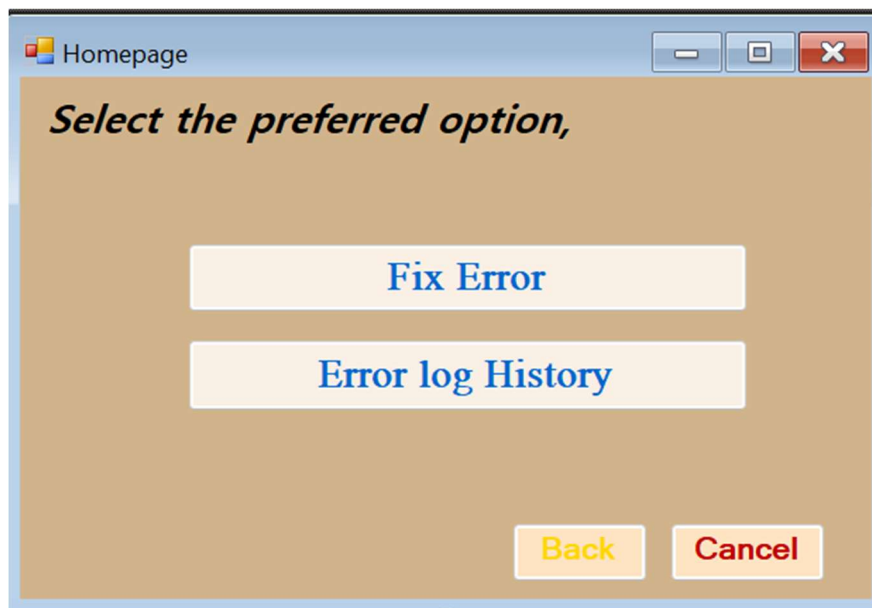


Figure 55: Homepage for Error logs.

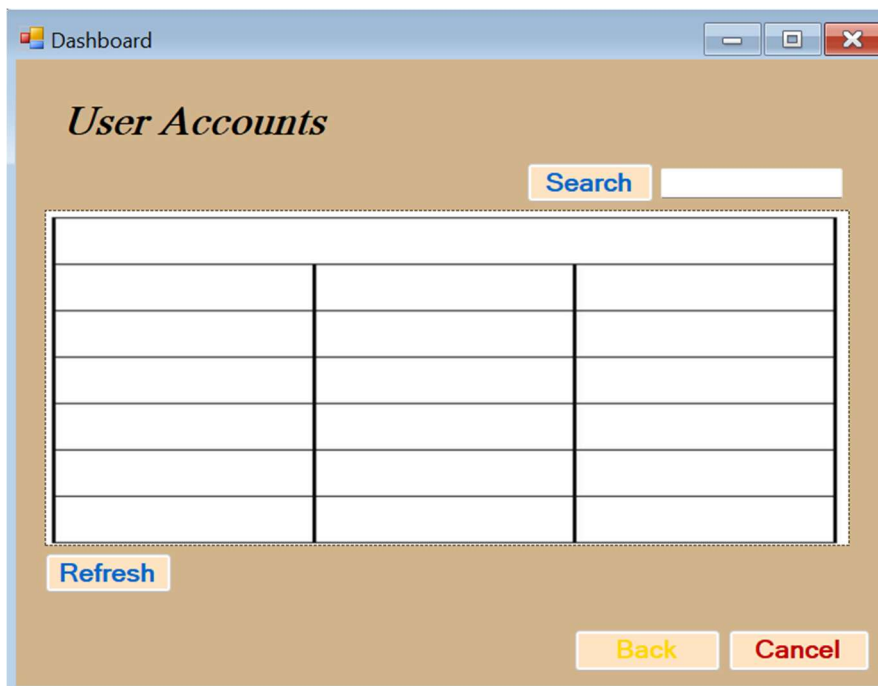


Figure 56: User Account history page.

Dashboard

User Accounts

User ID

First name

Last Name

Address

Contact Number

Gender ☐ Male ☐ Female

Email

Figure 57: New user accounts page.

Error log

Error Logs

Figure 58: Fixing Error log page.

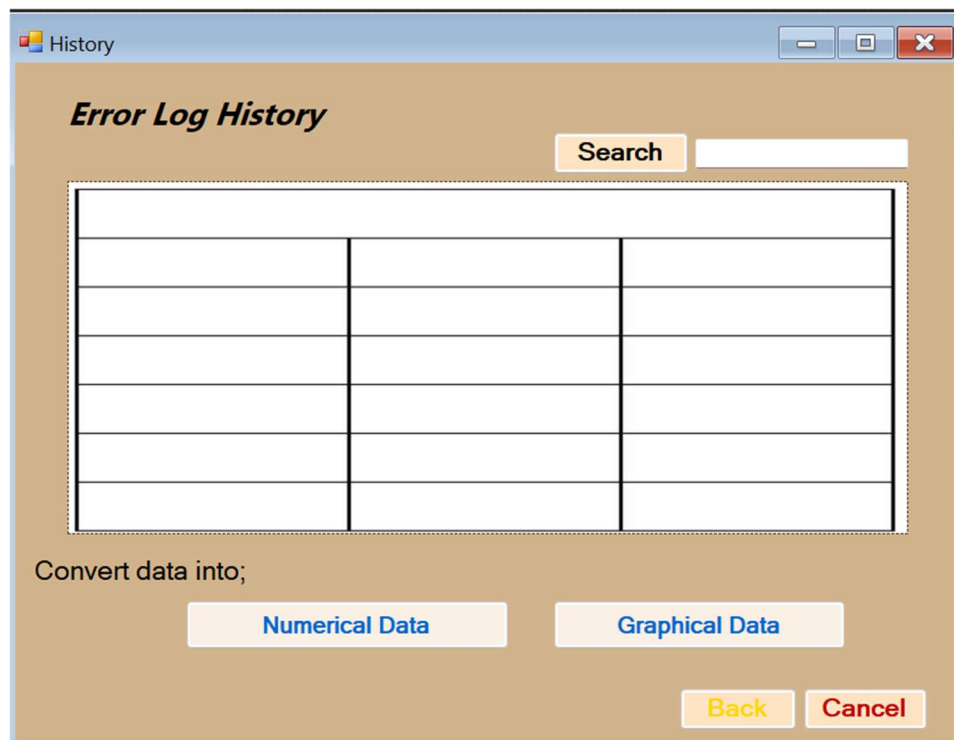


Figure 59: Error log history page.

4.3.3 Interfaces for Financial Manager.

After the financial manager logs into the system by entering the correct username and password, the system will display the financial manager homepage. It contains the activities the manager is permitted to do, like monitoring currency transactions and updating currency rates. If the manager selects the currency transaction option, it will show options like currency transaction history and currency transaction report. The manager can select his/her preferred option. On the other hand, if the manager chooses the currency rate option, it will also ask the manager to choose if he/she wants to update currency rate or view currency history, which will then direct the admin to its respective pages.

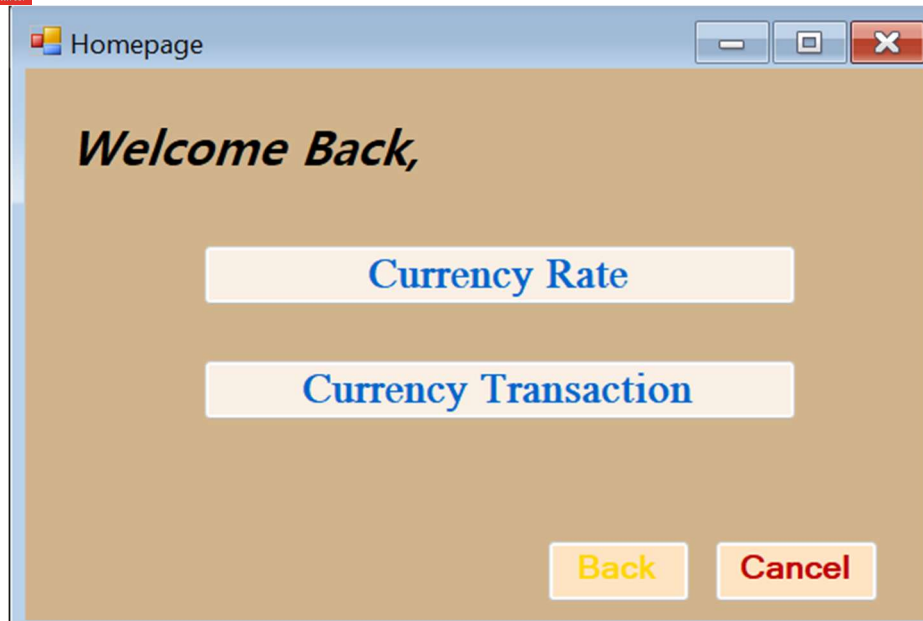


Figure 60: Financial manager homepage.

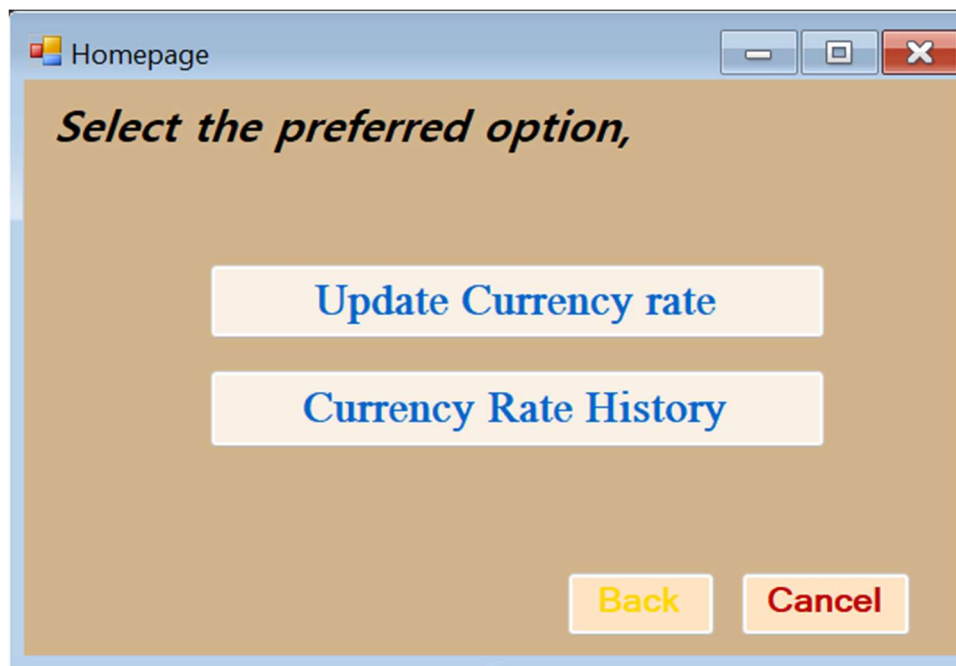


Figure 61: Currency rates homepage.

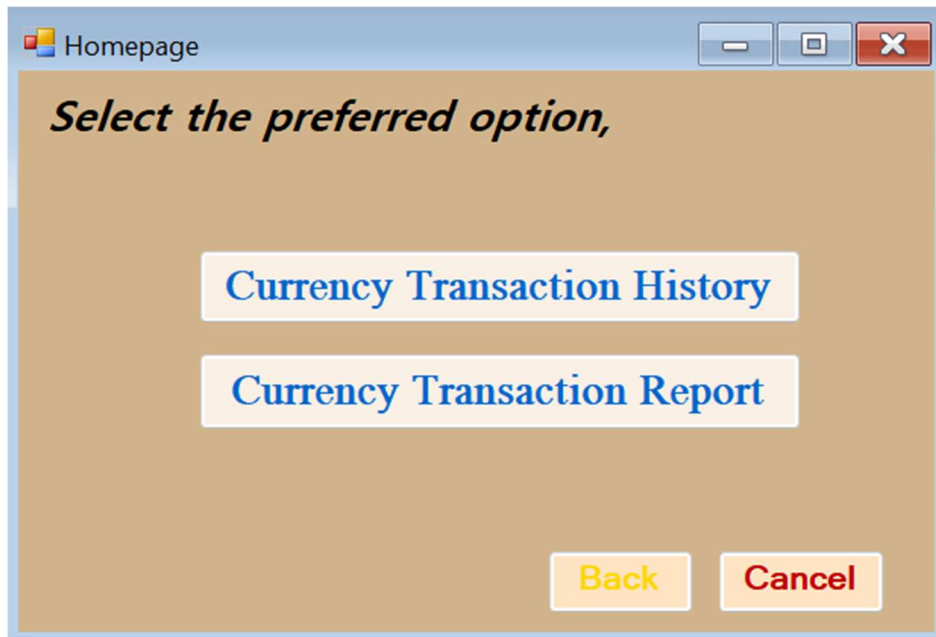


Figure 62: Currency transactions homepage.

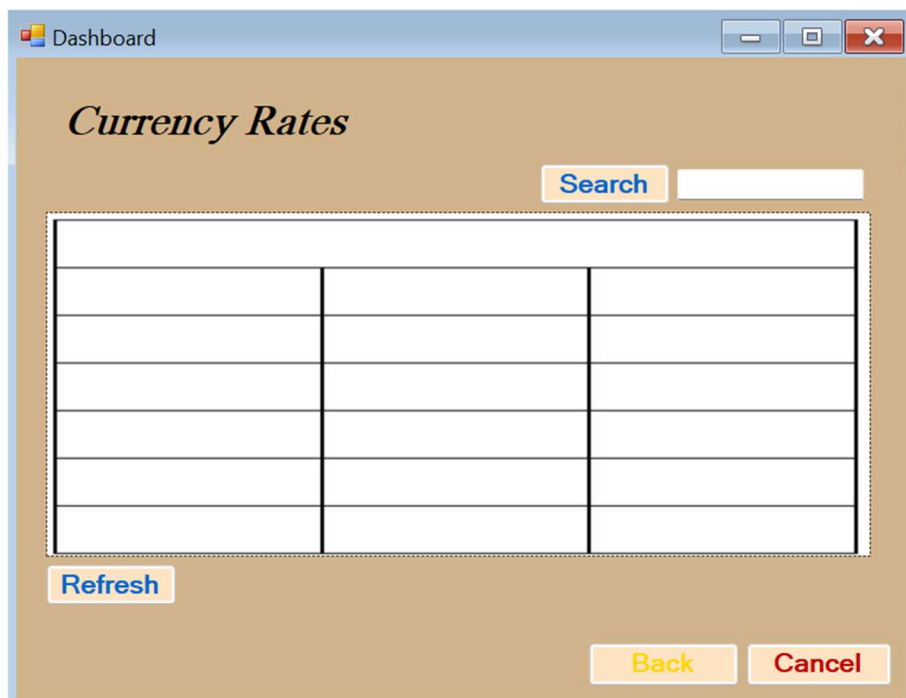


Figure 63: Currency rate history page.

Dashboard

Currency Rate

Rate ID **Search**

Financial Manager ID

Currency name

Last Updated

Rate for 1GBP

Rate Validity ☐ Valid ☐ Not valid

REGISTER **UPDATE** **DELETE**

Back **Cancel**

Figure 64: Updating currency rate logs page.

History

Currency Transaction History

Search

Convert data into;

Numerical Data **Graphical Data**

Back **Cancel**

Figure 65: Currency rate history page.

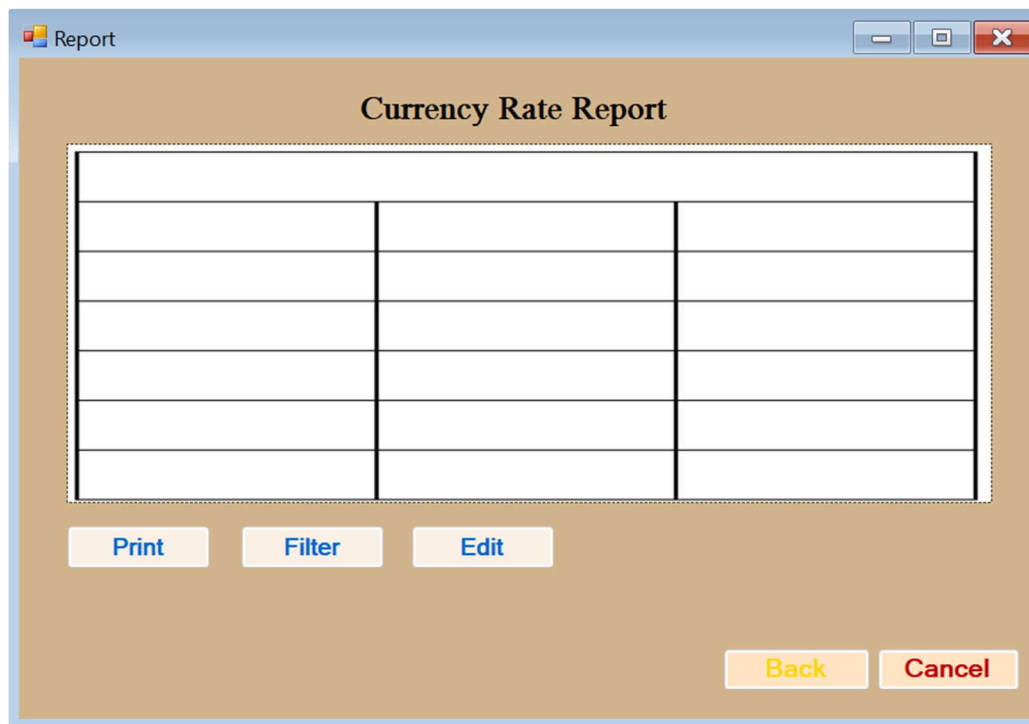


Figure 66: Currency transaction report.

4.3.4 Interfaces for Investment Specialist.

After the Investment specialist logs into the system by entering the correct username and password, the system will display the investment specialist homepage. It contains the activities the specialist is permitted to do, like approving quotes, updating Investment plans and generating reports. If the specialist selects the Investment quote option, it will show options like Investment quote requests and Investment quote history. The specialist can select his/her preferred option. On the other hand, if the specialist chooses the investment plan option, he/she will be able to update or create new Investment plans.

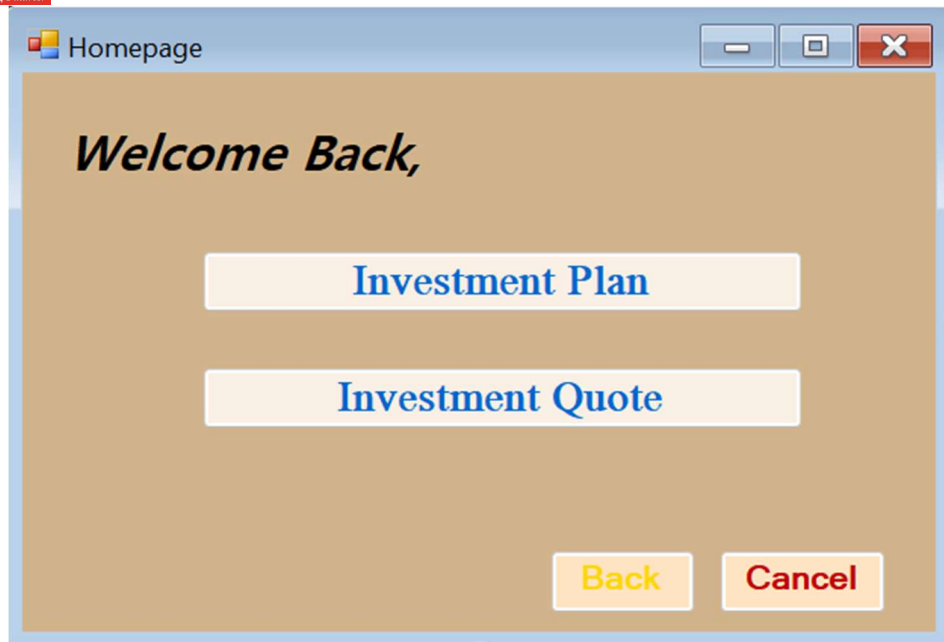


Figure 67: Investment Specialist homepage.

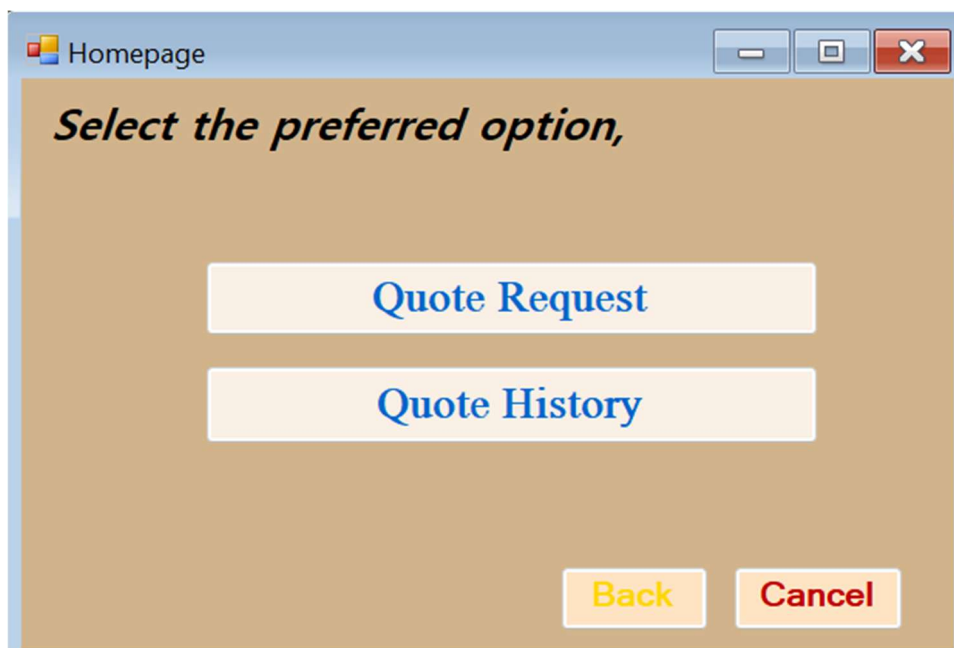


Figure 68: Investment quote homepage.

Dashboard

Investment Plan

Plan ID

Min Initial Lumpsum

Min Monthly Contribution

RBSX group per month

Estimated Tax

Duration

Max Predicted Return

Min Predicted Return

Max Investment per year

INSERT **UPDATE** **DELETE**

Back **Cancel**

Figure 69: Updating Investment Plans

History

Investment Quote History

Search

Convert data into;

Numerical Data **Graphical Data**

Back **Cancel**

Figure 70: Investment quote history.



The screenshot shows a web application window titled "Dashboard" with a standard Windows-style title bar (minimize, maximize, close buttons). The main content area is titled "Investment Quote Generation". It contains two columns of input fields. The left column includes: "Quote ID" (text input), "User ID" (text input), "Plan ID" (text input), "Plan Type" (dropdown menu), "Duration" (dropdown menu), "Start Date" (calendar picker showing Friday, January), "End Date" (calendar picker showing Friday, January), "Initial Lumpsum" (text input), and "Monthly Contribution" (text input). The right column includes: "Max Predicted Return" (text input), "Min Predicted Return" (text input), "Total fee" (text input), "Total tax" (text input), and "Total profit" (text input). At the bottom left, there are two large buttons: "SUBMIT" (green text) and "RESET" (orange text). At the bottom right, there are two smaller buttons: "Back" (yellow text) and "Cancel" (red text).

Figure 71: Investment quote generation form.

4.3.5 Interfaces for the user (client)

After the user logs into the system, it displays the homepage of the user. It contains options like currency transaction, investment plan and investment quote. The investment plan will lead to the selection of investment plan type which is desired by the user. The investment quote option will further lead to choices like new Investment quote and saved quotes, while the currency transaction button will lead to further options like new transaction and transaction history.

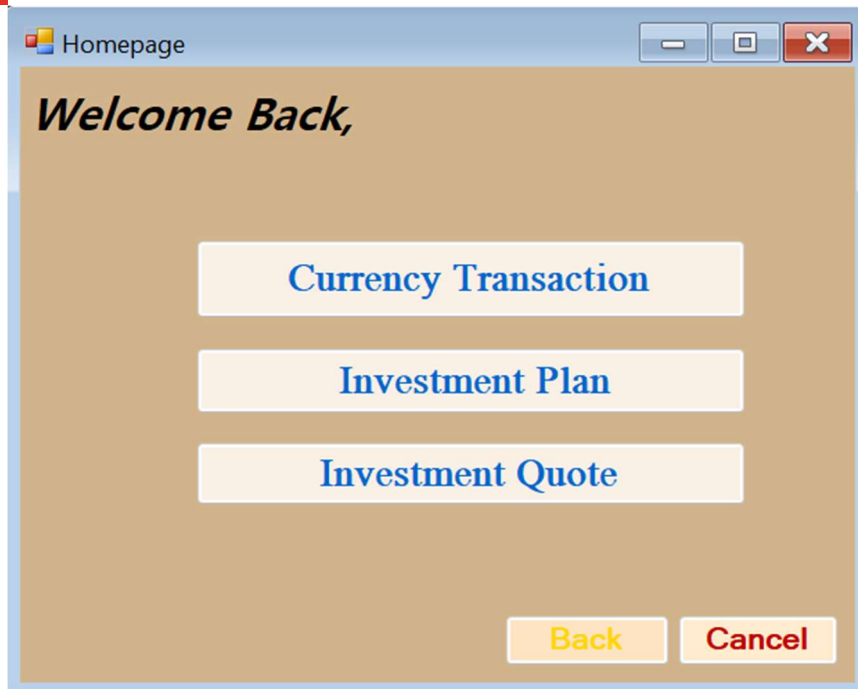


Figure 72: Homepage for user.

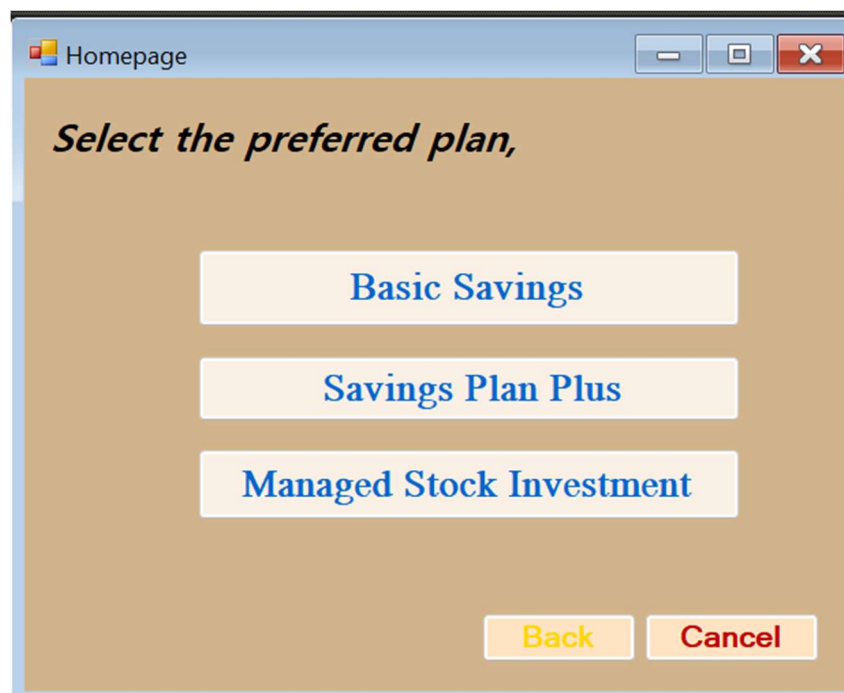


Figure 73: Homepage for Investment Plan.

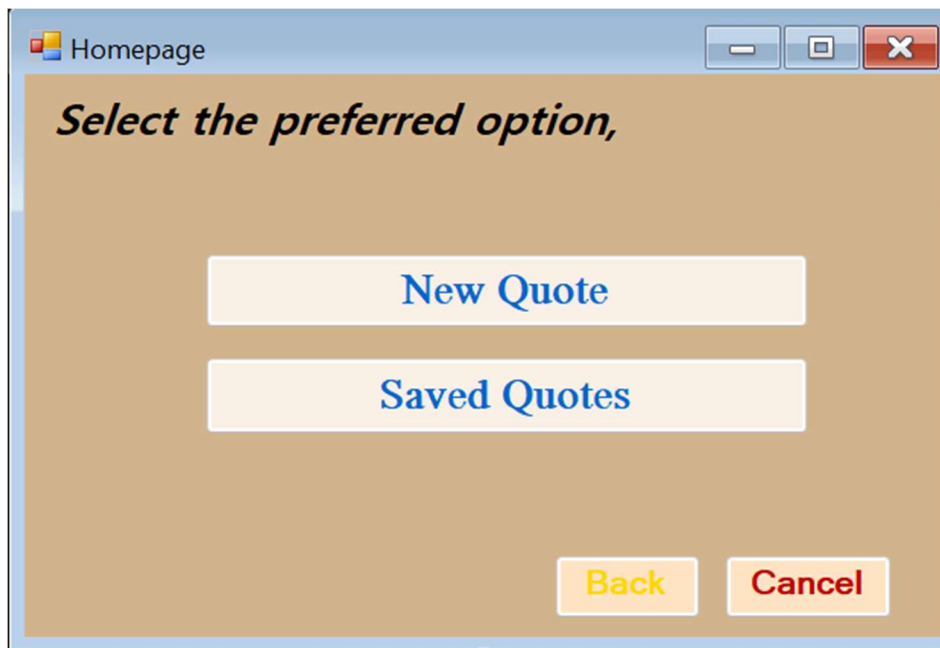


Figure 74: Homepage for Investment quotes

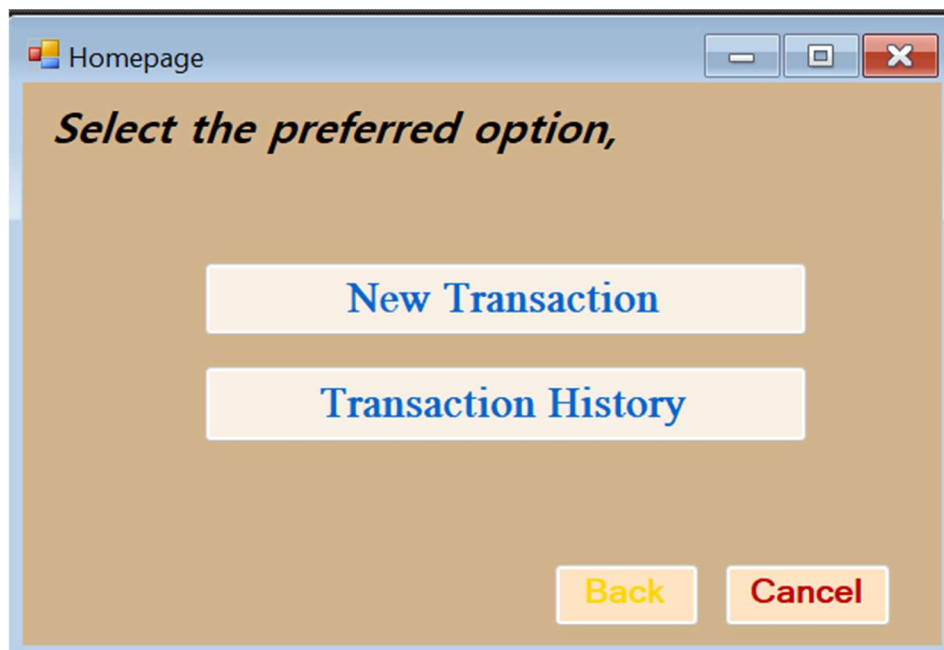


Figure 75: Homepage for Currency Transaction.

Dashboard

Saved Quotes

Search

Refresh

Back Cancel

Figure 76: Saved quotes page.

Dashboard

Investment Quote Generation

Quote ID

User ID

Plan ID

Plan Type

Duration

Start Date Friday , January

End Date Friday , January

Initial Lumpsum

Monthly Contribution

Max Predicted Return

Min Predicted Return

Total fee

Total tax

Total profit

SUBMIT RESET

Back Cancel

Figure 77: Quote generation form.

Dashboard

Currency Transaction

Transaction ID

User ID

Initial Currency

Target Currency

Converting Amount

Converted Amount

Transaction Timestamp

Transaction Fee

Transaction Status

SUBMIT **RESET**

Back **Cancel**

Figure 78: Transaction form.

History

Currency Transaction History

Search

Convert data into;

Numerical Data **Graphical Data**

Back **Cancel**

Figure 79: Transaction history page.

4.4 Feedback for my designs.

❖ Feedback Form Questions

Rating the Interface Designs of Enomy Finances.

B I U

Once you have finished looking at the PDF document containing all the interface designs of Enomy Finances that is sent to you along with this form, Please spare some of your time to fill out this form.

Name *

Short-answer text

Are you a technical person or a non-technical person *

☐ Technical

☐ Non-technical

Figure 80: Feedback form Questions- 01

How intuitive is the overall interface design? *

1 2 3 4 5

Very confusing ☐ ☐ ☐ ☐ ☐ Very Intuitive

Rate the navigation elements (menus, buttons, and links) in terms of easiness to locate and understand? *

1 2 3 4 5

☐ ☐ ☐ ☐ ☐

Does the interface maintain consistency in layout, colors, and typography across all pages? *

☐ Yes

☐ No

☐ Maybe

Figure 81: Feedback Form Questions-02

Is the content and information hierarchy clear and easy to follow? *

1 2 3 4 5

Very unclear ☐ ☐ ☐ ☐ ☐ Very clear

...

Rate the interface's effectiveness in balancing aesthetics and functionality? *

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Is the design accessible and inclusive (e.g., readable fonts, sufficient contrast, support for screen readers)? *

☐ Yes

☐ No

☐ Maybe

Figure 82: Feedback Form Questions – 03

Rate the interface's tailoring to the target audience and their needs? *

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Rate the use of interactive elements (buttons, forms, dropdowns) in designing the interface? *

1 2 3 4 5

Very Poor ☐ ☐ ☐ ☐ ☐ Excellent

What aspects of the interface do you find most effective or impressive? *

Long-answer text

...

What areas of the interface could be improved or redesigned? *

Long-answer text

Figure 83: Feedback Form Questions-04.

❖ Feedback Form Responses.

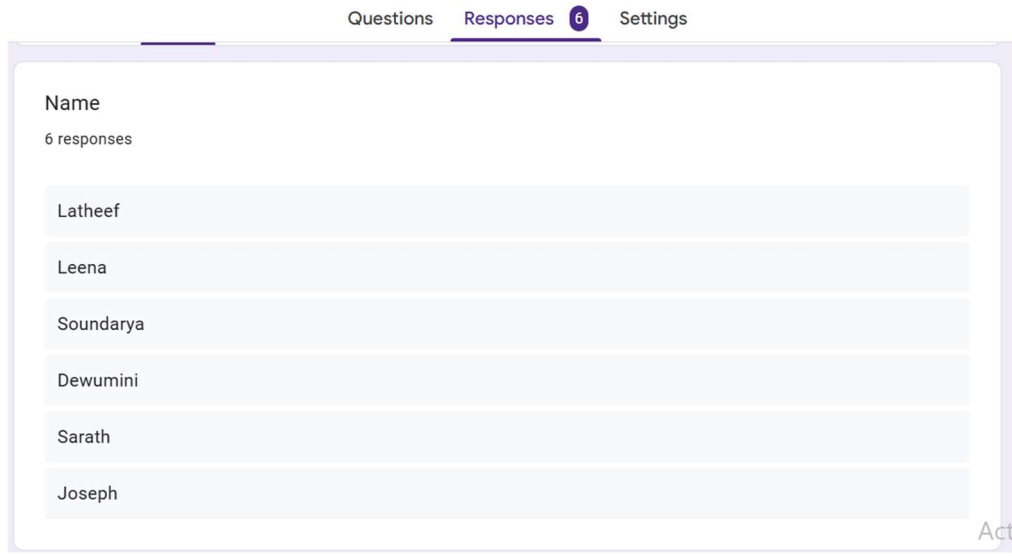


Figure 84: Feedback form response-01

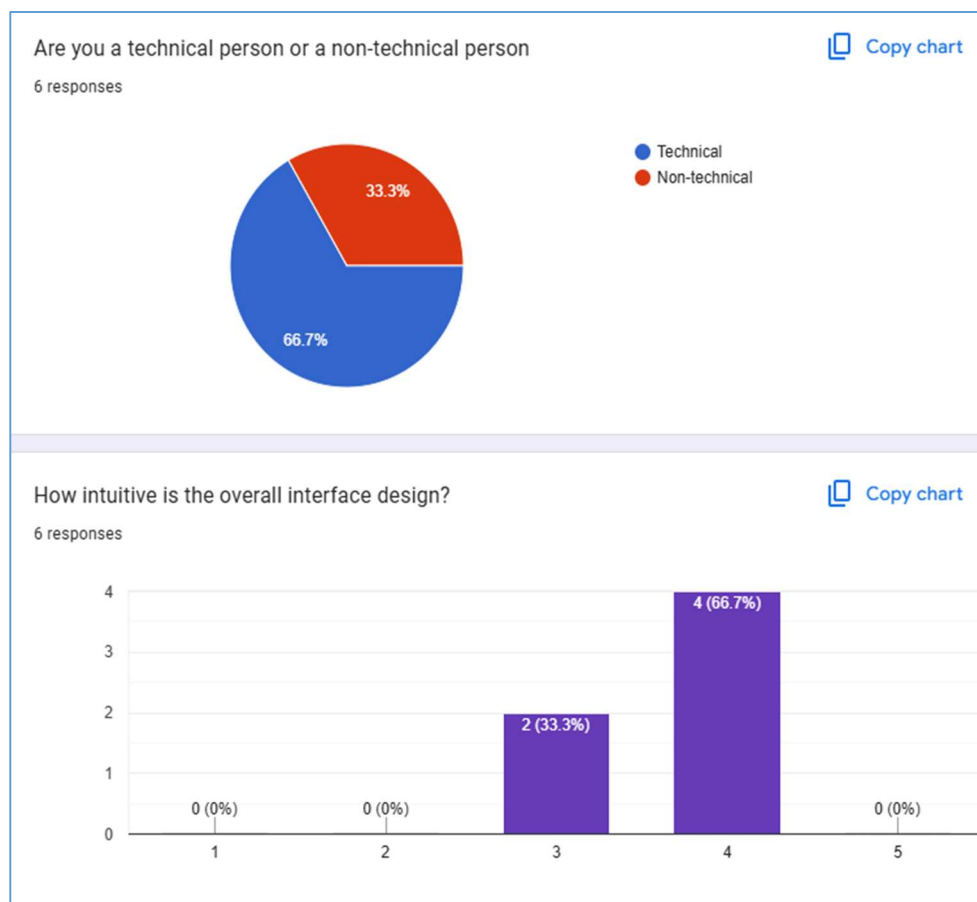


Figure 85: Feedback form response--02

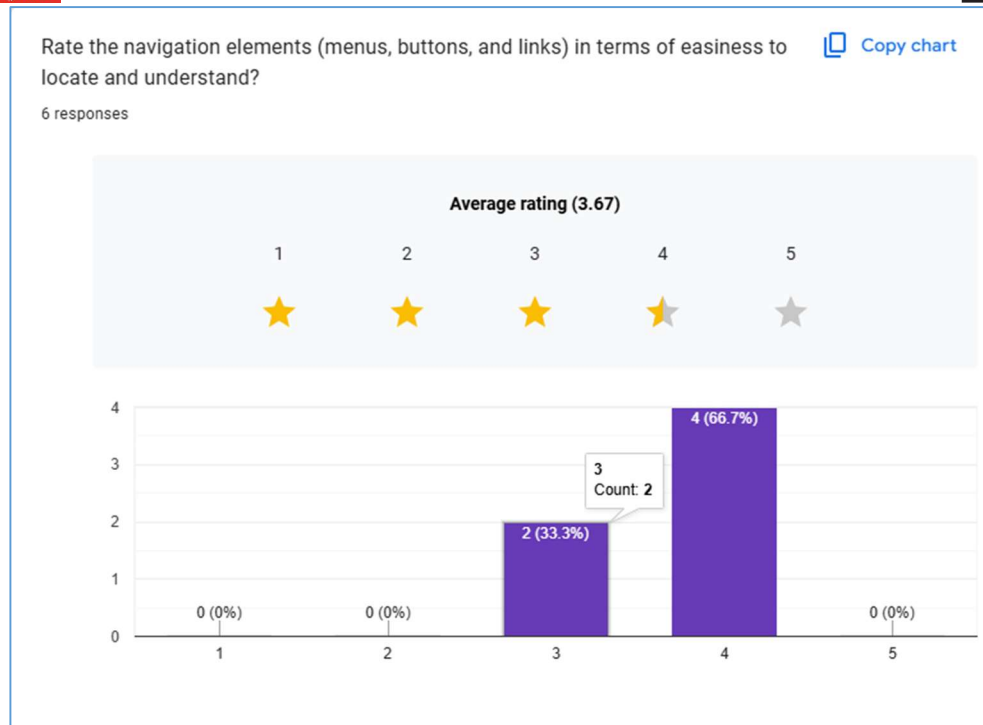


Figure 86: Feedback form response-03.

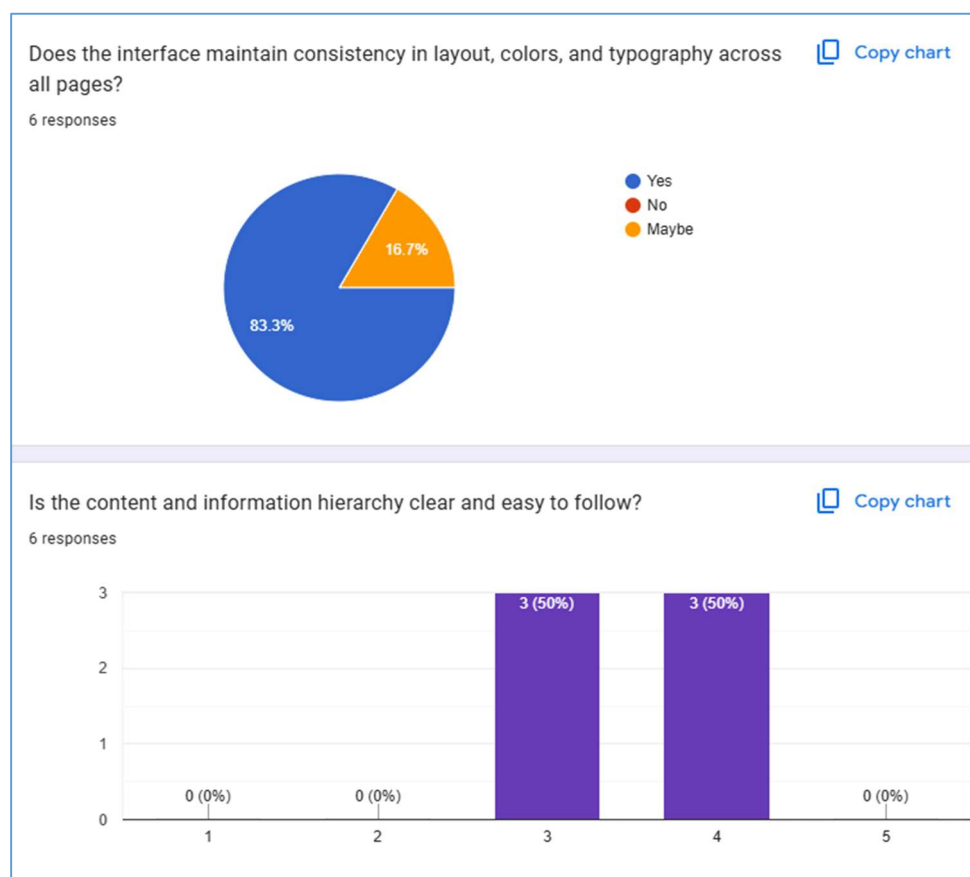


Figure 87: Feedback form response-04

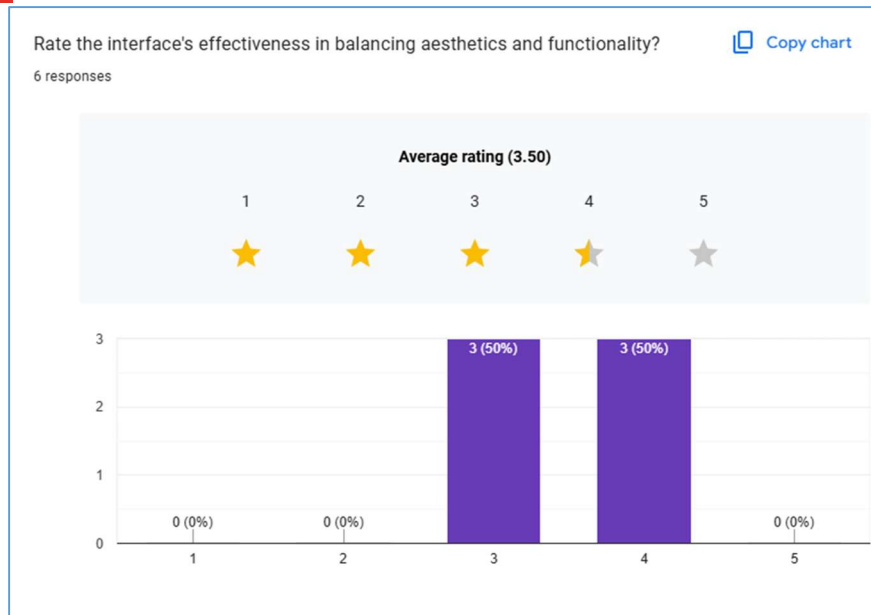


Figure 88: Feedback form response-05

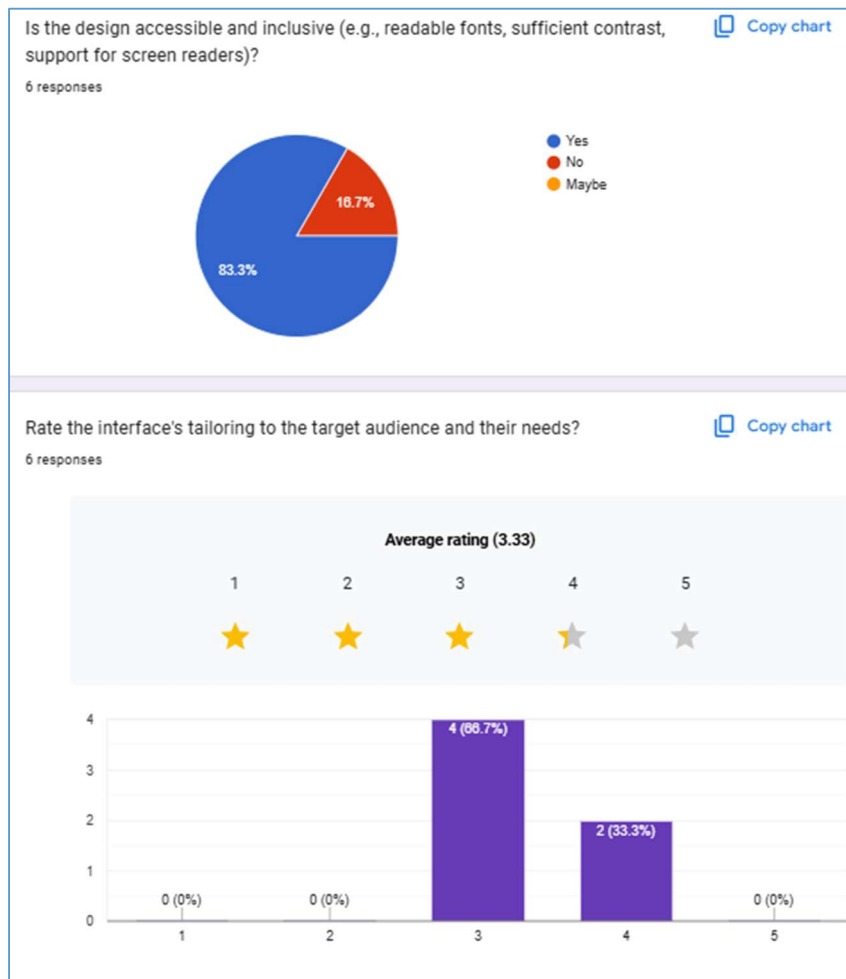


Figure 89: Feedback form response- 06



Figure 90: Feedback form response-07

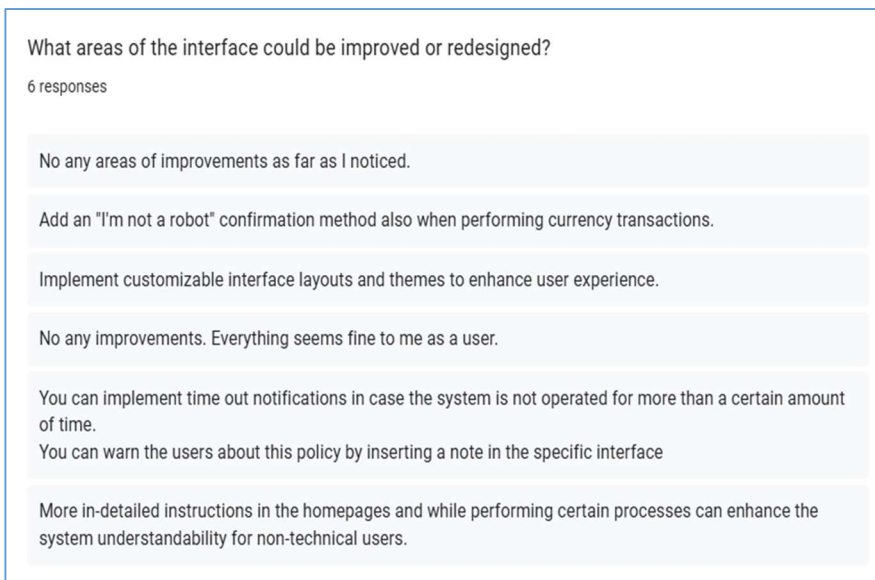


Figure 91: Feedback form response-08

4.4.1 Analysis of the Feedback.

The feedback is gained from both technical and non-technical users in order to capture the opinions of people with diverse range of knowledge in IT and Security. We can observe that the feedback gained suggests that the interface design is organized and effective, with users rating its intuitiveness and navigational simplicity. A majority (68.7%) scored the interface design at 4 out of 5, while the navigation components (menus, buttons, and links) obtained an average rating of 3.67, demonstrating that users usually find it straightforward to locate and understand, but there is space for improvement.

Uniformity in layout, color schemes, and font over all pages received commendation, with 83.3% of participants confirming its effective maintenance. However, 16.7% expressed uncertainty, suggesting that certain aspects may require additional development for a more consistent experience. The clarity of material and information hierarchy received ratings between 3 and 4, indicating that although the content is predominantly comprehensible, enhancements might be made to increase readability and organization. The equilibrium between aesthetics and functionality received an average score of 3.50, indicating largely favorable feedback, but some people perceive potential for improvement in this area.

The interface's accessibility and inclusivity gained a positive response too, with 83.3% affirming its accessibility; however, 16.7% of users expressed doubt, suggesting that enhancements for features such as screen readers could be beneficial. The customization of the interface for the target population received a rating of 3.33, indicating that it partially satisfies user requirements, however additional personalization methods could enhance the experience. The incorporation of interactive features, like buttons and dropdowns, received a score of 3.33, suggesting that although they are available, their usability and efficacy can be improved.

Users found various parts of the interface impressive, including simple navigation pathways, role-based access controls for greater security, a calming color scheme, and a separate error log panel to improve operational efficiency. Security elements such as read-only data fields were valued for their contribution to data authenticity.

In terms of improvement areas, users proposed adding CAPTCHA verification to currency transactions for increased security, implementing timeout warnings for idle sessions, and offering more extensive instructions on homepages and during complex processes to aid non-technical users. Additionally, suggestions like changeable layouts and themes were also made to improve user experience. Overall, the feedback reveals that the system is effective and well-received, with particular strengths in navigation,

security, and design consistency. However, implementing the suggested modifications could further increase the system's usability, security, and user-friendliness.

4.5 Interpreting if my designs are compliant in terms of security and maintainability.

4.5.1 Security

a) Secure login.

The system is designed such that, each time a user has to enter into the system, the user has to enter his/her username along with the password. This ensures that only authorized users can access to the system. Moreover, only three attempts are granted for the login process at a given time. The system will automatically get locked denying access, if all three attempts fails.

b) Role-based access

The system initially displays a user type selection screen, where users have to select their choice. This also enables users to gain limited access to only the functions that could be permitted for them. The system also contain separate interfaces for each type of users, which other type of users are denied access to. This keeps all the sensitive data secured and builds trust among the users.

c) Secure data inputs.

At instances where data inputs are required. For instances at currency transaction and Investment quotes generation processes, input validation mechanisms are implemented in order to prevent SQL injection or XSS. This is done by the use of radio buttons, combo boxes and date time pickers rather than using text boxes. Allowing read only options and allowing automated fills for some of the data, like transaction timestamps and total tax for the investment also ensures secure and valid data inputs.

d) Multi-factor authentication

The system has implemented OTP systems while performing transactions and generating quotes. Only when the OTP is verified by the system, will the transaction proceed or the quote generation request will be sent. This enhances the security and validity of the processes performed within the system.

e) Log-out options

In each and every interface of the system, a cancel button is attached. Upon clicking the cancel button, the user will be given the option to log out of the system. Also, if the user just leaves the system, an automatic log out from the system is also implemented.

4.5.2 Maintainability

a) Separate sections for different processes

The system contains separate sections for different processes. For instance, the user must access different sections to perform transactions, view investment plans and generate quotes. This type of mechanism helps in easy updating and debugging processes.

b) Search and filter options.

The system provides with advanced search and filter mechanisms. This benefits in quicker locating of data or settings.

c) Tracking and troubleshooting.

The system contains error panels where the admin can visit and fix the errors. The error panel contains details about the error type, its severity, location, etc. The system will also automatically locate to the specific error when a certain error is clicked and locate button is pressed.

d) Consistent UI elements

The system uses reusable design components like buttons, texts and interfaces. This helps in reducing redundancy and simplify updates.

4.6 Development process using SDLC models.

The development of Enomy Finances, a financial management system, should be done systematically and methodically to ensure that every phase is well-planned and then executed. The Waterfall Model is best for this project because of its linearity and sequential nature. This can ensure clarity, predictability, and tight control during the development process. Enomy Finances is concerned with correct financial calculations, security, and compliance, and therefore it needs a systematic approach that minimizes changes to the previous stages. This is also a reason why waterfall method is best suited for the development process of Enomy finances.

The analysis of requirements is the first step for development. It involves a complete gathering and recording of the user and system requirements. That includes analyzing what financial functionality is necessary, the security protocols required, and what compliance standards need to be met for this project. Key deliverables include comprehensive requirement documents outlining supported currencies, limits on transactions, fee structures, real-time updates, and reporting tools. This will provide a reference for subsequent stages to keep the stakeholders and developers on the same page.

The system design phase comes after the requirement analysis. During this phase, transformation of the requirements into an elaborative architecture is takes place. Designers create wireframes and mockups to represent how the user interface would look, the navigation flow, and the critical features to be included in the system, such as dashboards, transaction history, and reports. This phase ensures that the structure of the system will support scalability, maintainability, and security.

The next stage is implementation. It gives way to actual coding and development. Here, the developers start building code based on the designed solution, per the best industry practices to ensure security and performance. For Enomy Finances, user authentication, data encryption, and real-time update implementations are crucial. Further, modular coding methods are followed. That is, for processes like the currency conversion or the reporting module, testing is done one by one instead of testing after entire system development. Hence, this benefits in preventing propagation of faults across the system.

Once the implementation is complete, the project moves into the testing phase. Extensive testing is done to ensure that the system meets all the functional and non-functional requirements. The test cases range from unit testing to integration, system, and user acceptance testing. In the case of a financial management system such as Enomy Finances, a lot of emphasis is placed on testing the accuracy of calculations, secure handling of transactions, and performance of the system under huge load. Any issues or challenges identified are resolved before moving into the deployment stage.

The deployment stage is the process of delivering the final system to the end-users. Enomy Finances system is deployed in a live environment, ensuring all configurations are set up for smooth operation. Training sessions and user guides are provided to guide the users to use the system effectively. Finally, the process of maintenance keeps the system running and up-to-date. Updates are made regularly in order to increase functionality, eliminate bugs, and repair security holes. Periodic audits, continuous monitoring, error detection systems and feedbacks help make sure the system will evolve with users' changing needs over time.

Activity 05

5.1 Tracing Software Requirement throughout the lifecycle

The Enemy Finance project is aiming to deliver a strong and secure financial management system that meets user needs, management requirements, and compliance standards. Tracing the requirements of the software throughout the lifecycle is very much important. It ensures that the objectives of the project are aligned at each stage of development, right from the initial requirement gathering to the final implementation.

In this requirement analysis phase, the software requirements are gathered through interviews with stakeholders, board meetings, and surveys. Interviews were conducted with the main management committee and IT team to track the drawbacks and requirements of the project. Board meetings helped in brainstorming and prioritize features, and surveys were conducted among a small group of randomly picked end-users. The survey was focused in discussing specific aspects of the system like the user interface and access control practices. All these requirement gathering processes outlined priorities that included easy-to-use user interfaces, comprehensive system security, and capability for detailed financial reporting. Functions of currency conversion, real-time exchange rate updating, and transaction tracking were also some other vital requirements.

Besides that, system security was one of the top concerns. This can be solved by featuring encryption for transactions, multi-factor authentication, and role-based access control. Reporting capabilities were also an important feature to be considered. The management wanted reports on financial records, summaries of transactions, and regulatory compliance. All these requirements were captured in detail in a Software Requirements Specification (SRS) document serving as the single source of truth for the project.

To ensure all these requirements remained central throughout the lifecycle, a Requirements Traceability Matrix (RTM) was employed. The RTM mapped each requirement to specific deliverables, such as design elements, test cases, and implementation modules. This tool provided a clear way to track the progress of requirements across all phases of the project.

During the design phase, requirements identified during the requirement analysis phase were translated into system designs such as wireframes and architectural diagrams. By referring and following

the RTM, it was ensured that all functional and non-functional requirements were included in the designs and nothing was left out.

In the prototyping phase, a basic working model of the system was developed to validate the important functionality alignment with the requirements. For example, prototypes of the dashboard and transaction records have been tested with stakeholders to ensure they meet the expectations on usability and reporting. The RTM helped verify that each prototype addressed specific user and system needs as documented in the SRS.

In the Testing Phase, All the testing included the traceability matrix, whereby test cases were linked directly with the initial requirements. Functional tests-cases like ensuring currency conversion and financial reporting were interconnected with security tests like ensuring the implementation of encryption and access control requirements. As the RTM allowed for simple confirmation that each requirement was tested, making sure that all standards of the project were met was done without much effort.

Finally, during Implementation and Deployment, mapping of developed components back to RTM at implementation was done in order to ensure completeness. Additionally, final checks on the deployed system were done to assure all requirements would work as expected by the management.

Besides the RTM, a requirement checklist- a list of requirements with status and dependencies was also used to keep the project in track with its requirements. Also, collaboration tools, like project management softwares supported to maintain visibility of requirement statuses across the entire team, thereby enabling tracking and communication among the team effectively.

5.2 Approaches to improve Software Quality

Software quality is the most crucial success factor of the Enomy Finances project. The following are some ways in which the quality of the software can be improved.

- Quality Assurance (QA) Testing

The main role of QA testing is finding and resolving issues in the earliest stages of development. This helps to minimize costs by preventing faults from reaching the end users. Also, extensive and detailed documentation of testing techniques and outcomes provides a good way for continuous improvements. Given below are some of the testings that can be conducted in the QA testing criteria.

1. **Functional Testing:** This is the testing of all the identified software requirements to check and make sure whether the functions are working as expected. For example, testing how currency conversion, transaction limits, or fee generation work will come under functional testing. Manual as well as the use of automated testing technologies like Selenium or Postman-can help validate functionality at ease.
2. **Security Testing:** Security testing is a must because of the sensitivity of the information stored in the system. This can be done through penetration testing, vulnerability assessments, and verification of proper working of encryption methods. This test is vital in order to ensure that no data breaches or unauthorized access can take place to the system. Features like multi-factor authentication and secure APIs should be extensively verified to maintain user trust and reputation of the organization.
3. **Regression Testing:** Regular regression testing ensures none of the new additions or changes has affected already existing functionalities. Automated regression tests confirm that a system remains stable and reliable while the system keeps evolving.

- Code Reviews and Peer Feedback

Code reviews and peer contributions are two group activities that vastly improve software quality and maintainability by setting up a secure and scalable basis for further development. In the Enomy Finances project, such techniques yield several advantages like,

1. **Bug Detection:** The primary idea behind this is having team members go through each other's code. It helps identify various problems, such as logical errors, coding standards violated, or missed edge cases. However, evaluation with bias may be a concerned disadvantage.
2. **Information Sharing:** Code reviews encourage information sharing among team members, creating a deeper understanding of the system and improves team spirit. Junior developers can get the opportunity to learn from older developers, while experienced team members can gain insights into fresh perspectives from the junior members.
3. **Consistency and Standards:** Peer reviews ensure that coding standards are followed, making the code more readable and maintainable. For example, following consistent naming conventions for variables and principles of modular design make the software easier to manage and scale up.
4. **Team Collaboration:** A culture of feedback promotes team collaboration and accountability.

5.3 Evaluation of Solution Suitability for Enomy Finances

The Enomy-Finances solution proves its accuracy by corresponding to some of the big project objectives. The system should be designed to be simple and modular. Main functionalities like currency conversion tools, transaction history, and reporting should be visually available and logically laid out to guarantee that users can quickly navigate the site, find information, and perform tasks with a minimum of effort. Features like interactive dropdown menus for selecting currencies, searching facilities to locate particular transactions, and filtering to create customized reports enhance the ease of use and minimize human input in processes. Real-time validation checks at the time of data input like warning of invalid entries or error messages on transactions exceeding the limit will further enhance data accuracy and user satisfaction.

When focusing on security of the system, advanced mechanisms like Multi-factor authentication and the verification of user identity Role Based Access Controls, and with the help of protocols that assures data encryption, can contribute significantly to compliance and data protection regulations while preventing the leakage of customer data. The main functionality also includes an advanced reporting capability that allows the user to build customized reports on currencies and transaction categories. Numerical and Graphical representation options for the reporting category provides users to take informed and clear decisions. While export options in PDF and Excel make it easy to share and record.

Real-time capabilities like immediate updates of currency exchange rates and transaction statuses, ensure that the system remains accurate and updated, hence taking care of the dynamic nature of the financial sector. Effortless integration into the wireframe design delivers solid information to consumers for vital decision-making. By embracing these features, the Enomy-Finances solution not only meets but outperforms the project objectives, showing its capacity to handle complicated financial activities while putting security, usability, and adaptability in the service of varied stakeholder needs.

5.3.1 Examples from the Interface Design that proves meeting project needs.

- Currency transaction and conversion functionality

Dashboard

Currency Transaction

Transaction ID

User ID

Initial Currency

Target Currency

Converting Amount

Converted Amount

Transaction Timestamp

Transaction Fee

Transaction Status

SUBMIT **RESET**

Back **Cancel**

Figure 92: Currency Transaction functionality.

- Able to view Transaction History.

History

Currency Transaction History

Search

Convert data into;

Numerical Data **Graphical Data**

Back **Cancel**

Figure 93: View transaction history.

- Reporting Options.

Print Filter Edit

Back Cancel

Figure 94: Reporting Options.

- Using of interactive components like Dropdown menu and Radio buttons.

Quote ID Max Predicted Return

User ID Min Predicted Return

Plan ID Total fee

Plan Type Total tax

Duration Total profit

Start Date Friday, January Approval Status ☐ Approved

End Date Friday, January ☐ Declined

Initial Lumpsum

Monthly Contribution

SUBMIT RESET

Back Cancel

Figure 95: Usage of interactive components.

- Search bars for locating options

Dashboard

Currency Rate

Rate ID **Search**

Financial Manager ID

Currency name

Last Updated

Rate for 1GBP

Rate Validity ☐ Valid ☐ Not valid

REGISTER **UPDATE** **DELETE**

Back **Cancel**

Figure 96: Implementation of search bars.

- Filter tools to generate specific reports.

Report

Investment Plan Report

Print **Filter** **Edit**

Back **Cancel**

Figure 97: Implementation of filter tools.

5.3.2 Examples from the Interface Design that proves requirement alignments.

- Role based access controls



Figure 98: Role based access controls.

- Customizable reports (Edit option)

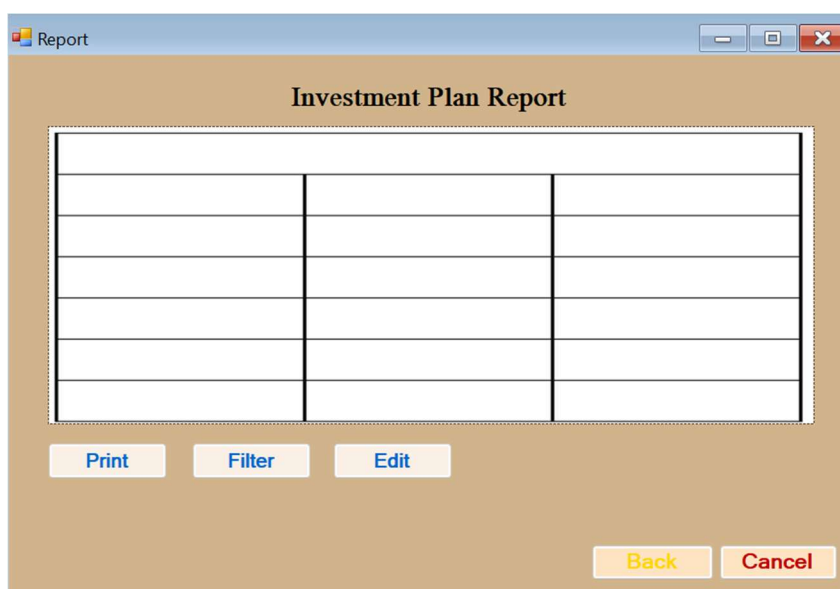


Figure 99: Customizable reports.

- Textual, Numerical and Graphical data representation options (Textual data representation on default)

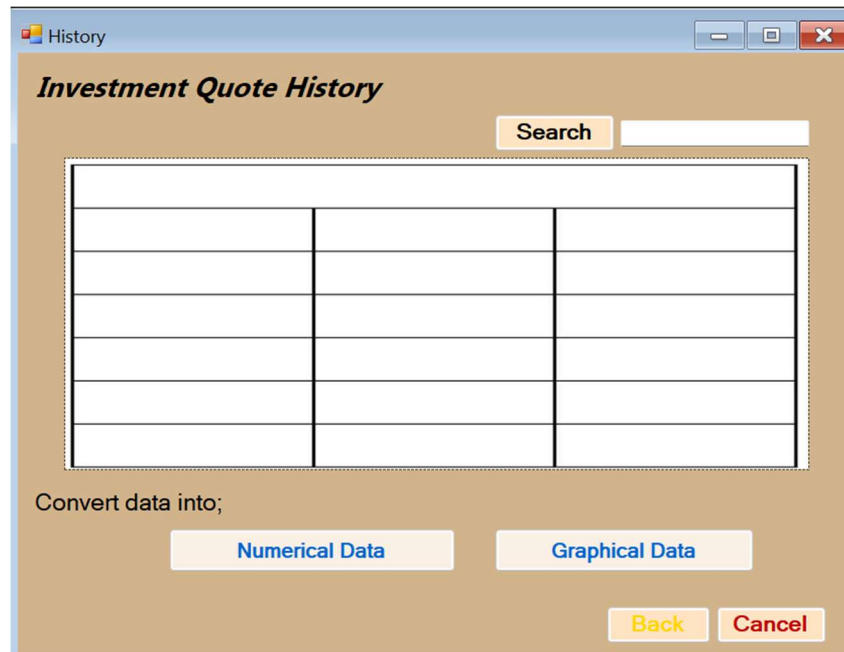


Figure 100: Data representation in many forms.

- Report exporting options (Print).

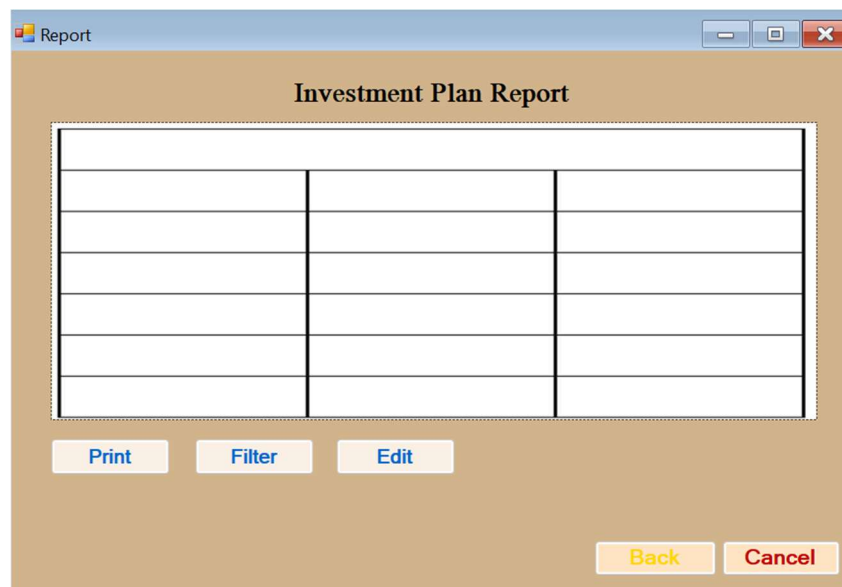


Figure 101: Report exporting options.

5.4 Suitability of chosen tools and techniques.

The tools and methodologies for Enomy-Finances project were carefully picked to be appropriate for meeting the project objectives concerning visualization, functionality, and ease of use. Figma or Adobe XD was used for wireframing the interface because it provides drag-and-drop features and allows vast collaboration options. These capabilities enabled the creation of beautiful, interactive designs that captured the flow and placement of major components. It also allows for quick prototyping which enables iterative changes in response to client input. Also, flowchart tools such as Lucidchart and Microsoft Visio provided clear illustration of algorithmic workflows. These technologies provided more clarity in it's appearance, thus making it more easily understood by the stakeholders and confirming the logical design of the system. When comparing to alternative tools, considerable strengths existed with the chosen tools. For instance, Balsamiq is another well-known wireframing tool, its focus on low-fidelity designs made it less suitable for visualization of interactive elements and aesthetic qualities. Similarly, while open-source options like Dia could have been used for flowcharts, their limited functionality and more complex interfaces would have made the design process slower and more complicated. Figma's cloud-based functionality and Adobe XD's extensive plugin support outperformed alternatives by enabling real-time collaboration, streamlined integrations with development tools, and efficient handoffs to developers. Visual Studio Code was found to be very helpful in algorithmic and data design for creating and debugging the underlying code base. Its extensions functions, advanced debugging, and version control make it superior to simple IDEs like Notepad++ or Sublime Text in handling complex logics and data processing that the project needed. All of these tools and methodologies put together created the best mix of functionality, usability, and flexibility. Hence proving that they were the most adequate choice for the Enomy-Finances project.

5.5 System Investigation and Design Process

The initial investigation of needs for the Enomy Finances serves as a cornerstone in establishing a structured and effective design process. During the requirement analysis phase, a thorough analysis of the user and technical requirements, and the possible operational challenges was conducted. By gathering input from the stakeholders and analyzing the drawbacks of the existing system, the team got a comprehensive ideas of both, the non-functional as well as the functional requirements. By the end of the investigation, some key needs were observed; like seamless currency conversion tools, personalized investment managements, robust security and an intuitive user interface.

These insights that was gained were meticulously translated into a structured design. The need for a intuitive interface was addressed by incorporating clean and simple layouts along with easy navigation paths into the design. Concerns on data security were met by embedding secure login protocols, encryption mechanisms and role based control levels. Moreover, the need for efficiency in handling calculations for currency conversions and investment quotes was solved using the implementation of optimized algorithms and modular design principles.

This structured design approach not only enhanced system reliability, but also ensured adaptability for future enhancements. The modular designs incorporated supports the system to add new currencies and investment plans while maintaining the core functionality of the system. Also, the implementation of user feedback during design phase allowed for iterative adjustments, thereby aligning the final design more close to user expectations.

Therefore, we can conclude that the process of shaping an effective design for the Enomy finances was highly depended on the initial systems investigation. The systematic alignment between investigation and design phases significantly contributed to the solution's overall effectiveness.

5.4 Risk Mitigation

The table given below highlights the risks that were identified during the investigation process for the Enemy Finances project and the mitigation strategies employed during the design phase.

Risk	Description	Mitigation Strategy
Data Security	Risks associated with unauthorized access, data breaches and leaking of sensitive information	Implement secure login protocols, Encrypted data storages, Role-based access controls along with regular security audits.
User Accessibility	Problems like the system being complicated to use for the individuals with disabilities, or individuals with less technical literacy.	Design an accessible user interface that could be supported by all screen sizes with a simple layout. Implement features like screen readers and adjustable texts.
User Satisfaction	Risks related to users being dissatisfied with the interface due to confusing navigation paths and lack of functionalities.	Conduct frequent user feedback sessions and usability testing to align the design with user preferences.
Currency value accuracy	Risks related to performing currency conversions with outdated or incorrect currency rates.	Implementing a fallback mechanism for offline conversion rates. Gain updates about the currency rates from a well trusted and updated source.
Budget Overrun	Risk of exceeding the allocated budget due to the occurrences of unexpected changes and incidents	Establish a clear scope, conducting in-detail cost estimations, allocating a certain amount from the budget for unexpected emergencies.
Regulatory Compliance	Risk due to non-compliance with financial regulations and data protection laws.	Embedded compliance checks during design, aligned database management with GDPR standards, and consulted legal advisors.
System Reliability	Risks like system crashes, slow performance, and inability to handle a heavy load	Utilize modular design principles, load testing, and robust error-handling mechanisms.

5.5 Advantages of Data-driven Softwares

The data-driven aspects of the Enomy Finances system are vital for its effectiveness in ways that enhance reliability, responsiveness, and capability while meeting the core financial needs of the company and its customers. Real-time data access ensures that all currency conversions, savings estimates like taxes and fees, and investment calculations are done with the most accurate and up-to-date information and methods possible. Application Programme Interfaces pull refreshed exchange rates and market data into the system for end-user decision-making. It reduces the probability of errors made based on data that has gone stale, therefore maintaining confidence in the system. An integral feature of all data-driven system is its graceful handling of faults. It needs to cope well with these errors without affecting users' trust. Effective error-handling systems ensures that occurrences of unexpected errors or disruptions do not affect the user experience or affect financial data integrity. Diagnostic logging, automatic data caching, and fail-safe methods allow users to easily restart their work in case of disruptions, making the system highly reliable. For instance, when a client is filling an investment quote form and there is disruption to the network, the client's inputs are stored locally and resynchronized with the database upon reconnecting again. Moreover, data-driven approaches also supports for decision-making for both staff and clients. Dynamic, real-time reports allow customers to gain meaningful insights into financial trends, account summaries, investment performance, and transaction records.

Additionally with the help of data-driven systems, the clients will be able to see their forecasted returns for 1 year, 5 years, and 10 years, plus detailed breakdowns of the fees, taxes, and earnings. The staff will be able to create custom reports summarizing client portfolios to help them provide more focused and meaningful financial guidance. This functionality will not only enhance operational efficiency but also contribute to the business in realizing its mission of delivering proper and clear services. Besides, the data-driven system also supports scalability and smoothness. As the Enomy Finances system scales up, due to the centralized database and modular architecture, new feature like more investment plans, new currency rates and enhanced features for reporting can be easily integrated without disturbing ongoing processes and functionalities. This capability of the system, to adapt with changing user needs and industry advancements while maintaining data accuracy guarantees sustainability over a long period.

In a nutshell, we can say that the data-driven architecture of the Enomy Finances system is helpful in achieving its financial goals. Real-time access to data promotes accuracy and speed, strong error handling improves reliability and availability, and dynamic reporting raises the user's decision-making capabilities.

These benefits guarantee that the system will not only meet the current needs of its users but will also foster the long-term growth of Enomy Finances and its operational excellence.

5.6 Performance Review against Requirements.

The performance of the Enomy Finances system was carefully assessed in order to ensure that it fulfills the defined requirements, functionalities, usability, security, and user experience.

1. **Functionality-** The system successfully meets important functional requirements such as currency transactions, individual investment plans and investment plans, and dynamic report generation. The currency conversion tool provides accurate values into two decimal places across different currencies, including GBP, USD, EUR, TRP and more, enabling updated currency transactions. The savings and investments module provides thorough breakdowns of returns, fees, and taxes for various investment quotes that is created according to the user inputs. Investment plans provide the user a guidance on which type of investments will suit them according to their incomes and outgoings.

2. **Usability-** A user-friendly interface offers accessibility for clients and staff. The design features with straightforward navigation, clear labels, and interactive elements guides the users through complex financial operations. Different types of data representations like Textual, numerical, and graphical significantly helps in proper and clear decision-making for clients.

3. **Data Security-** Data security is a major focus when it comes to designing software. Incorporating comprehensive features like secure login procedures, encrypted data storage, and role-based access controls; system data like transaction histories and personal information is protected by compliance with industry security standards, ensuring the system fits with modern cybersecurity procedures.

4. **Error Handling-** Error-handling techniques were tested thoroughly, guaranteeing that the system fails gracefully under unforeseen circumstances. Real-time data caching prevent data loss allowing users to continue their tasks easily following recovery.

5. Dynamic Reporting- Reports provided by the system are accurate, configurable, and suited to user requirements, enabling both clients and staff to make educated decisions. For example, staff can provide client portfolio summaries, while clients can envision long-term financial goals through detailed savings and investment projections.

➤ **Successes**

- The integration of real-time data and user-friendly design dramatically boosts usability and efficiency.
- Feedback from stakeholders enhanced the accuracy of calculations and the significance of the facts given.
- Robust security measures ensured high confidence in data protection and system reliability.

➤ **Limitations**

- Some limitations were noticed, such as slight delays in data synchronization during high network loads. These events were infrequent but could harm user experience during peak demand.
- Initial onboarding comments revealed that advanced users may benefit from additional configurable analytics capabilities, which could be addressed in future versions.

In summary, the Enomy Finances system demonstrates high success in completing the objectives, with major accomplishments in functionality, usability, and security. Addressing the mentioned shortcomings in future will significantly enhance the system's effectiveness and user happiness.

5.7 Acting on Feedbacks to improve Solutions.

Feedback is an important aspect in refining any system, as it works to align the design with the expectations of users while improving functionality and usability. While developing the Enomy Finance system, feedback from peers and stakeholders was useful to identify areas of strength and suggestions for change. These served to create a more robust, friendly, and safe platform.

The system was also appreciated for having a clear navigational path, role-based access controls, and segregation of processes, which made it more user-friendly and secure. The attractive color scheme used and the provision of read-only options in forms for data entry were also appreciated for enhancing user experience and data integrity. The dedicated panel for error logs was appreciated as a very useful feature for enhancing operational efficiency. These were retained since these features met user needs in the right manner.

Feature enhancements were desired to enhance safety and usability. For this, following modifications were carried out. Employed a money transaction process was modified by implementing a CAPTCHA mechanism which prevents fraudulent operation. To enhance the security further, session timeout notifications were developed in which the user was notified when the system was idle for a certain period of time. Lastly, clear instructions were given on the home page and steps necessary to make the system usable by a non-technical user. Through acting on these comments, Enomy Finances got polished into a safe, accessible, and user-oriented platform.

5.8 Future Recommendations.

To guarantee the Enomy-Finances system remains a stable and adaptable solution for its customers, it is necessary to explore future modifications that solve present constraints and fulfill evolving requirements. These recommendations focus on increasing functionality, accessibility, security, and scalability to better alignment of the system with the organization's long-term objectives.

One notable enhancement is the attachment of complex data analytics tools, such as predictive modeling and trend analysis. By giving deeper insights into market habits and financial prospects, these tools can guide clients to make better informed decisions and assist staff in finding profitable patterns and tactics. This will not only improve the decision-making processes of the users but also promote and portray Enomy-Finances as a forward-thinking leader in its respective sector.

Improving the system's user interface (UI) accessibility is another crucial area for improvement. Adding capabilities like as screen reader compatibility and keyboard navigation would guarantee the platform is comprehensive and consistent with modern accessibility standards. This would increase the system's accessibility to persons with diverse abilities, indicating the organization's dedication to vastness and user happiness thereby increasing its demand and reputation.

To address concerns regarding data security, the system could benefit by adding additional security measures, such as multi-factor authentication (MFA), biometric controls and AI-driven anomaly detection. These changes would improve the system's security against illegal access, data breaches and evolving cyber threats. Enhanced security would also increase user trust and compliance with industry requirements.

Another idea is the development of more configurable reporting tools. Providing possibilities for customers to specify specific metrics, choose various data visualization formats, and export reports in different kinds of files would boost productivity and efficiency for staff while enhancing the overall user experience.

Lastly, enhancing the system's scalability and performance is necessary to plan for future expansion. Implementing load-balancing techniques and utilizing scalable cloud infrastructure will ensure the system functions well during peak demand and can accommodate a rising client base. This would support the organization's expansion ambitions while maintaining a high quality of performance and reliability. These changes together addresses significant areas for improvement in the current system. Advanced analytics would give useful data, while accessibility modifications promote inclusivity and broader usability. Enhanced security measures prevent hazards, configurable reporting features respond to various user needs,

and scalability upgrades prepare the system for future demands. By implementing these ideas, Enomy-Finances may present itself as a highly-efficient, client-focused organization capable of handling the demands of a developing financial market.